

Measurement of inclusive and differential fiducial cross sections of Higgs boson production in the $H \rightarrow 4\ell$ decay channel in proton-proton collisions at 7 TeV and 8 TeV

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<https://twiki.cern.ch/twiki/bin/viewauth/CMS/HIG14028QA>
- Journal target TBD, goal is paper submission before summer conferences (EPS, HH, LHCP)

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Abstract

In this document we present measurements of inclusive and differential fiducial cross sections of the Higgs boson production in proton collisions at center-of-mass energies of $\sqrt{s} = 7$ TeV and $\sqrt{s} = 8$ TeV using the $H \rightarrow 4\ell$ decays ($\ell = e, \mu$). Measurements are based on 5.1 fb^{-1} (19.7 fb^{-1}) of proton 7(8) TeV collision data, recorded with the CMS detector at LHC. The measurements of differential cross section are performed in bins of the transverse momentum and rapidity of the four-lepton system, invariant masses of the two lepton pairs, five helicity angles formed by the leptons in the Collins-Soper frame, jet multiplicity, transverse momentum of the leading jet, and distance in rapidity between the Higgs boson candidate and the leading jet. Measurement of the cross section of the Z boson production in $Z \rightarrow 4\ell$ decay channel and its ratio to the Higgs boson production cross section in $H \rightarrow 4\ell$ channel is also performed. All results are corrected for the detection efficiency and resolution effects. The inclusive fiducial cross section for the Higgs boson production in $H \rightarrow 4\ell$ decay channel is measured to be $0.56^{+0.67}_{-0.44} \text{ (stat.)}^{+0.21}_{-0.06} \text{ (sys.)}$ fb for proton collisions at 7 TeV and $1.11^{+0.41}_{-0.35} \text{ (stat.)}^{+0.14}_{-0.10} \text{ (sys.)}$ fb for proton collisions at 8 TeV. The results of all measurements are compared with theoretical calculations based on Standard Model and no significant deviation is observed.

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Measurement of inclusive and differential fiducial cross sections of Higgs boson production in the $H \rightarrow 4\ell$ decay channel in proton-proton collisions at 7 TeV and 8 TeV

The CMS Collaboration

Abstract

Measurements of the inclusive and differential fiducial cross sections of Higgs boson production in proton-proton collisions at center-of-mass energies of $\sqrt{s} = 7$ TeV and $\sqrt{s} = 8$ TeV using $H \rightarrow 4\ell$ decays ($\ell = e, \mu$) are presented. Measurements are based on 5.1 fb^{-1} (19.7 fb^{-1}) of 7(8) TeV proton-proton collision data, recorded with the CMS detector at the LHC. The measurements of differential cross section are performed in bins of the transverse momentum and rapidity of the four-lepton system, jet multiplicity, transverse momentum of the leading jet, and distance in rapidity between the Higgs boson candidate and the leading jet. A measurement of the cross section of Z boson production in the $Z \rightarrow 4\ell$ decay channel and its ratio to the Higgs boson production cross section in the $H \rightarrow 4\ell$ channel is also performed. The inclusive fiducial cross section for Higgs boson production in the $H \rightarrow 4\ell$ decay channel is measured to be $0.56^{+0.67}_{-0.44} \text{ (stat.)}^{+0.21}_{-0.06} \text{ (sys.)}$ fb for proton-proton collisions at 7 TeV and $1.11^{+0.41}_{-0.35} \text{ (stat.)}^{+0.14}_{-0.10} \text{ (sys.)}$ fb for proton-proton collisions at 8 TeV. The results of all measurements are compared with theoretical calculations based on the Standard Model and no significant deviation is observed.

- An important property to measure of the discovered Higgs boson is its (differential) fiducial cross section
 - Important test of SM predictions and probe of BSM effects
- Similar measurements (nearly) available:
 - ATLAS $H \rightarrow \gamma\gamma$ ([JHEP09\(2014\)112](#)) 4ℓ ([Phys. Lett. B 738 \(2014\) 234–253](#))
 - CMS $H \rightarrow \gamma\gamma$ (HIG-14-016, CWR ended)
- To minimize model dependence, we perform the measurement in a fiducial space close to experimental acceptance
- The definition can be reproduced by theorists/phenomenologists, such that the experimental results can be compared to any theoretical calculation
 - RIVET code will be provided

- Inclusive fiducial cross sections (*for paper*):
 - Inclusive $H \rightarrow 4\ell$ cross section (7 and 8 TeV)
 - Inclusive $Z \rightarrow 4\ell$ cross section (8 TeV)
 - Ratio of $H \rightarrow 4\ell$ and $Z \rightarrow 4\ell$ cross sections (8 TeV)
 - Ratio of $H \rightarrow 4\ell$ (7 TeV) and $H \rightarrow 4\ell$ (8 TeV) cross sections
- Differential cross sections for $H \rightarrow 4\ell$ (8 TeV) :
 - Production related observables (*for paper*):
 $p_T(H)$, $y(H)$, $N(\text{jets})$, $p_T(\text{leading jet})$, $|y(H)-y(\text{leading jet})|$
 - Decay related observables (*for public twiki*):
 $m(Z_1)$, $m(Z_2)$, $|\cos\theta^*|$, $|\cos\theta_1|$, $|\cos\theta_2|$, $|\Phi|$, $|\Phi_1|$
- Differential cross sections of same observables for $Z \rightarrow 4\ell$ (8 TeV)
(for public twiki)

Datasets and Event Selection



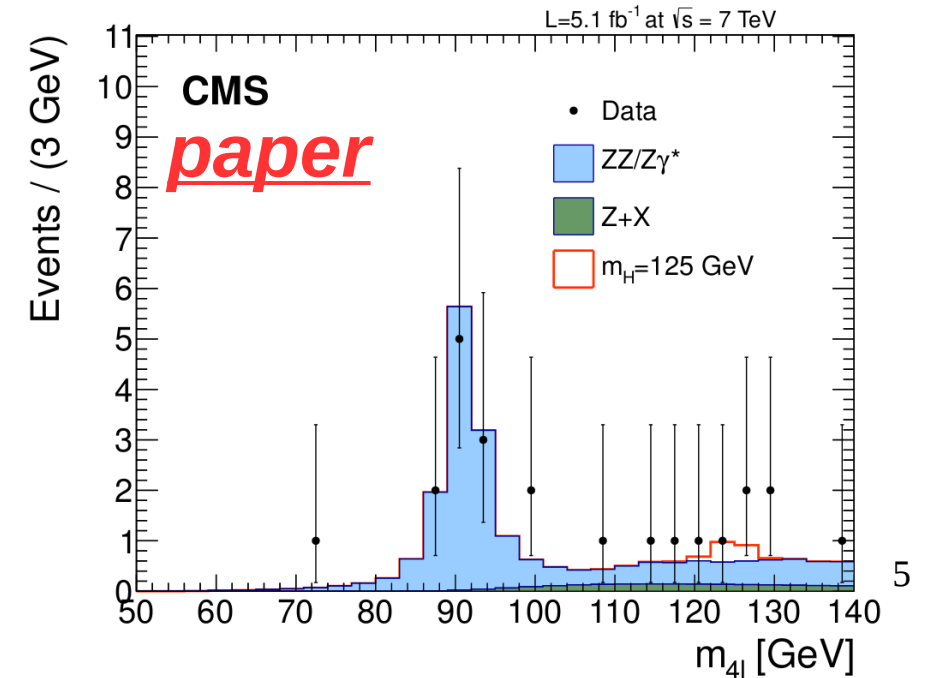
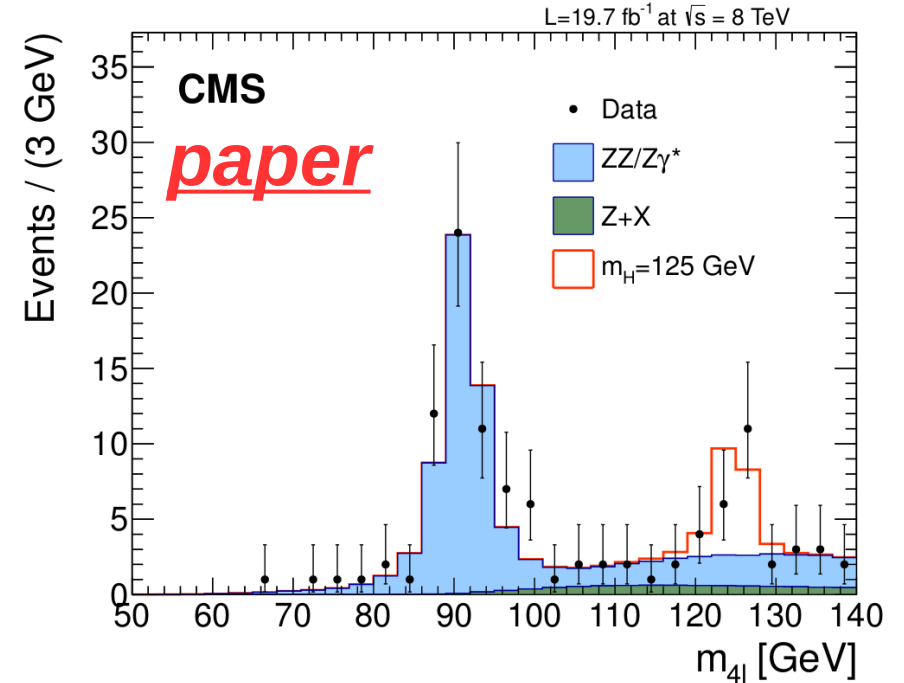
- Datasets, triggers, and event selection follows the legacy $H \rightarrow 4\ell$ analysis

Phys. Rev. D 89 (2014) 092007

paper

Channel	4e	4 μ	2e2 μ
19.7 fb ⁻¹ @ 8 TeV			
$q\bar{q} \rightarrow ZZ$	3.05 ± 0.35	7.62 ± 0.49	9.03 ± 0.69
$Z+X$	1.52 ± 0.34	1.18 ± 0.48	4.16 ± 1.06
$gg \rightarrow ZZ$	0.20 ± 0.05	0.41 ± 0.10	0.50 ± 0.13
Total background expected	4.77 ± 0.71	9.21 ± 0.69	13.69 ± 1.29
$H \rightarrow 4\ell$ ($m_H = 125.0$ GeV)	2.93 ± 0.45	5.63 ± 0.67	7.29 ± 0.93
Observed	9	15	15
5.1 fb ⁻¹ @ 7 TeV			
$q\bar{q} \rightarrow ZZ$	0.84 ± 0.10	1.80 ± 0.11	2.24 ± 0.28
$Z+X$	0.34 ± 0.09	0.22 ± 0.09	1.03 ± 0.29
$gg \rightarrow ZZ$	0.03 ± 0.01	0.06 ± 0.02	0.07 ± 0.02
Total background expected	1.21 ± 0.14	2.08 ± 0.14	3.37 ± 0.40
$H \rightarrow 4\ell$ ($m_H = 125.0$ GeV)	0.70 ± 0.11	1.24 ± 0.14	1.67 ± 0.26
Observed	1	3	6

Event counts in the range
 $105 \text{ GeV} < m(4\ell) < 150 \text{ GeV}$



Fiducial Volume Definition

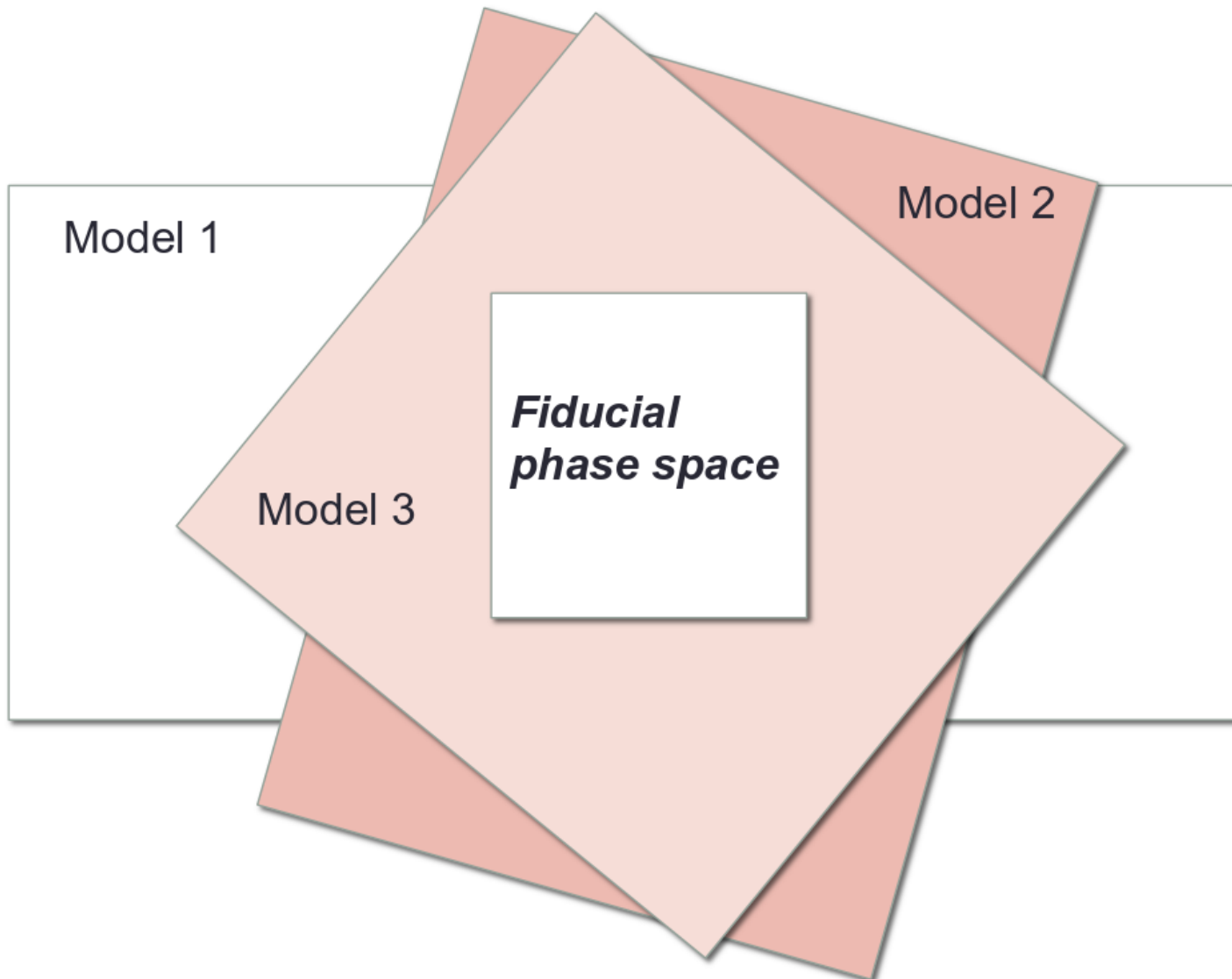
- The fiducial volume definition is chosen to closely match the reconstruction level selection, defined using leptons at the hard scattering level

<i><u>paper</u></i>	Requirements for the $H \rightarrow 4\ell$ fiducial phase space	
Lepton kinematics and isolation		
leading lepton p_T		$p_T > 20 \text{ GeV}$
next-to-leading lepton p_T		$p_T > 10 \text{ GeV}$
additional electrons (muons) p_T		$p_T > 7(5) \text{ GeV}$
pseudorapidity of electrons (muons)		$ \eta < 2.5(2.4)$
p_T sum of all stable particles within $\Delta R < 0.4$ from lepton		less than $0.4 \cdot p_T$
Event topology		
existence of at least two SFOS lepton pairs, where leptons satisfy criteria above		
inv. mass of the Z_1 candidate		$40 \text{ GeV} < m(Z_1) < 120 \text{ GeV}$
inv. mass of the Z_2 candidate		$12 \text{ GeV} < m(Z_2) < 120 \text{ GeV}$
distance between selected four leptons		$\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$
inv. mass of any opposite sign lepton pair		$m(\ell^+ \ell'^-) > 4 \text{ GeV}$
inv. mass of the selected four leptons		$105 \text{ GeV} < m_{4\ell} < 140 \text{ GeV}$
the selected four leptons must originate from the $H \rightarrow 4\ell$ decay		

- For jets, ak5 gen-jets with $p_T > 30 \text{ GeV}$ and $|\eta| < 4.7$ (cleaned from isolated leptons)
- A crucial point is the inclusion of isolation in the fiducial selection
 - Does not include neutrinos or FSR photons
 - without isolation, the difference in efficiency between production modes can be more than 50%

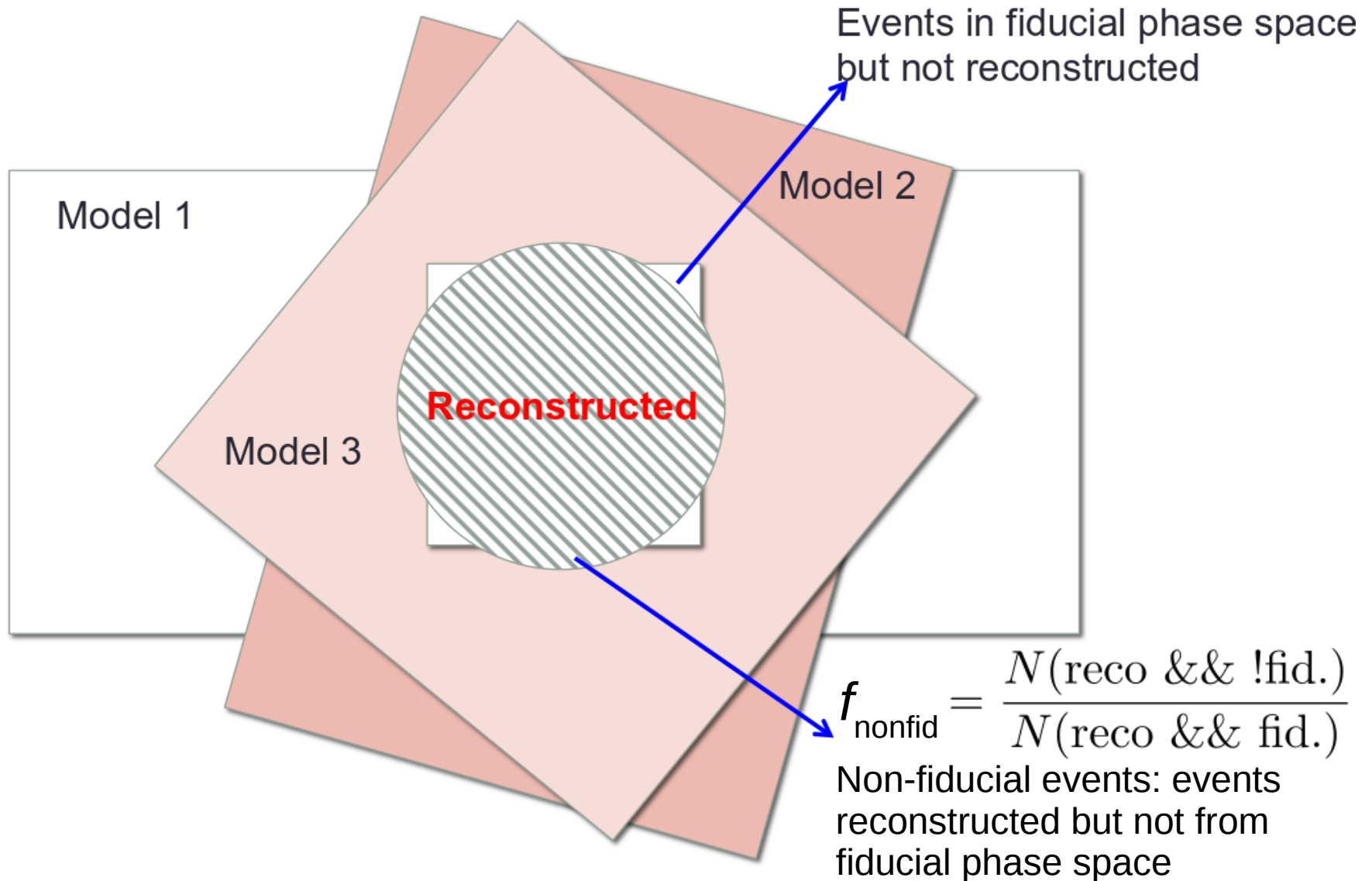
Fiducial Cross Section Overview

- Fiducial cross sections are necessary because acceptance has a strong model dependence, for example between SM production modes by up to 60%



Fiducial Cross Section Overview

$$\varepsilon = \frac{N(\text{reco} \ \&\& \ \text{fid.})}{N(\text{fid.})}$$



Analysis Strategy

The signal component is extracted from a fit to the $m(4\ell)$ distribution

Fiducial Signal Shape (P_{res}):

Double Sided Crystal Ball

→ norm proportional to σ_{fid}

→ efficiencies (ϵ) from simulation

“Non-Fiducial” Signal shape:

Same as fiducial shape

→ fraction of fiducial signal (f_{nonfid})

→ fraction depends on model

“Non-resonant” Signal Shape (P_{nonres}):

One or more leptons not directly from H decay (e.g. WH, ZH, ttH)

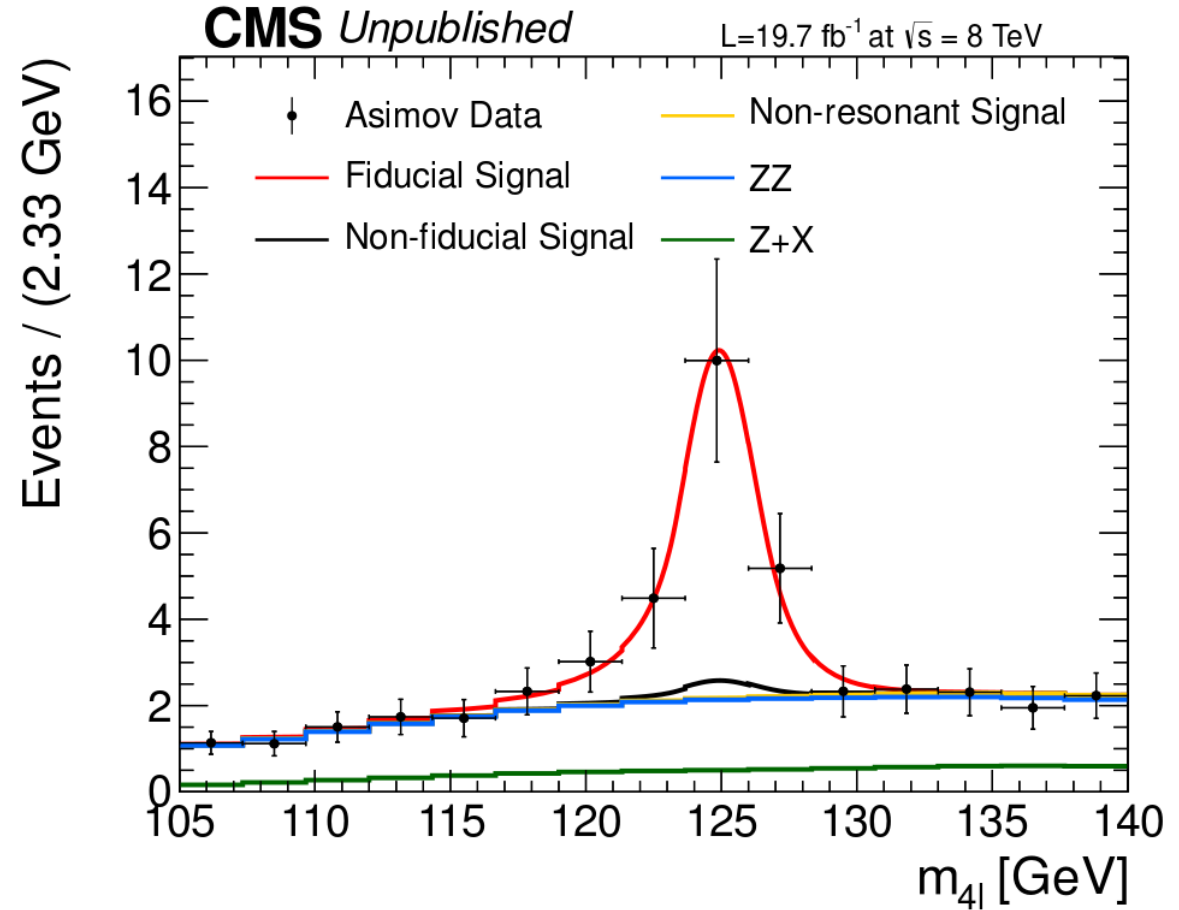
→ Landau distribution

→ shape constrained from simulation

→ norm +/- 10 times SM (uniform prior)

qqZZ,ggZZ shape and norm from MC

Z+X shape and norm from control regions in data



$$N_{obs}^{f,i}(m_{4\ell}) = N_{fid}^{f,i}(m_{4\ell}) + N_{nonres}^{f,i}(m_{4\ell}) + N_{nonfid}^{f,i}(m_{4\ell}) + N_{bkg}^{f,i}(m_{4\ell})$$

$$= \left(1 + f_{nonfid}^{f,i}\right) \cdot \sigma_{fid}^{f,j} \cdot \epsilon_{i,j}^f \cdot \mathcal{L} \cdot \mathcal{P}_{res}(m_{4\ell})$$

$$+ N_{nonres}^{f,i} \cdot \mathcal{P}_{nonres}(m_{4\ell}) + N_{bkg}^{f,i} \cdot \mathcal{P}_{bkg}(m_{4\ell}),$$

f = final state
 i = observable bin
at reco level

Analysis Inputs: Signal (Inclusive)

- The inclusive efficiency and non-fiducial ratios are determined from simulation
→ corrections derived from data are applied to the MC efficiencies
- Mild dependence on the production mode ($\sim 7\%$) of the factor $(1+f_{\text{nonfid}})\epsilon$ is a result of the definition of the fiducial volume

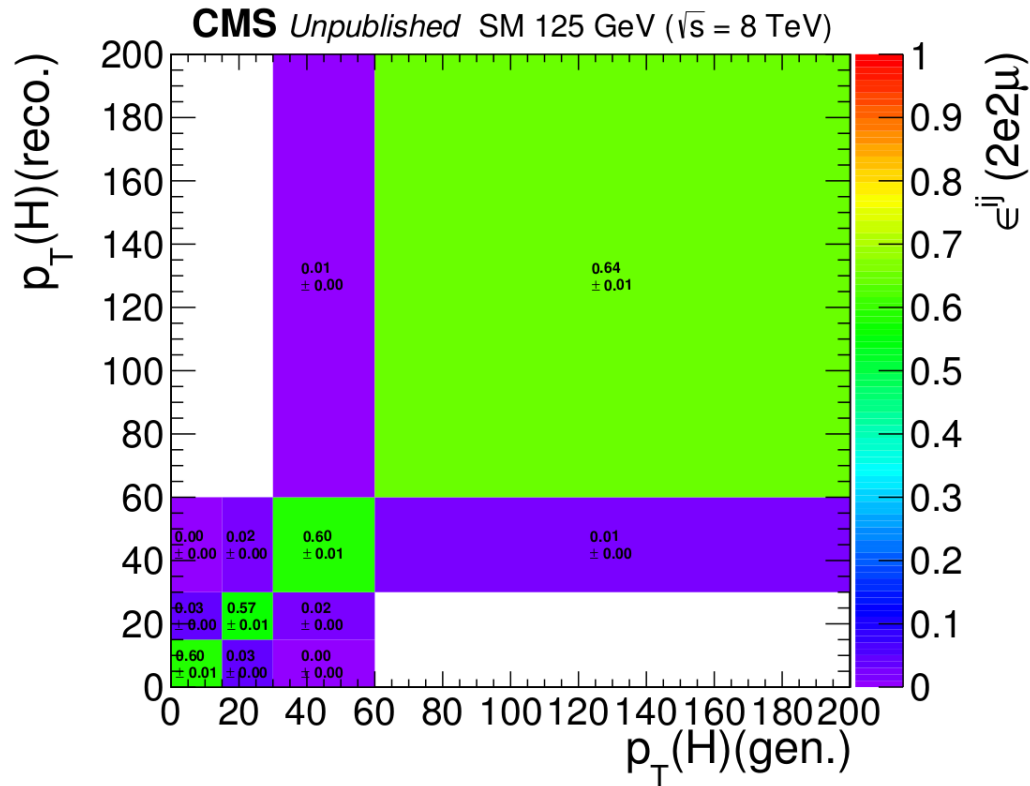
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Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
gg $\rightarrow$ H (POWHEG+JHUGEN)	0.422 ± 0.001	0.647 ± 0.002	0.053 ± 0.001	0.681 ± 0.002
VBF (POWHEG)	0.476 ± 0.003	0.652 ± 0.005	0.040 ± 0.002	0.678 ± 0.005
WH (PYTHIA)	0.342 ± 0.002	0.627 ± 0.003	0.072 ± 0.002	0.672 ± 0.003
ZH (PYTHIA)	0.348 ± 0.003	0.634 ± 0.004	0.072 ± 0.003	0.679 ± 0.005
t $\bar{t}$ H (PYTHIA)	0.250 ± 0.003	0.601 ± 0.008	0.139 ± 0.008	0.685 ± 0.010
Some characteristic models of Higgs-like boson with exotic decays and properties				
q $\bar{q}$ $\rightarrow$ H($J^{\text{CP}}=1^-$) (JHUGEN)	0.238 ± 0.001	0.609 ± 0.002	0.054 ± 0.001	0.642 ± 0.002
q $\bar{q}$ $\rightarrow$ H($J^{\text{CP}}=1^+$) (JHUGEN)	0.283 ± 0.001	0.619 ± 0.002	0.051 ± 0.001	0.651 ± 0.002
gg $\rightarrow$ H $\rightarrow$ Z γ^* (JHUGEN)	0.156 ± 0.001	0.622 ± 0.002	0.073 ± 0.001	0.667 ± 0.002
gg $\rightarrow$ H $\rightarrow$ $\gamma^*\gamma^*$ (JHUGEN)	0.188 ± 0.001	0.629 ± 0.002	0.066 ± 0.001	0.671 ± 0.002

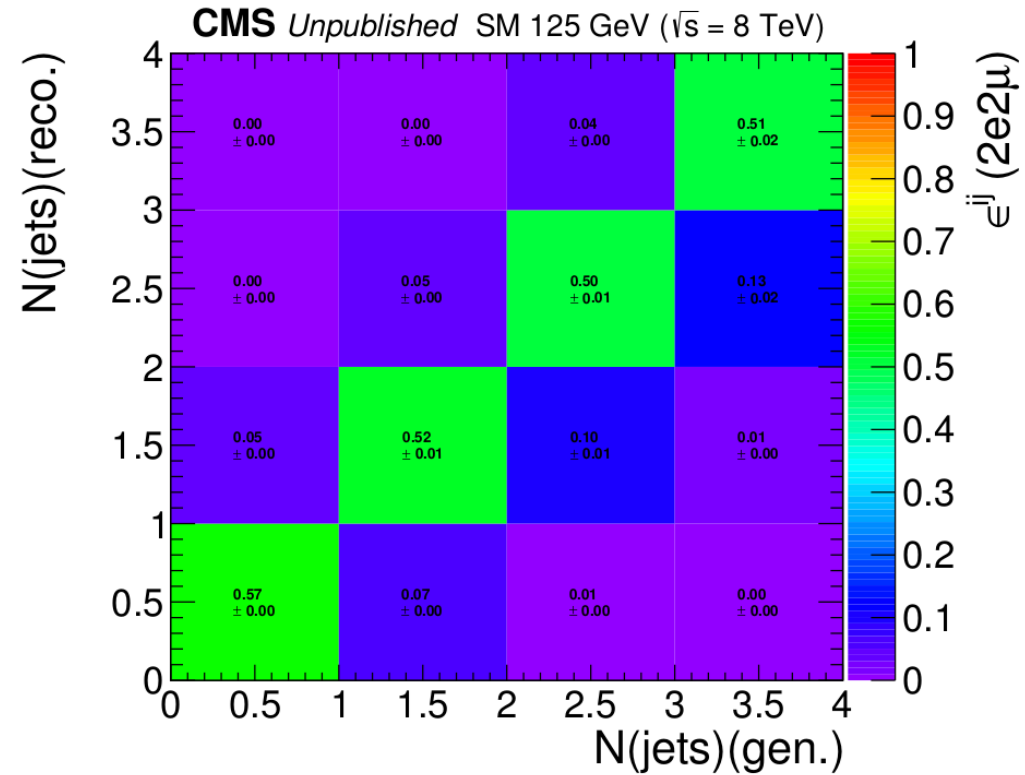
$$\begin{aligned}
 N_{\text{obs}}^{\text{f},i}(m_{4\ell}) &= N_{\text{fid}}^{\text{f},i}(m_{4\ell}) + N_{\text{nonres}}^{\text{f},i}(m_{4\ell}) + N_{\text{nonfid}}^{\text{f},i}(m_{4\ell}) + N_{\text{bkg}}^{\text{f},i}(m_{4\ell}) \\
 &= \left(1 + f_{\text{nonfid}}^{\text{f},i}\right) \cdot \sigma_{\text{fid}}^{\text{f},j} \cdot \epsilon_{i,j}^{\text{f}} \cdot \mathcal{L} \cdot \mathcal{P}_{\text{res}}(m_{4\ell}) \\
 &\quad + N_{\text{nonres}}^{\text{f},i} \cdot \mathcal{P}_{\text{nonres}}(m_{4\ell}) + N_{\text{bkg}}^{\text{f},i} \cdot \mathcal{P}_{\text{bkg}}(m_{4\ell}),
 \end{aligned}$$

Analysis Inputs: Signal (Differential)

twiki



twiki

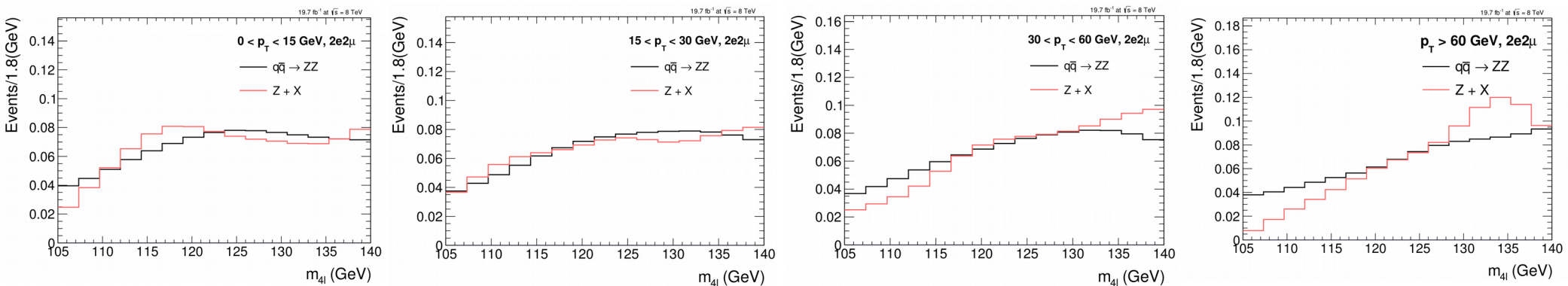


- For differential measurements, the efficiency becomes a generalized “**detector response matrix**” (different for every model)
- The response matrix is included in the likelihood such that we directly fit for the fiducial cross section at fiducial level taking into account bin migration effects

Analysis Inputs: Background

- For differential measurements, the background shape and normalization is estimated for each differential bin
- Large normalization uncertainty also covers uncertainty on the shape

Background shapes per differential bin:



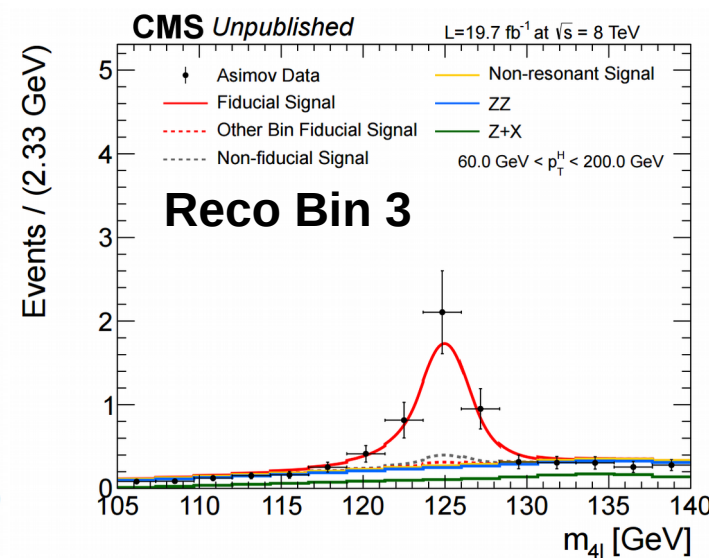
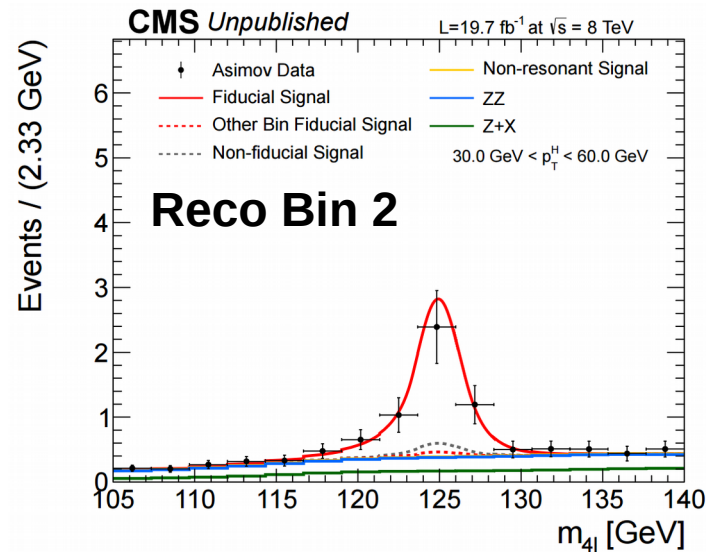
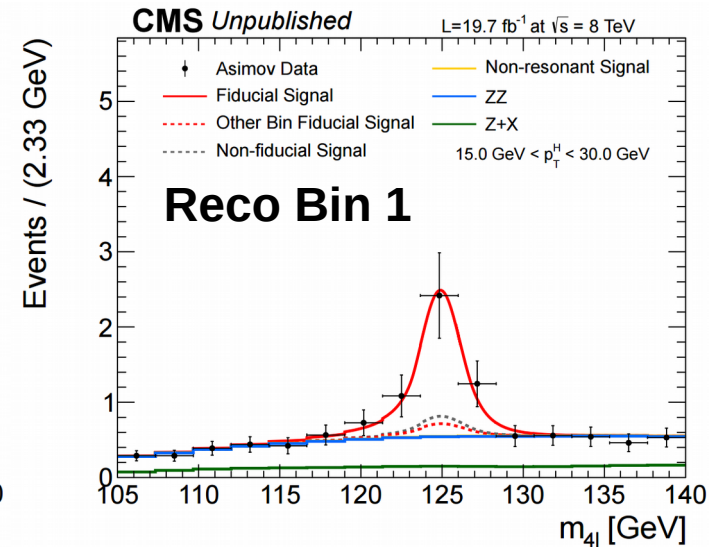
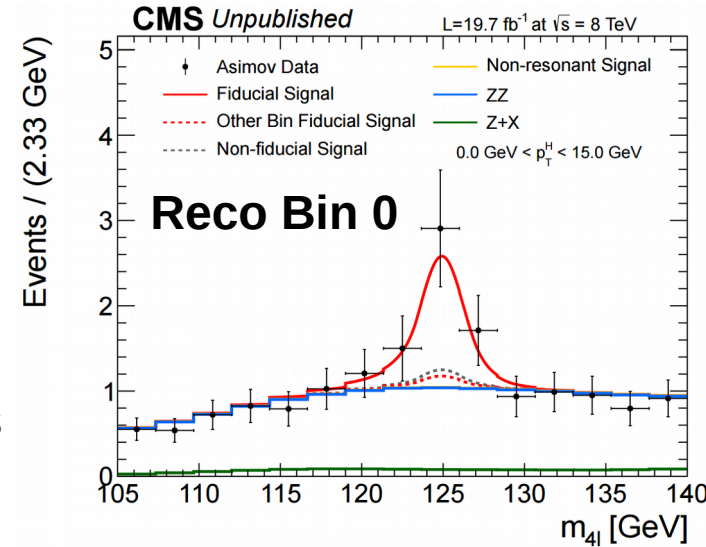
Background	final state	$0 \leq p_T(4\ell) < 15$	$15 \leq p_T(4\ell) < 30$	$30 \leq p_T(4\ell) < 60$	$60 \leq p_T(4\ell) < 200$
qqZZ	2e2μ	0.569	0.231	0.138	0.061
qqZZ	4e	0.548	0.236	0.146	0.069
qqZZ	4μ	0.569	0.228	0.140	0.062
ggZZ	2e2μ	0.299	0.268	0.261	0.172
ggZZ	4e	0.290	0.271	0.272	0.167
ggZZ	4μ	0.323	0.265	0.260	0.152
Z+X (CR)	4l	0.153	0.271	0.373	0.203

Fraction of background per differential bin

Fitting / “Unfolding” Procedure



- **Simultaneous fit** of 4 differential bins and 3 final states (=12 channels) per differential measurement
- **1 POI (σ_{fid}) per bin** controls the normalization of the “Fiducial Signal” components
- 2 additional parameters per bin to **float the fractions of each final state**
- **Fiducial signal** from a particular differential bin at fiducial level **can contribute to all differential bins** at reconstruction level
→ controlled by the detector response matrix
- **Mass is fixed to 125.0 GeV**



Example fits on Asimov Data for $p_T(4\ell)$

Systematic Uncertainties



- Experimental systematic uncertainties mostly from Legacy paper:
 - Background estimation
 - QCD scale ($\sim 3\%$ qqZZ, $\sim 24\%$ ggZZ) and PDF ($\sim 3\%$ qqZZ, 7% ggZZ)
 - Reducible Background (20%-40%)
 - Lepton reconstruction efficiency (10% 4e, 4% 4mu)
 - Signal Shape
 - Lepton energy scale (0.3% 4e, 0.1% 4mu)
 - Lepton energy resolution (20%)
 - Non-resonant signal contribution
 - Effect on the final measurement is $\sim +4\%/-11\%$
 - Integrated Luminosity (2.2% at 7 TeV, 2.6% at 8 TeV)
- For observables involving jets, Jet Energy Scale
 - Correlated across differential bins to preserve unity
 - 3%-12% for signal, 2%-16% for background

Model Dependence



- Model dependence of the measurement procedure is estimated by repeating the measurement using the efficiencies and non-fiducial ratios from a range of different models of production, decay, and spin-parity (exotic models taken from anomalous HVV couplings paper HIG-14-018)
 - In the inclusive measurement the model dependence is less than 7%
 - For differential measurements, in particular jet related observables, can be up to 25% for a particular bin
- If only SM production modes are considered, along with their existing experimental constraints (HIG-14-009), the model dependence is at most 3% for any measurement, and typically much smaller
- We quote the full model dependence without any experimental constraints as a separate systematic effect

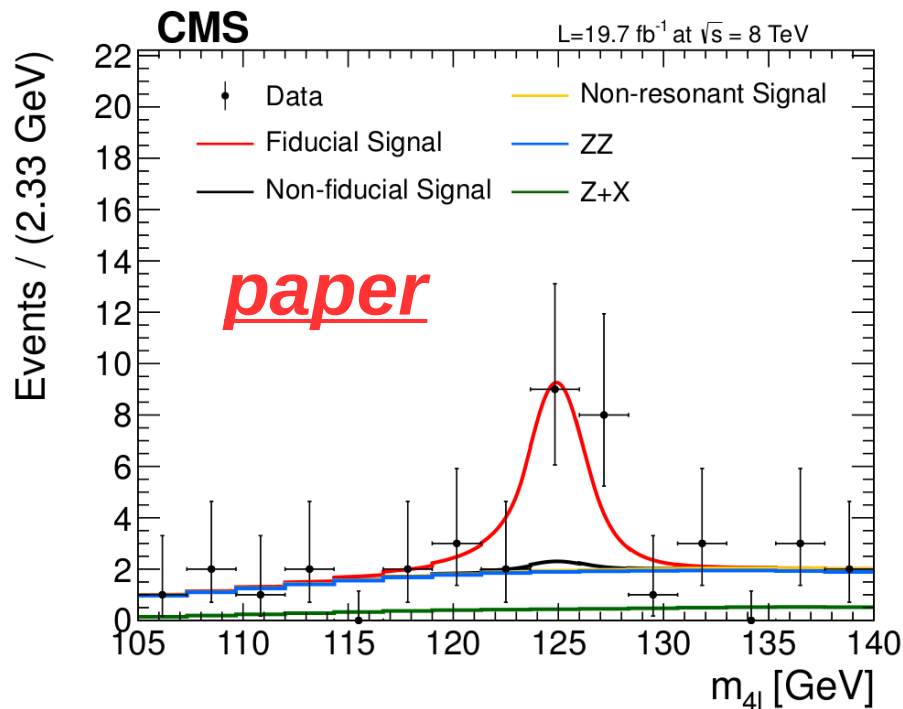
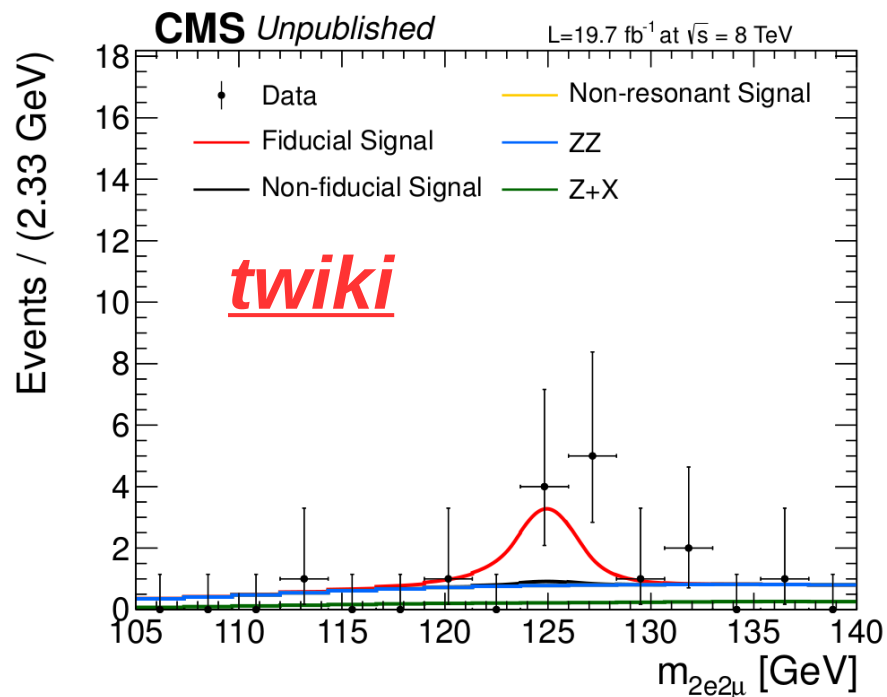
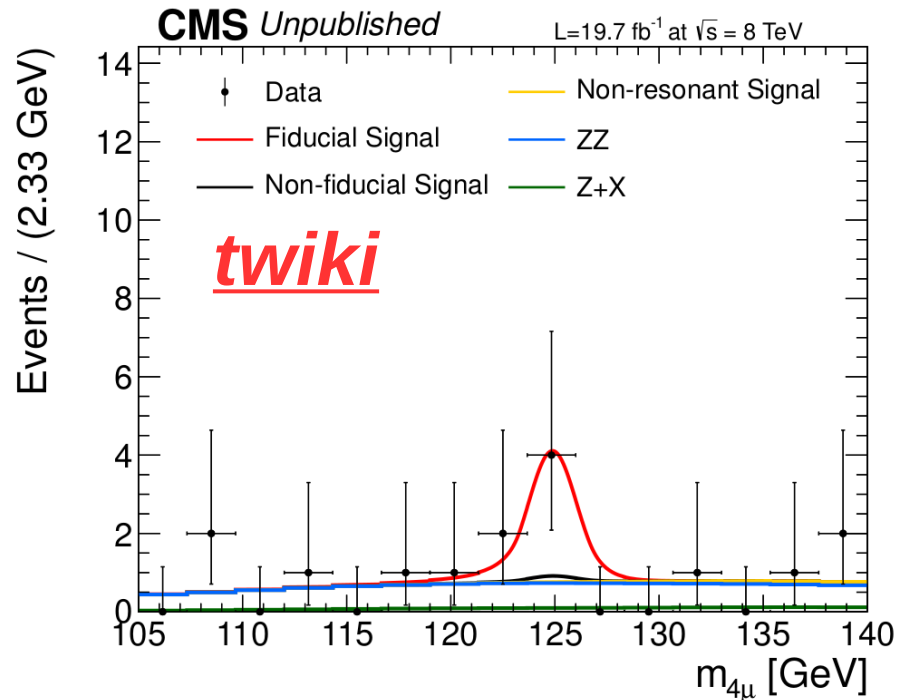
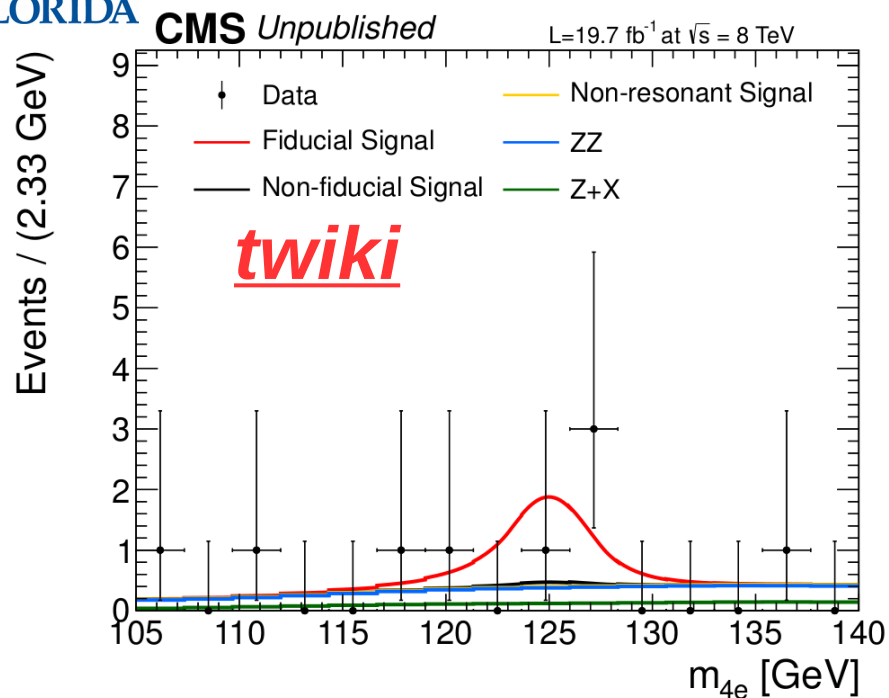
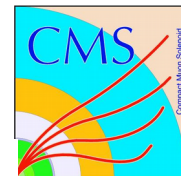
- ggH simulated with Powheg+JHUGen
 - NLO accuracy in QCD for 0-jets, interfaced to Pythia 6.4
 - Finite quark mass effects
- ggH simulated with Powheg+minloHJ
 - NLO accuracy in QCD for 0- and 1- jets, interfaced to Pythia 6.4
 - Finite quark mass effects
- ggH with HRes
 - NNLO in QCD + NNLL in resummation of soft gluon effects
 - Finite quark mass effects
 - Parton level generator, no interface to Pythia
 - Used to reweight Powheg+JHUGen+Pythia in a larger fiducial phase space
 - Plan to update reweighting to be at Powheg+JHUGen+Pythia level with Parton Shower but without Hadronization and UE (small effect)
- VBF simulated with Powheg
- WH, ZH, ttH simulated with Pythia 6.4
- All predictions normalized using cross section recommendations from the LHC Higgs Cross Section Working Group

- QCD scale and PDF/ α_s uncertainties on ggH production computed bin-by-bin for each differential observable
→ for minloHJ and HRes, taking into account events with negative weights
- For VBF, WH, ZH, ttH QCD scale and PDF/ α_s uncertainties taken as constant across bins and taken from LHCHSWG
- PDF/ α_s uncertainties correlated between VBF and VH, anti-correlated between ggH and ttH
- QCD scale uncertainties uncorrelated between production modes
- Uncertainty on acceptance (2%) and $H \rightarrow ZZ \rightarrow 4\ell$ branching ratio (2%) correlated across production modes
- For N(jets) measurement, use Stewart-Tackmann procedure

$$\Delta_N^2 = \Delta_{\geq N}^2 + \Delta_{\geq N+1}^2 \quad \text{arxiv:1107.2117}$$

Inclusive Results

Inclusive $m(4\ell)$ Fit Results

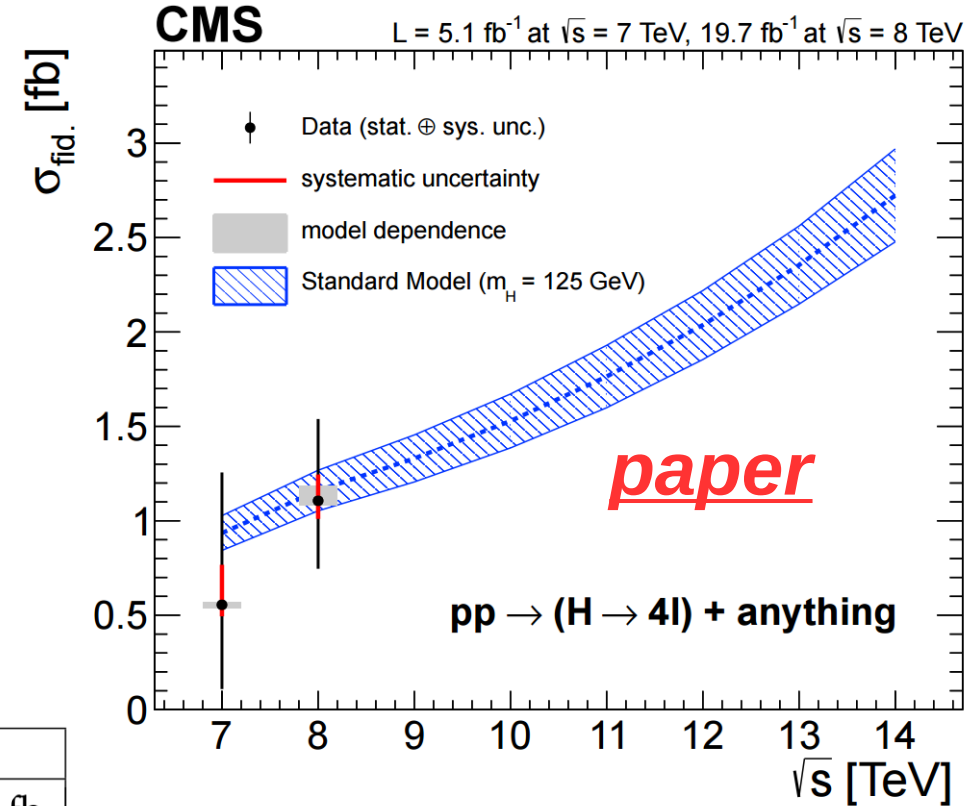


Inclusive $H \rightarrow 4\ell$ Results



- Powhag and HRes ggH samples generated from 7 – 14 TeV.
- At each energy we calculate for ggH:
 - Fiducial acceptance
 - Theoretical uncertainty
- All total cross sections taken from LHCHSWG

paper



Fiducial cross section $H \rightarrow 4\ell$ at 7 TeV

Measured $0.56^{+0.67}_{-0.44} \text{ (stat.) } ^{+0.21}_{-0.06} \text{ (sys.) } ^{+0.02}_{-0.02} \text{ (model) fb}$

$gg \rightarrow H(\text{HRES}) + XH$ $0.94^{+0.10}_{-0.10} \text{ fb}$

Fiducial cross section $H \rightarrow 4\ell$ at 8 TeV

Measured $1.11^{+0.41}_{-0.35} \text{ (stat.) } ^{+0.14}_{-0.10} \text{ (sys.) } ^{+0.08}_{-0.02} \text{ (model) fb}$

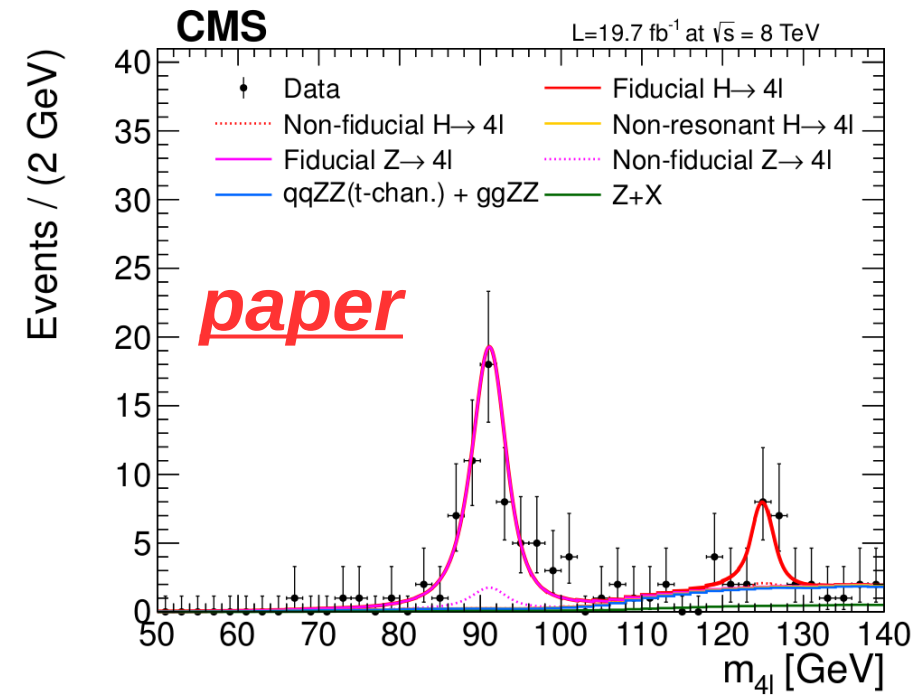
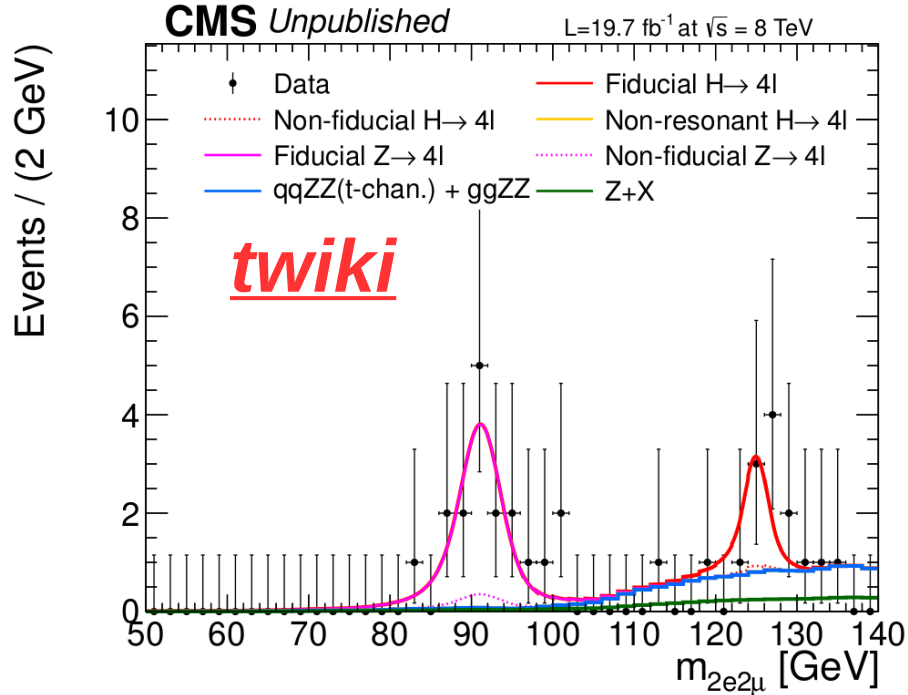
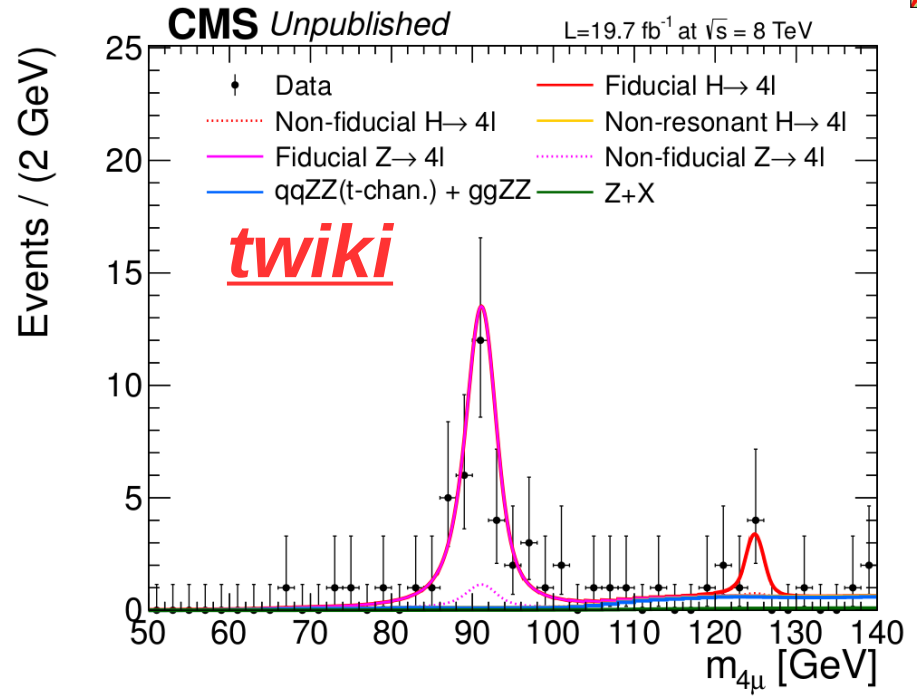
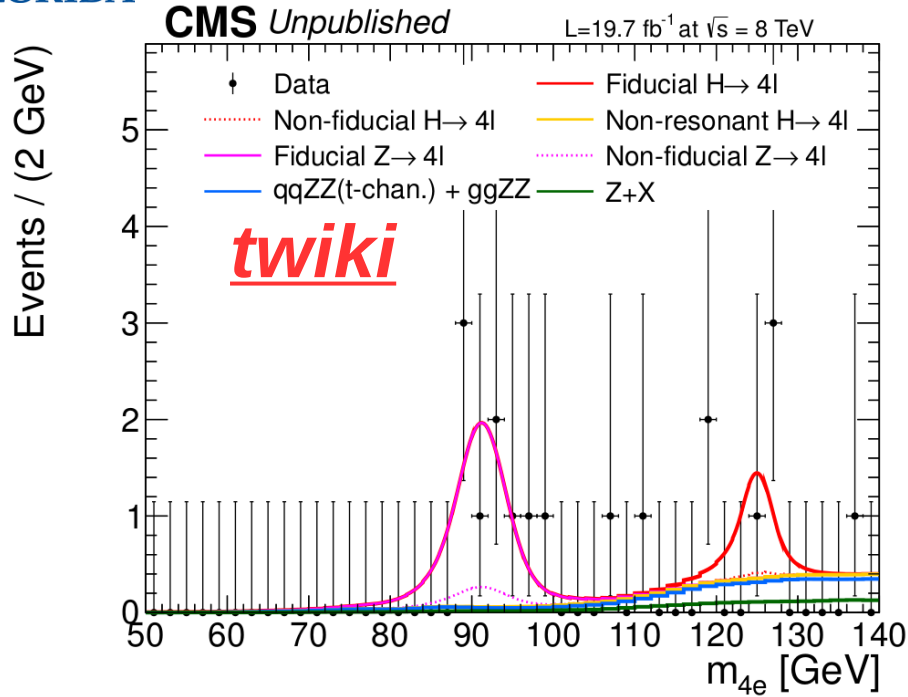
$gg \rightarrow H(\text{HRES}) + XH$ $1.16^{+0.11}_{-0.11} \text{ fb}$

Ratio of fiducial cross sections of $H \rightarrow 4\ell$ at 7 and 8 TeV

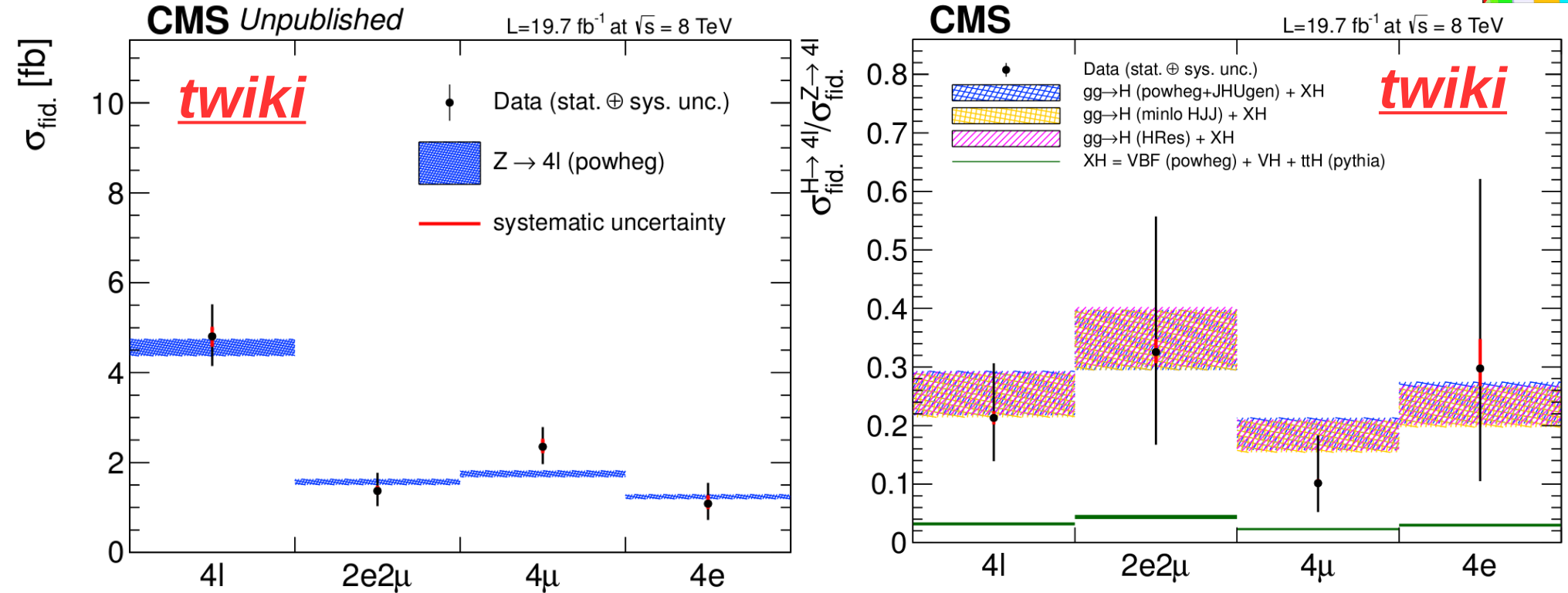
Measured $0.51^{+0.71}_{-0.40} \text{ (stat.) } ^{+0.13}_{-0.05} \text{ (sys.) } ^{+0.00}_{-0.03} \text{ (model)}$

$gg \rightarrow H(\text{HRES}) + XH$ 0.81

Simultaneous $m(4\ell)$ Fit Results



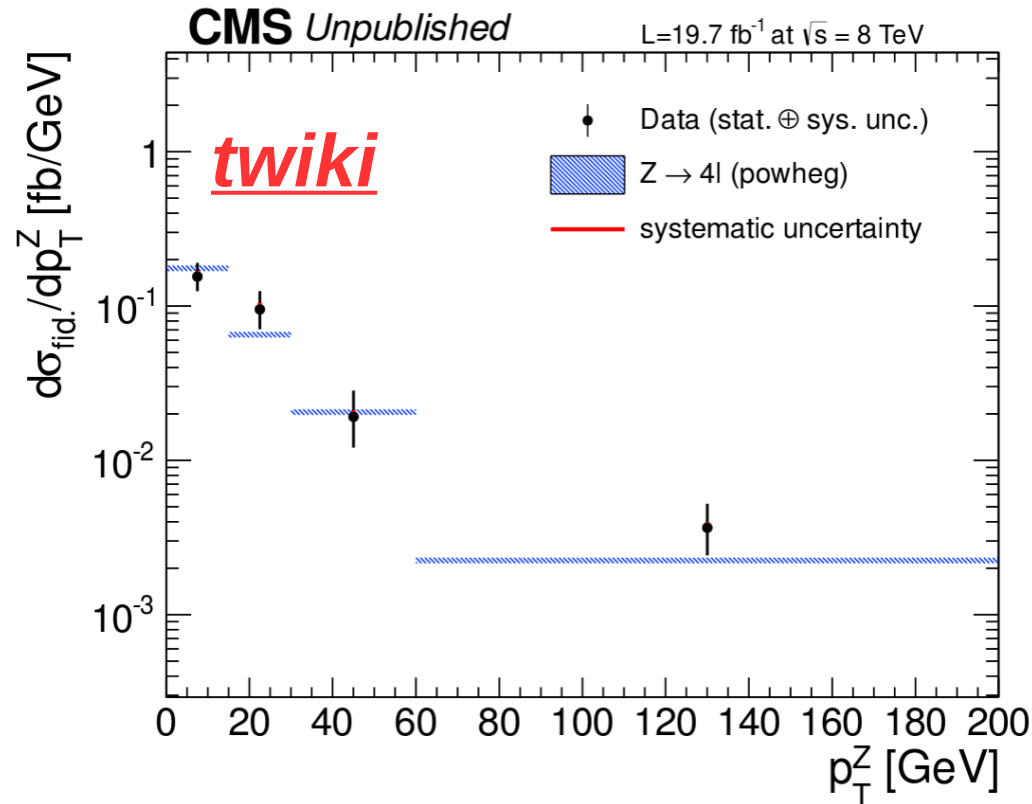
Inclusive $Z \rightarrow 4\ell$ and H/Z Results



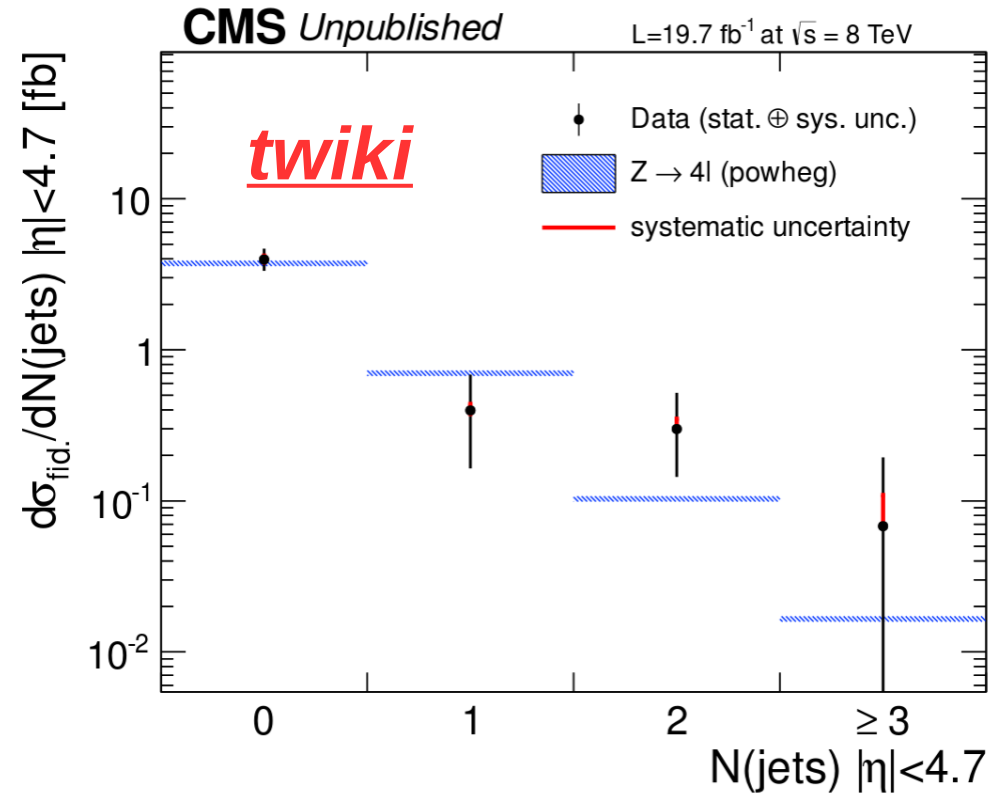
<u>paper</u>	Fiducial cross section $Z \rightarrow 4\ell$ at 8 TeV ($50 \text{ GeV} < m_{4\ell} < 105 \text{ GeV}$)	
Measured	$4.81^{+0.69}_{-0.63}(\text{stat.})^{+0.18}_{-0.19}(\text{sys.}) \text{ fb}$	
POWHEG	$4.56^{+0.19}_{-0.19} \text{ fb}$	
	Ratio of fiducial cross sections of $H \rightarrow 4\ell$ and $Z \rightarrow 4\ell$ at 8 TeV ($50 \text{ GeV} < m_{4\ell} < 140 \text{ GeV}$)	
Measured	$0.21^{+0.09}_{-0.07}(\text{stat.})^{+0.01}_{-0.01}(\text{sys.})$	
$gg \rightarrow H(\text{HRES}) + \text{XH}$ and $Z \rightarrow 4\ell$ (POWHEG)	$0.25^{+0.04}_{-0.04}$	

Differential Results

$Z \rightarrow 4\ell$ Differential Results



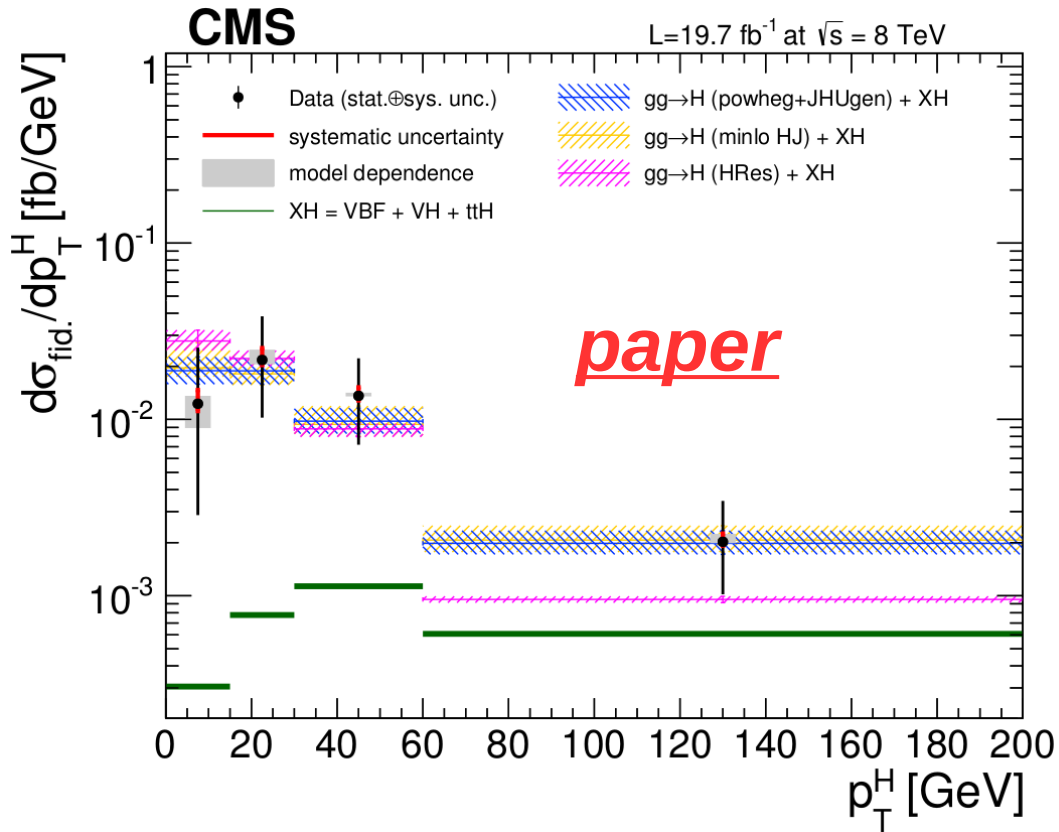
p-value = 0.51



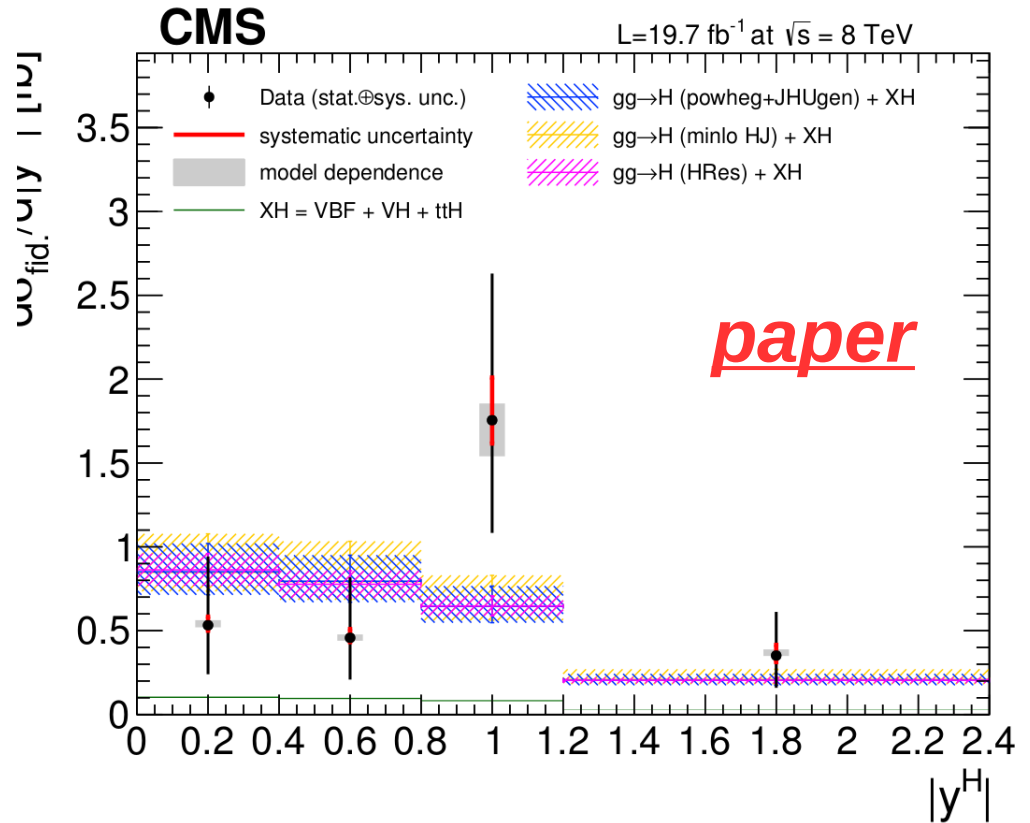
p-value = 0.39

- Differential measurements performed using $Z \rightarrow 4\ell$ resonance which has $\sim 4\times$ larger statistics
 - Serves as validation of the measurement procedure
 - more observables in the Extra Material slides

$p_T(H)$ and $y(H)$

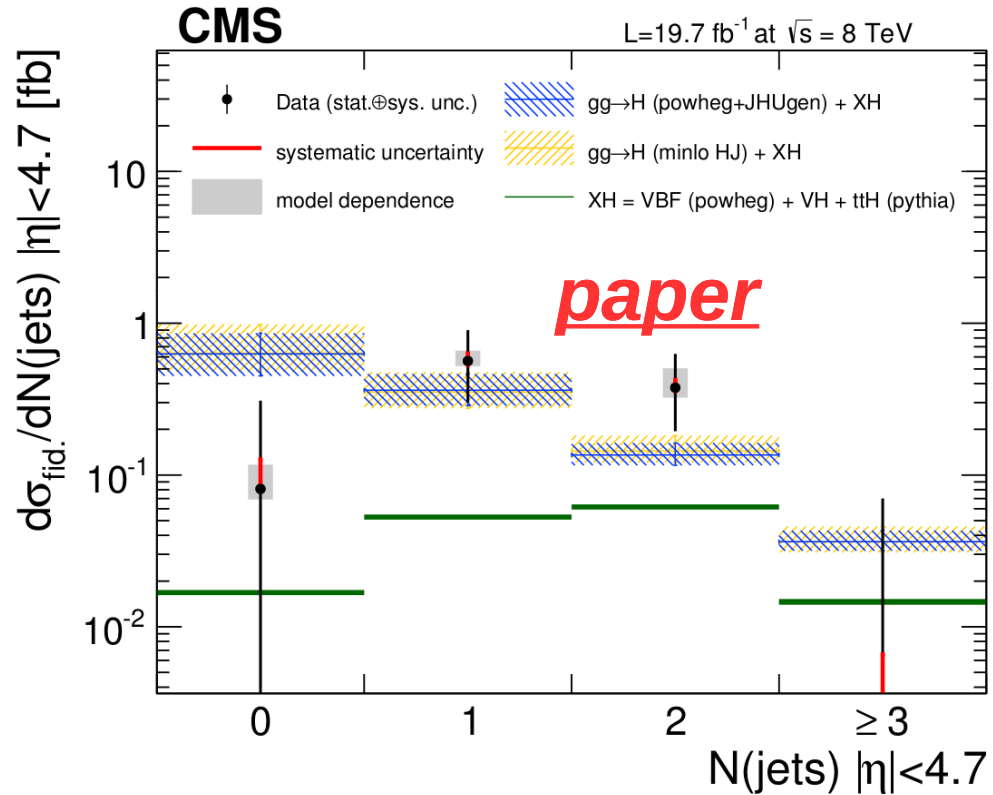


p-value = 0.95

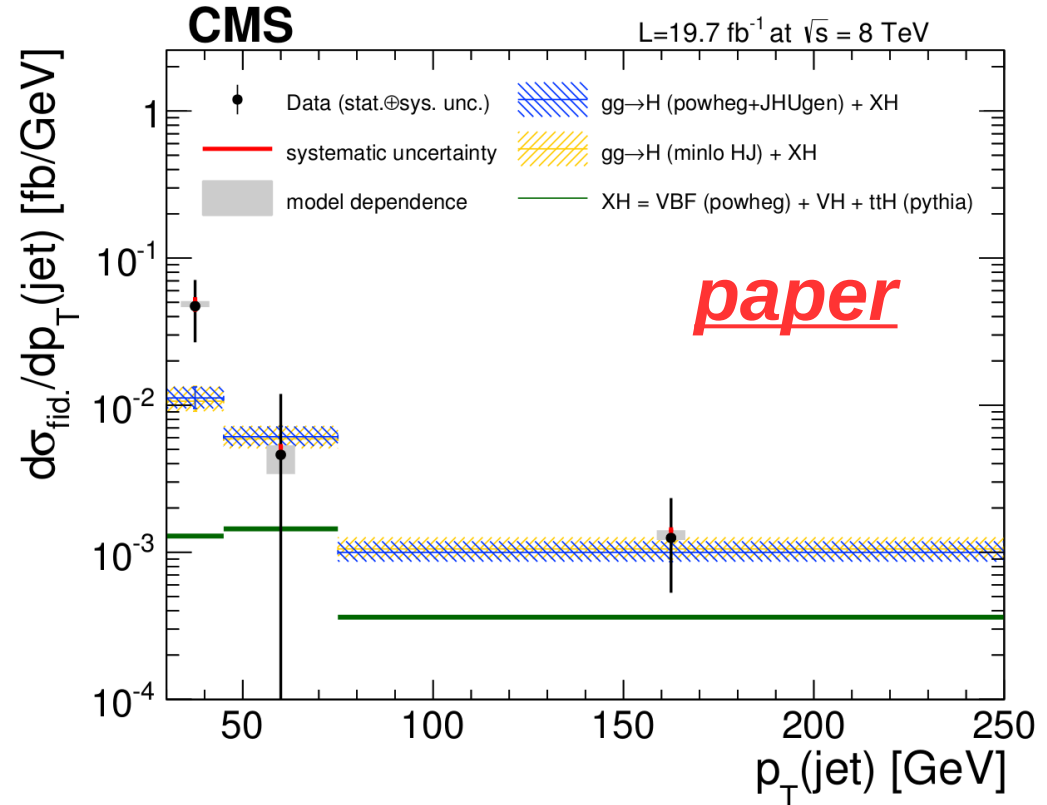


p-value = 0.24

N(jets) and p_T (leading jet)

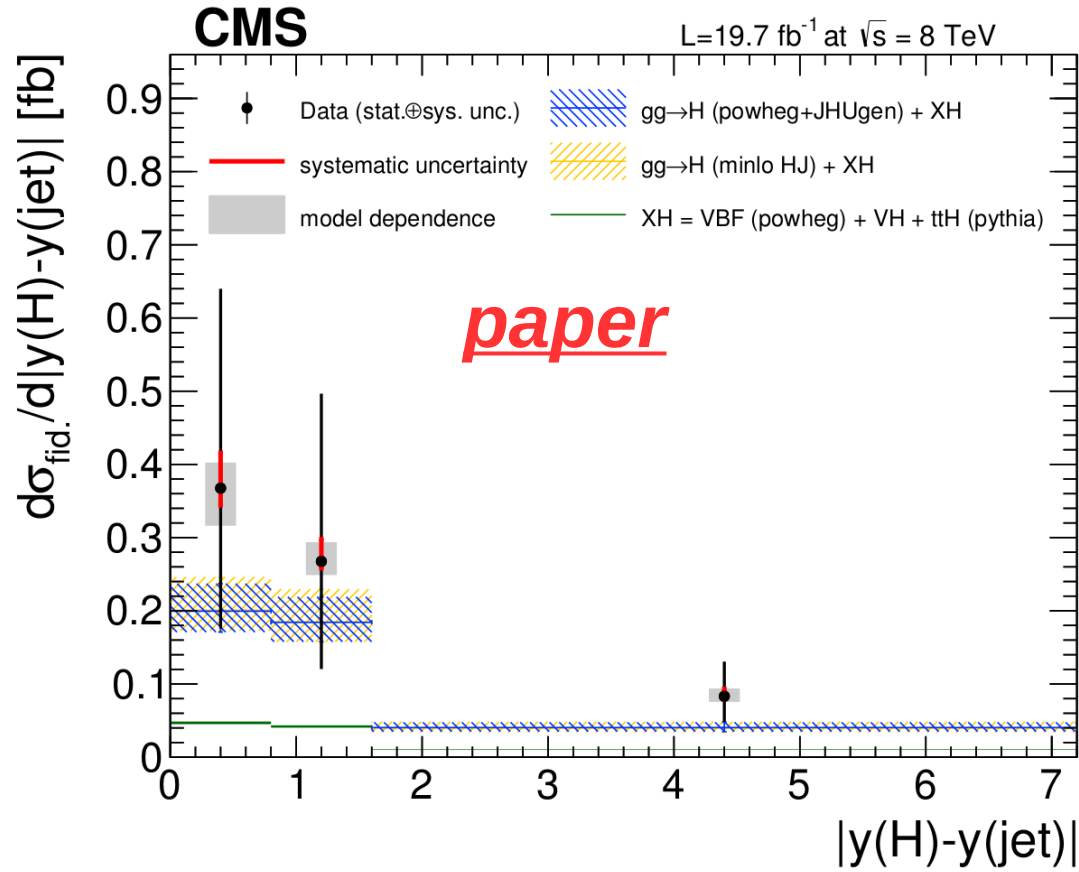


p-value = 0.13



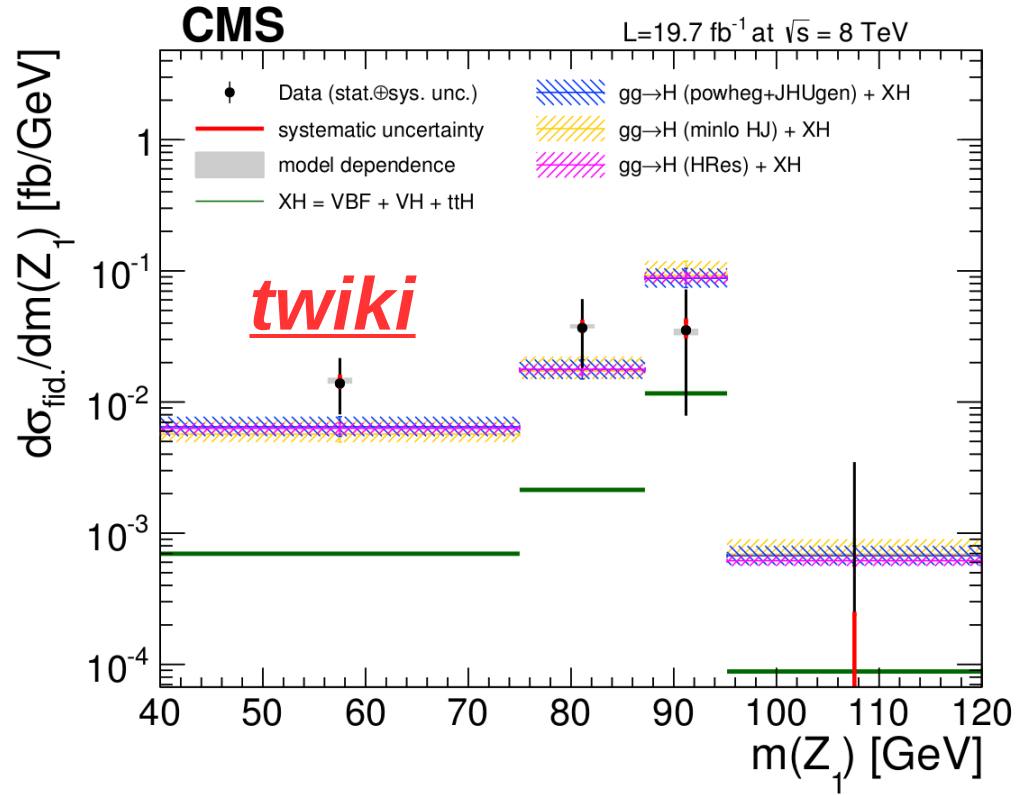
p-value = 0.14

$|y(H)-y(\text{leading jet})|$

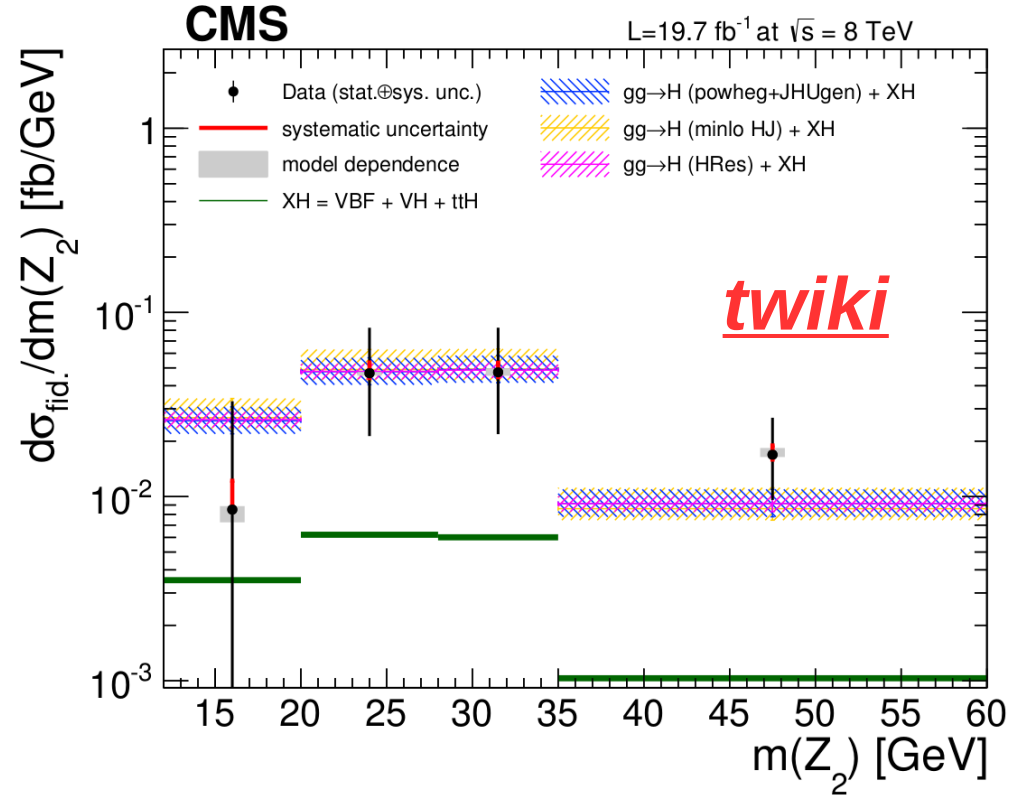


P-value = 0.21

$m(Z_1)$ and $m(Z_2)$

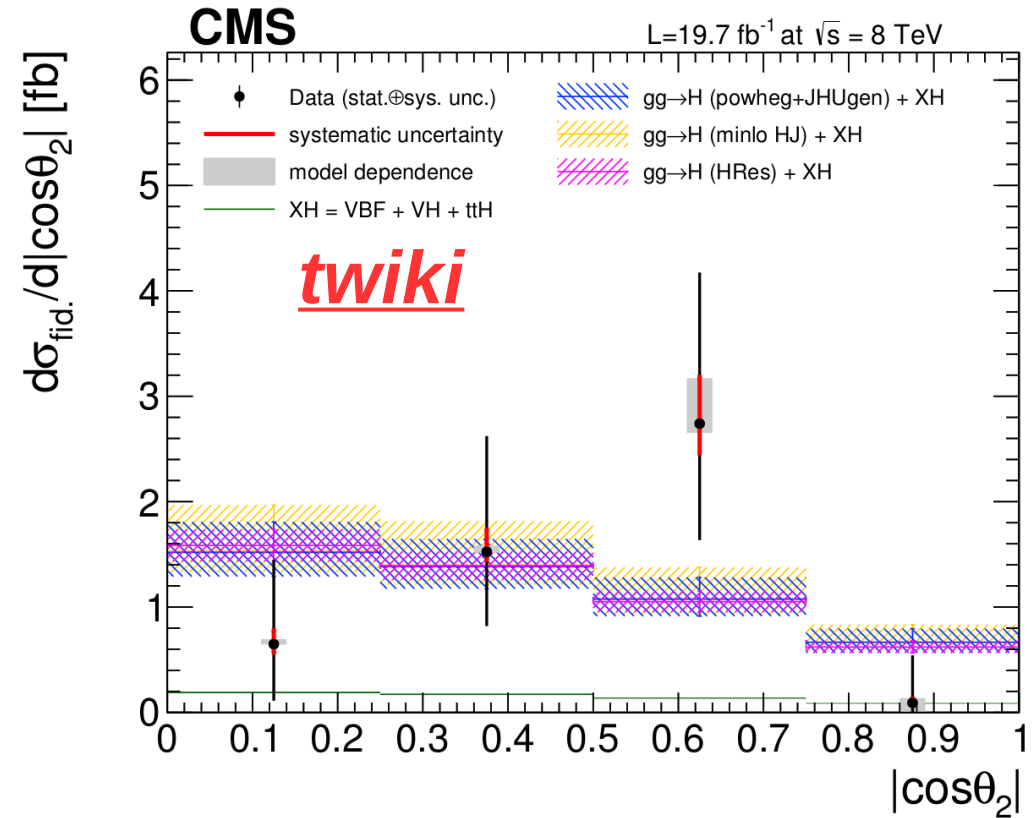
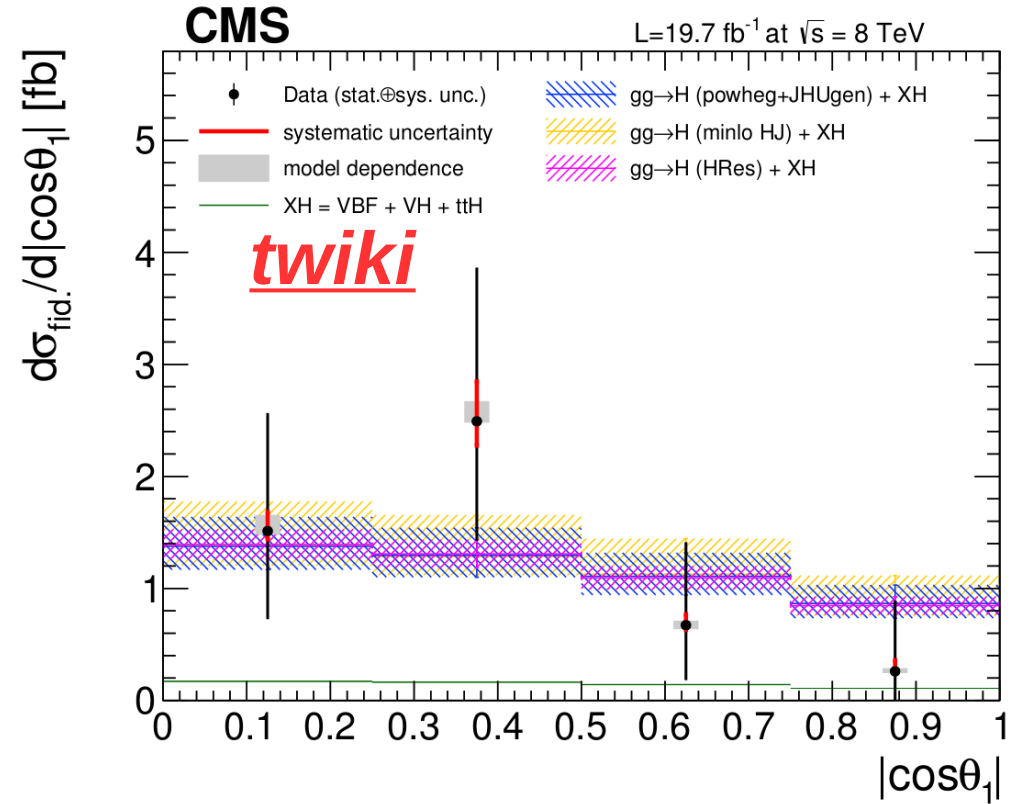


p-value = 0.31

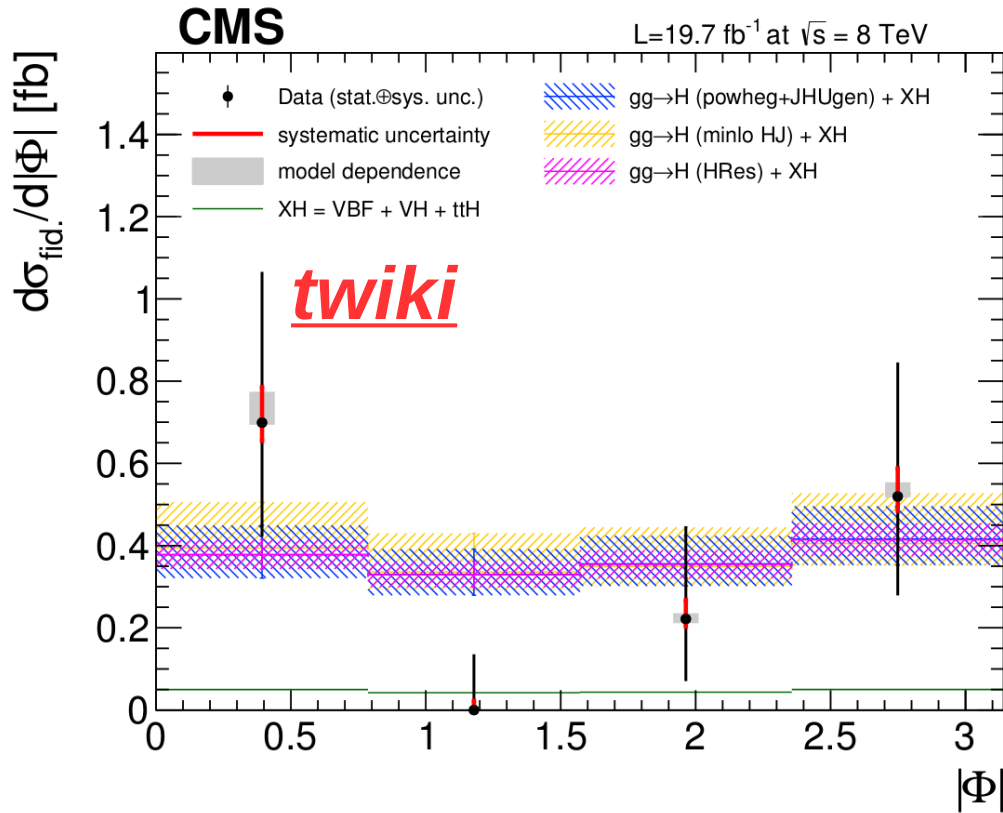


p-value = 0.79

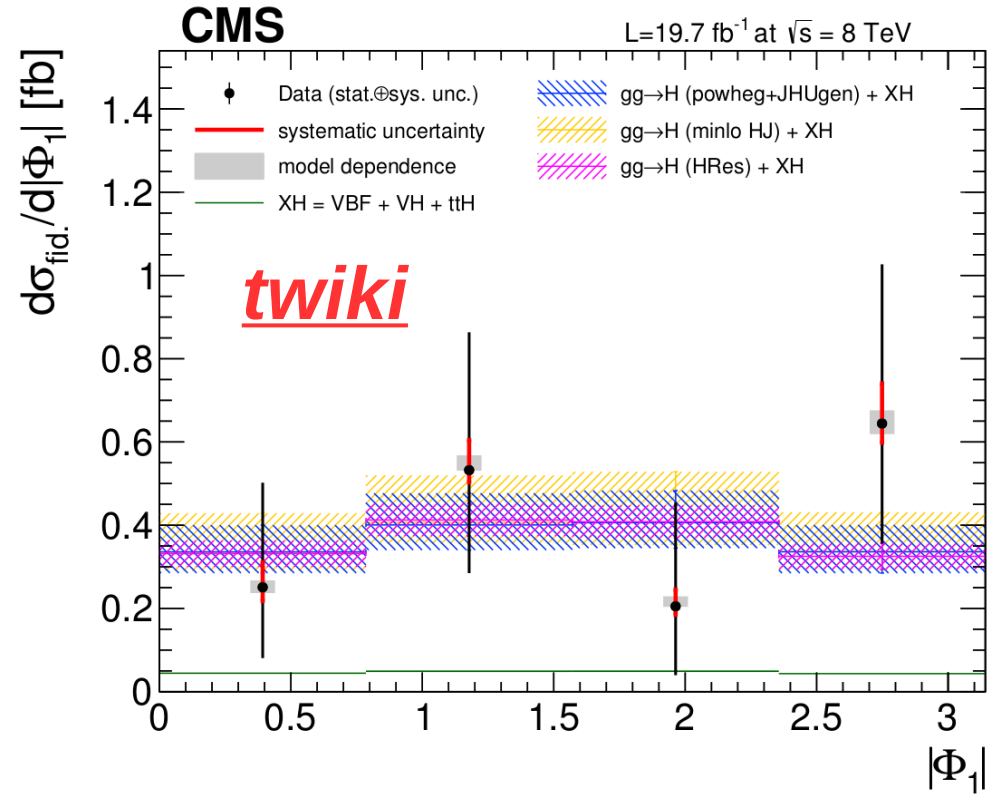
$\cos(\theta_1)$ and $\cos(\theta_2)$



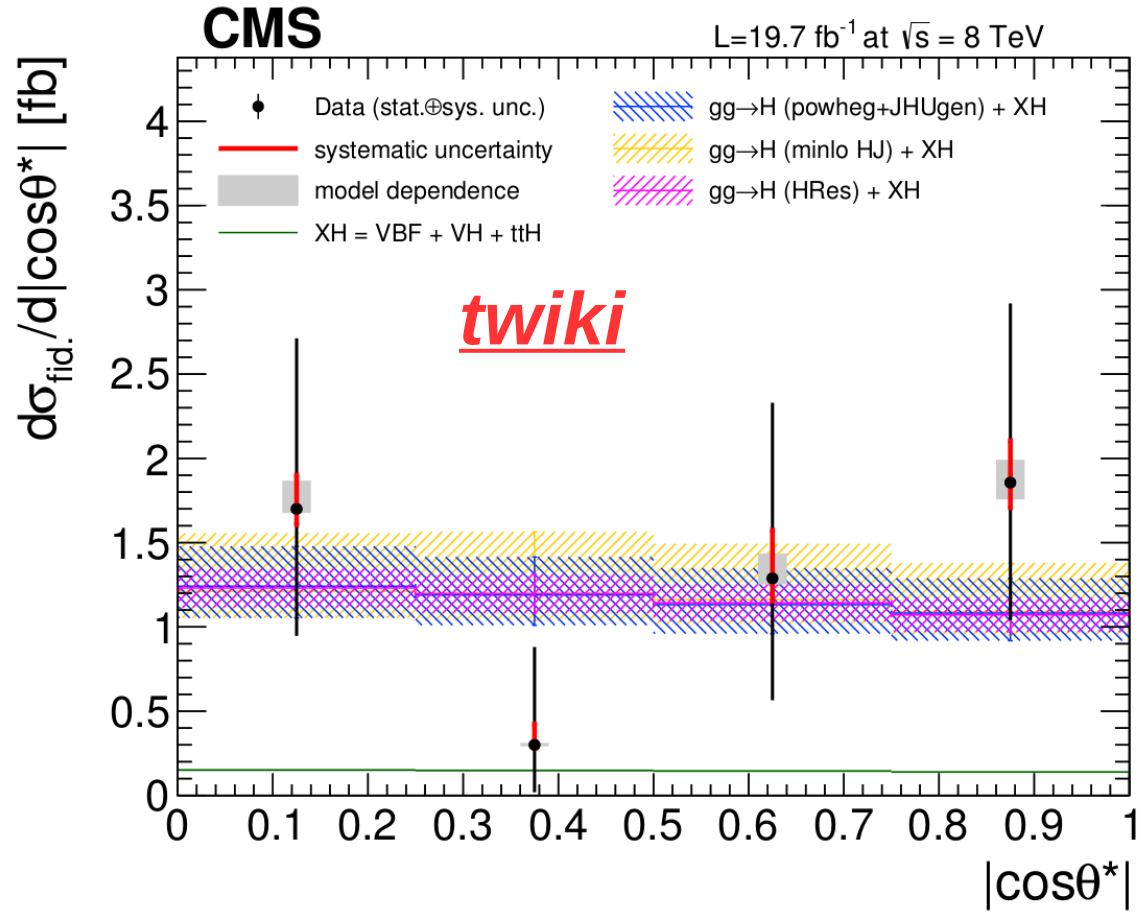
Φ and Φ_1



p-value = 0.30



p-value = 0.68



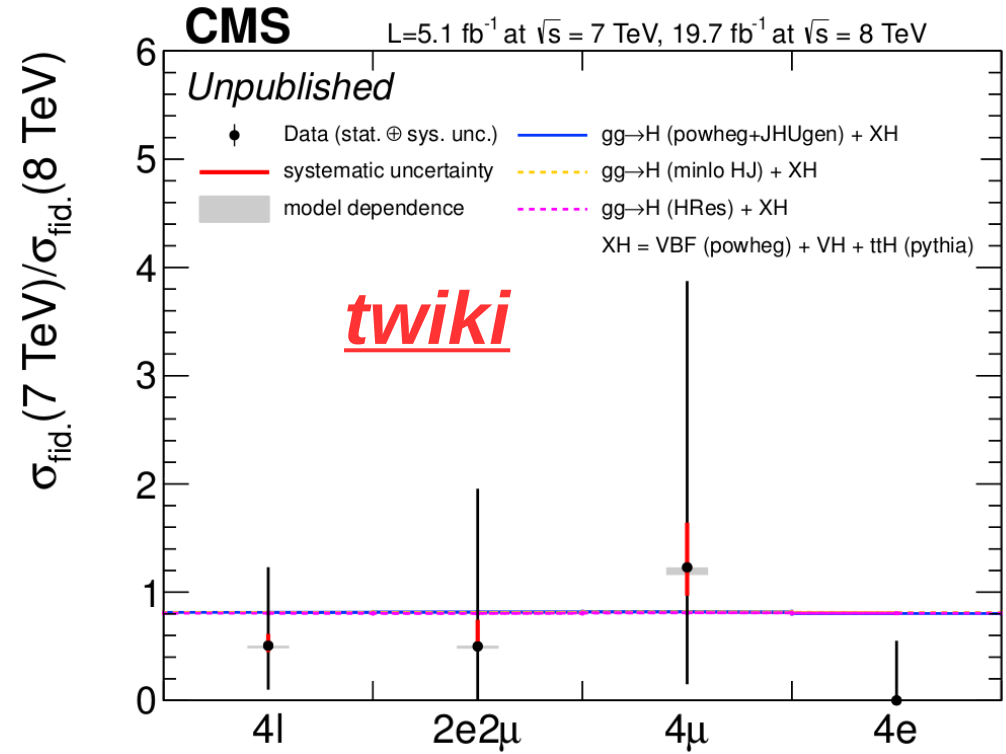
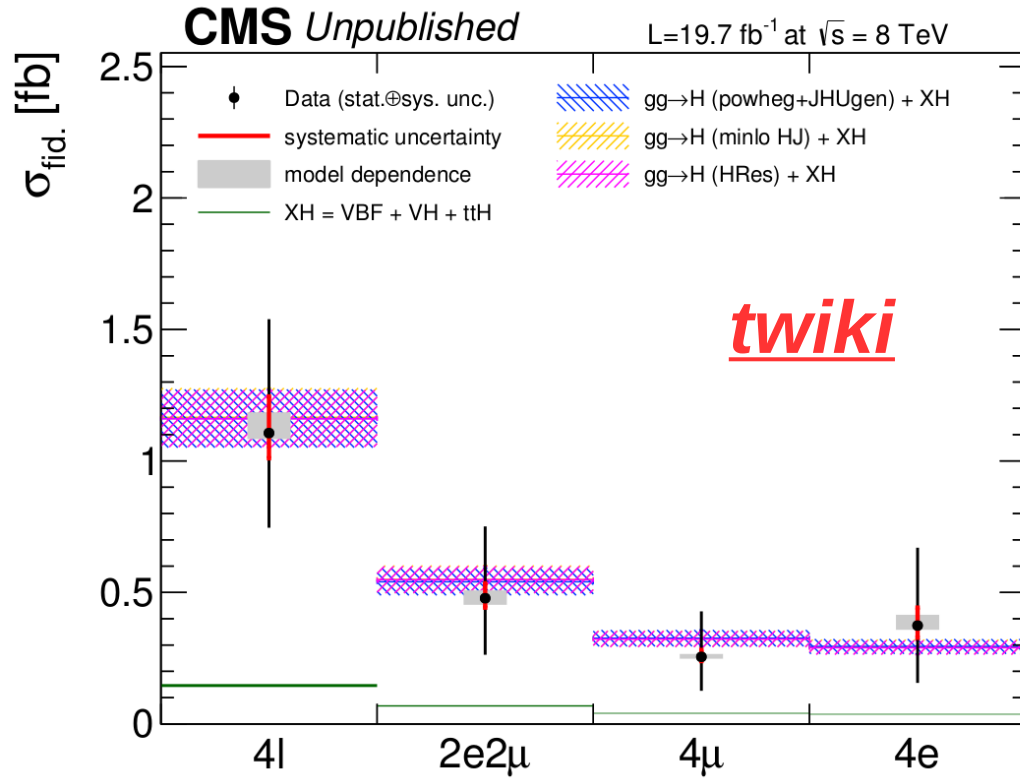
p-value = 0.50

Conclusions

- We have performed measurements of the inclusive and differential fiducial cross sections, as well as several cross section ratios, using the $H \rightarrow 4\ell$ decay channel
 - We have also measured analogous $Z \rightarrow 4\ell$ cross sections at 8 TeV as a validation of the measurement methodology
- We have estimated the model dependence of the measurement procedure using a wide range of models of $H \rightarrow 4\ell$ production
- We have compared to several theoretical predictions, and no significant deviations are observed in the current dataset
- ***We ask the collaboration to approve these results***
- We are looking forward to performing fiducial cross section measurements with 13 TeV data

Extra Material For Twiki

Inclusive Results per Final State



twiki

Fiducial cross section $H \rightarrow 4l$ (fb) at 8 TeV.

	$4l$	$2e2\mu$	4μ	$4e$
Measured (stat. $\oplus$ sys.)(model)	$1.106^{+0.426+0.08}_{-0.357-0.02}$	$0.478^{+0.273+0.03}_{-0.215-0.03}$	$0.255^{+0.173+0.01}_{-0.129-0.01}$	$0.374^{+0.296+0.04}_{-0.218-0.02}$
$gg \rightarrow H(\text{POHWHEG} + \text{JHUGEN}) + XH$	$1.16^{+0.12}_{-0.12}$	$0.54^{+0.05}_{-0.05}$	$0.33^{+0.03}_{-0.03}$	$0.30^{+0.03}_{-0.03}$
$gg \rightarrow H(\text{MINLOHJ}) + XH$	$1.15^{+0.11}_{-0.11}$	$0.54^{+0.05}_{-0.05}$	$0.32^{+0.03}_{-0.03}$	$0.29^{+0.03}_{-0.03}$

Fiducial cross section $H \rightarrow 4l$ (fb) at 7 TeV.

	$4l$	$2e2\mu$	4μ	$4e$
Measured (stat. $\oplus$ sys.)(model)	$0.555^{+0.694+0.016}_{-0.439-0.019}$	$0.234^{+0.583+0.009}_{-0.234-0.019}$	$0.323^{+0.437+0.012}_{-0.272-0.011}$	$0.000^{+0.169+0.000}_{-0.000-0.000}$
$gg \rightarrow H(\text{HRES}) + XH$	$0.935^{+0.104}_{-0.104}$	$0.441^{+0.049}_{-0.049}$	$0.261^{+0.029}_{-0.029}$	$0.233^{+0.026}_{-0.026}$
$gg \rightarrow H(\text{POHWHEG} + \text{JHUGEN}) + XH$	$0.945^{+0.102}_{-0.102}$	$0.441^{+0.048}_{-0.048}$	$0.267^{+0.029}_{-0.029}$	$0.237^{+0.026}_{-0.026}$
$gg \rightarrow H(\text{MINLOHJ}) + XH$	$0.924^{+0.1}_{-0.1}$	$0.434^{+0.047}_{-0.047}$	$0.26^{+0.028}_{-0.028}$	$0.231^{+0.025}_{-0.025}$

Fiducial cross section $Z \rightarrow 4l$ (fb) at 8 TeV

	$4l$	$2e2\mu$	4μ	$4e$
Measured (stat. $\oplus$ sys.)	$4.81^{+0.69+0.18}_{-0.63-0.19}$	$1.37^{+0.40+0.08}_{-0.34-0.05}$	$2.35^{+0.41+0.14}_{-0.37-0.11}$	$1.08^{+0.44+0.15}_{-0.35-0.08}$
POWHEG	$4.56^{+0.19}_{-0.19}$	$1.57^{+0.06}_{-0.07}$	$1.75^{+0.07}_{-0.07}$	$1.24^{+0.05}_{-0.05}$

twiki

Simultaneous Fit

Fiducial cross section $H \rightarrow 4l$ (fb)				
	$4l$	$2e2\mu$	4μ	$4e$
Measured (stat. $\oplus$ sys.)	$1.062^{+0.422}_{-0.342}$	$0.468^{+0.273}_{-0.206}$	$0.250^{+0.169}_{-0.125}$	$0.342^{+0.301}_{-0.204}$
$gg \rightarrow H(\text{HRES}) + XH$	$1.160^{+0.122}_{-0.122}$	$0.548^{+0.058}_{-0.058}$	$0.322^{+0.034}_{-0.034}$	$0.29^{+0.03}_{-0.030}$
$gg \rightarrow H(\text{POHWHEG} + \text{JHUGEN}) + XH$	$1.161^{+0.125}_{-0.125}$	$0.540^{+0.058}_{-0.058}$	$0.326^{+0.035}_{-0.035}$	$0.295^{+0.032}_{-0.032}$
$gg \rightarrow H(\text{MINLOHJ}) + XH$	$1.143^{+0.123}_{-0.123}$	$0.540^{+0.058}_{-0.058}$	$0.317^{+0.034}_{-0.034}$	$0.285^{+0.031}_{-0.031}$
Ratio of fiducial cross sections $H \rightarrow 4l / Z \rightarrow 4l$				
	$4l$	$2e2\mu$	4μ	$4e$
Measured (stat. $\oplus$ sys.)	$0.21^{+0.09+0.01}_{-0.07-0.01}$	$0.33^{+0.23+0.02}_{-0.16-0.02}$	$0.10^{+0.08+0.01}_{-0.05-0.00}$	$0.30^{+0.32+0.05}_{-0.19-0.03}$
$gg \rightarrow H(\text{HRES}) + XH$	$0.25^{+0.040}_{-0.040}$	$0.35^{+0.05}_{-0.05}$	$0.18^{+0.03}_{-0.03}$	$0.23^{+0.04}_{-0.03}$
$gg \rightarrow H(\text{POHWHEG} + \text{JHUGEN}) + XH$	$0.25^{+0.04}_{-0.04}$	$0.34^{+0.05}_{-0.05}$	$0.19^{+0.03}_{-0.03}$	$0.24^{+0.04}_{-0.03}$
$gg \rightarrow H(\text{MINLOHJ}) + XH$	$0.25^{+0.04}_{-0.04}$	$0.34^{+0.05}_{-0.05}$	$0.18^{+0.03}_{-0.03}$	$0.23^{+0.04}_{-0.03}$
Predicted fiducial cross section of $Z \rightarrow 4l$ (fb)				
	$4l$	$2e2\mu$	4μ	$4e$
POWHEG	$4.578^{+0.186}_{-0.191}$	$1.571^{+0.064}_{-0.066}$	$1.766^{+0.072}_{-0.074}$	$1.241^{+0.051}_{-0.052}$

Event Counts in Full Mass Range

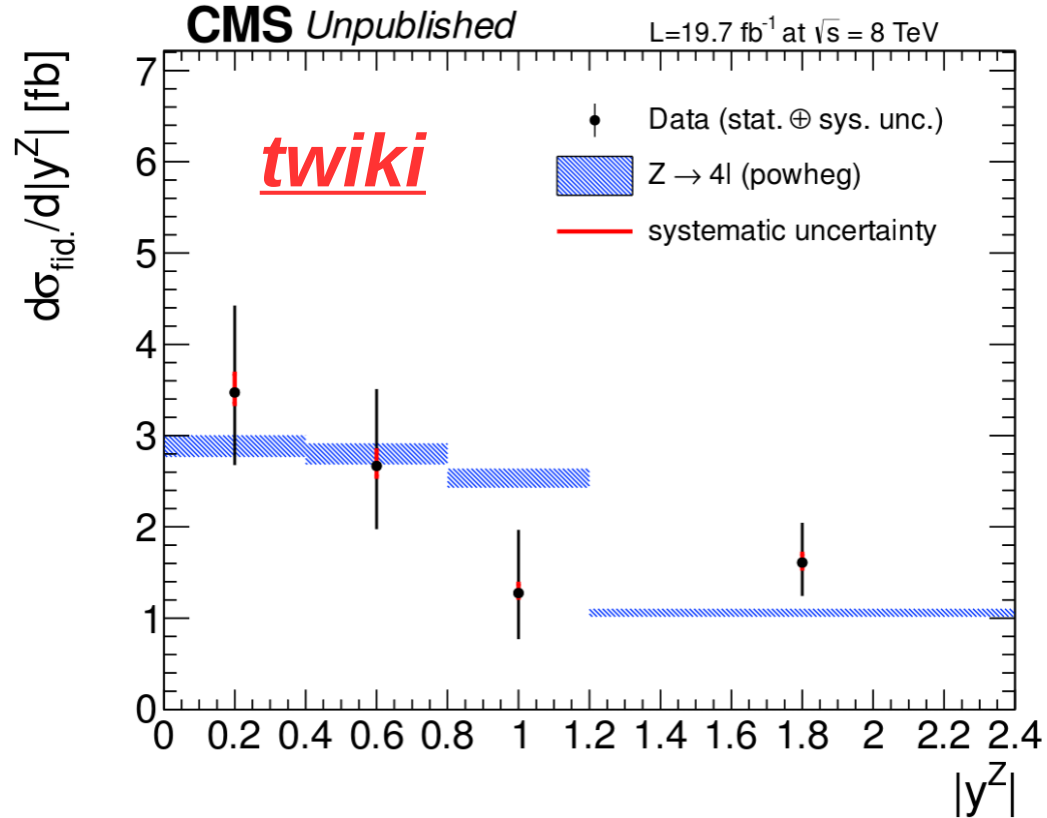
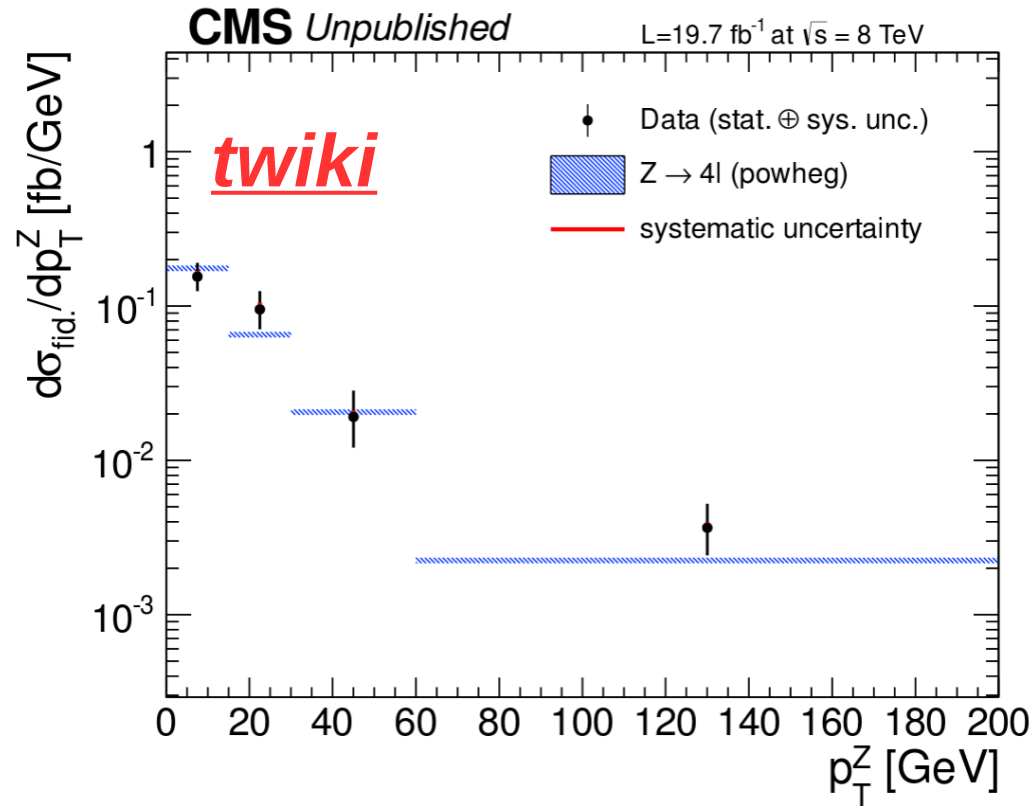
twiki Event counts in the range $50 \text{ GeV} < m(4\ell) < 105 \text{ GeV}$

Channel	4e	4 μ	2e2 μ
$q\bar{q} \rightarrow ZZ$ (t- and u-channels)	0.30 ± 0.02	1.03 ± 0.04	0.56 ± 0.05
Z+ X	0.43 ± 0.14	0.13 ± 0.03	1.10 ± 0.13
$gg \rightarrow ZZ$	0.15 ± 0.01	0.43 ± 0.02	0.10 ± 0.01
Total background expected	0.88 ± 0.14	1.59 ± 0.11	1.16 ± 0.44
$Z \rightarrow 4l$ ($q\bar{q} \rightarrow ZZ$ s-channel)	9.87 ± 0.11	29.16 ± 0.19	18.34 ± 0.23
Observed	9	42	18

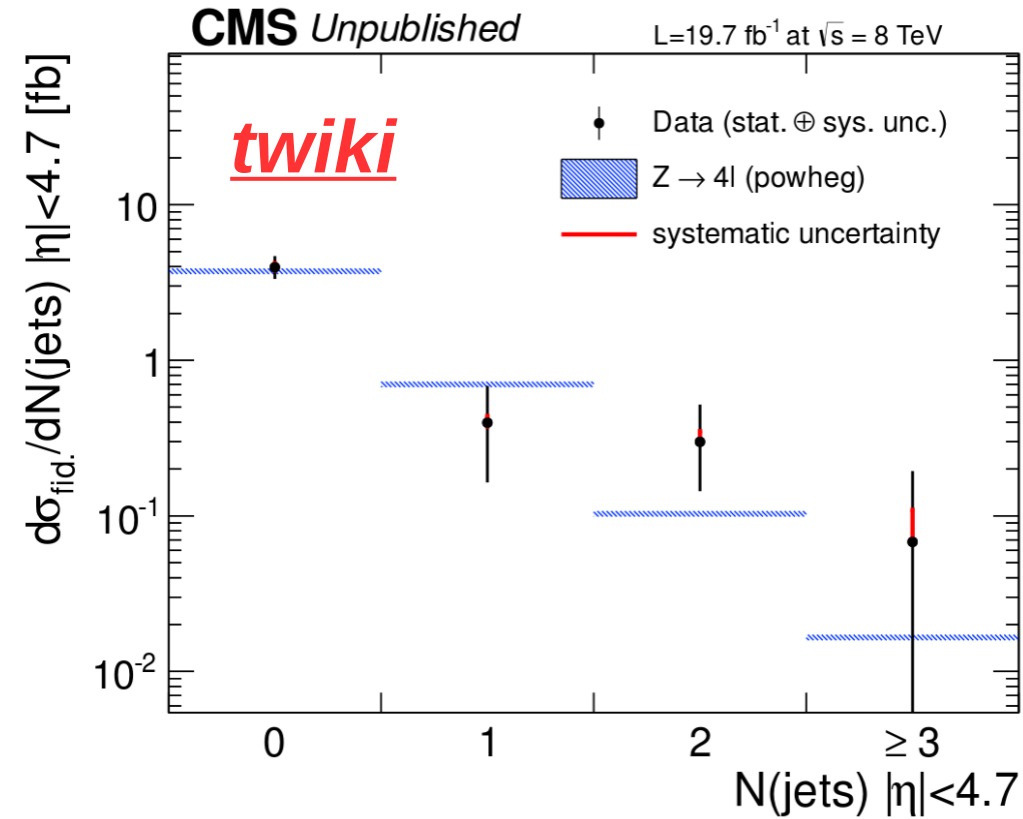
twiki Event counts in the range $50 \text{ GeV} < m(4\ell) < 140 \text{ GeV}$

Channel	4e	4 μ	2e2 μ
$q\bar{q} \rightarrow ZZ$ (t- and u-channels)	2.38 ± 0.06	6.78 ± 0.11	7.50 ± 0.20
Z+X	1.96 ± 0.44	1.31 ± 0.56	5.26 ± 1.38
$gg \rightarrow ZZ$	0.36 ± 0.02	0.84 ± 0.02	0.53 ± 0.02
All background expected	4.65 ± 0.44	8.93 ± 0.57	13.29 ± 1.39
$H \rightarrow 4l$ ($m_H = 125.0 \text{ GeV}$)	2.94 ± 0.03	5.68 ± 0.05	7.33 ± 0.06
$Z \rightarrow 4l$ ($qq \rightarrow ZZ$ s-channel)	10.41 ± 0.11	30.63 ± 0.21	19.80 ± 0.26
Observed	18	57	33

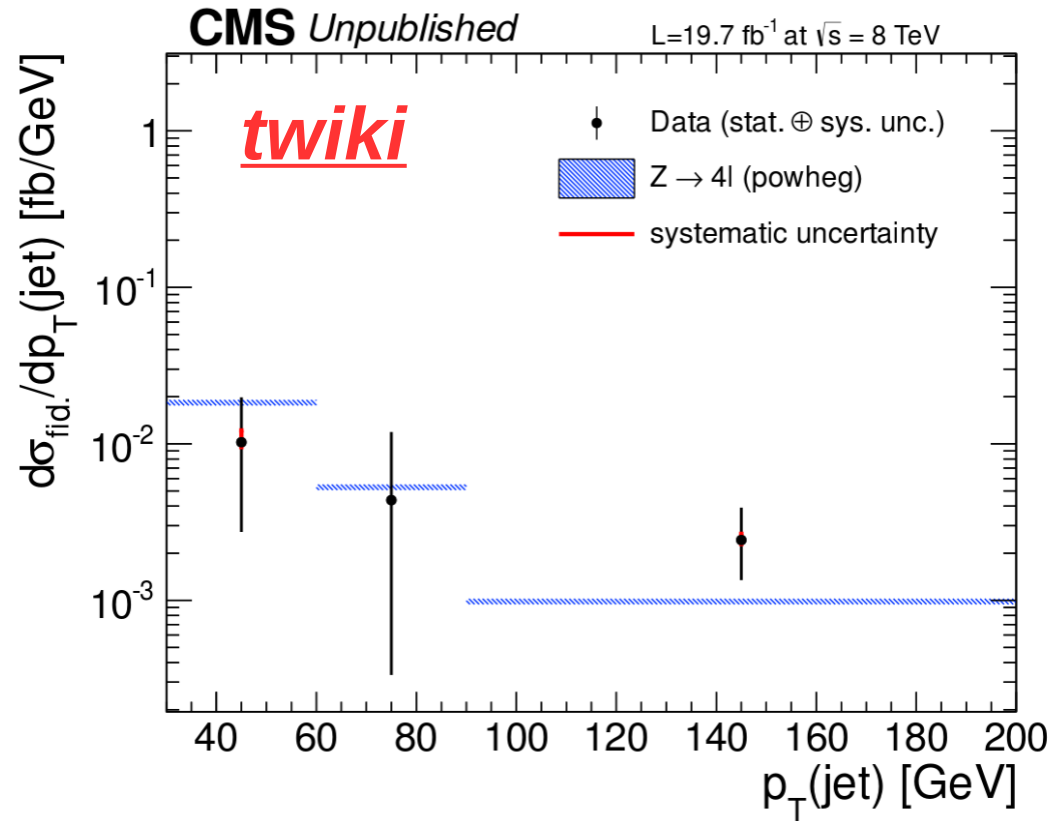
$Z \rightarrow 4\ell$ Differential Results



$Z \rightarrow 4\ell$ Differential Results

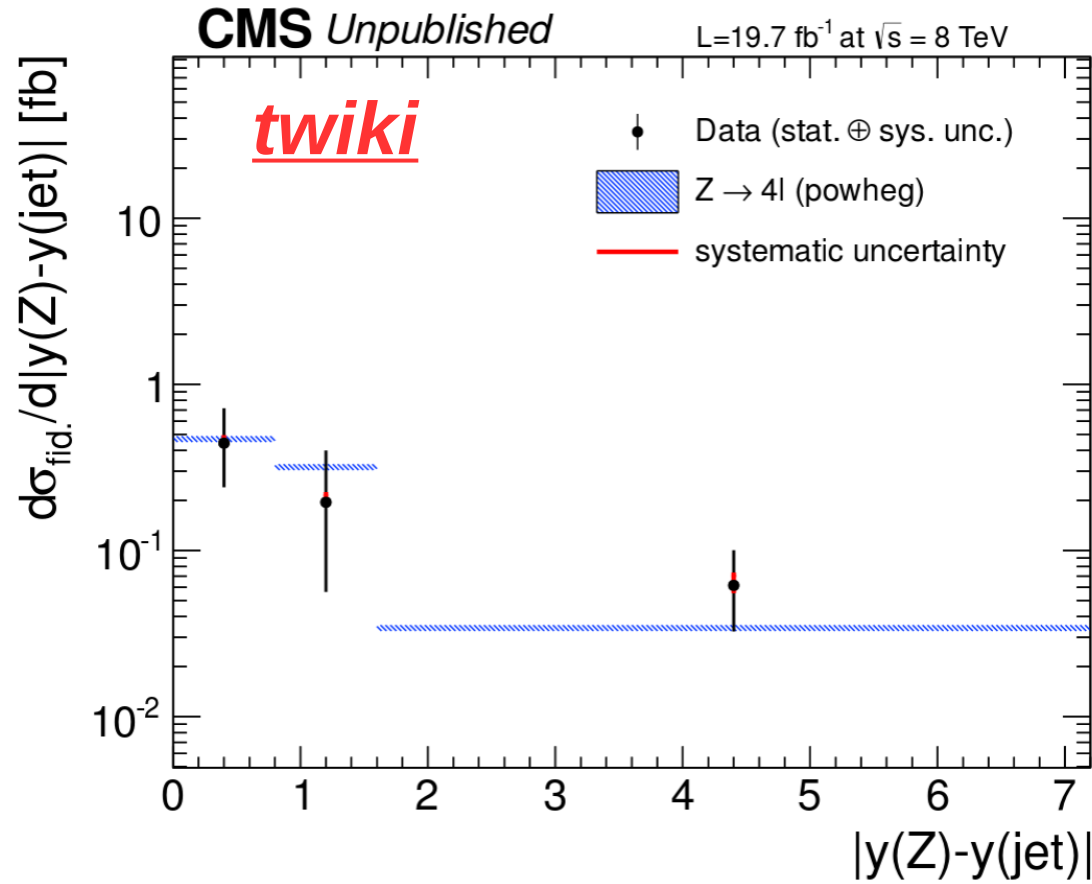


p-value = 0.39



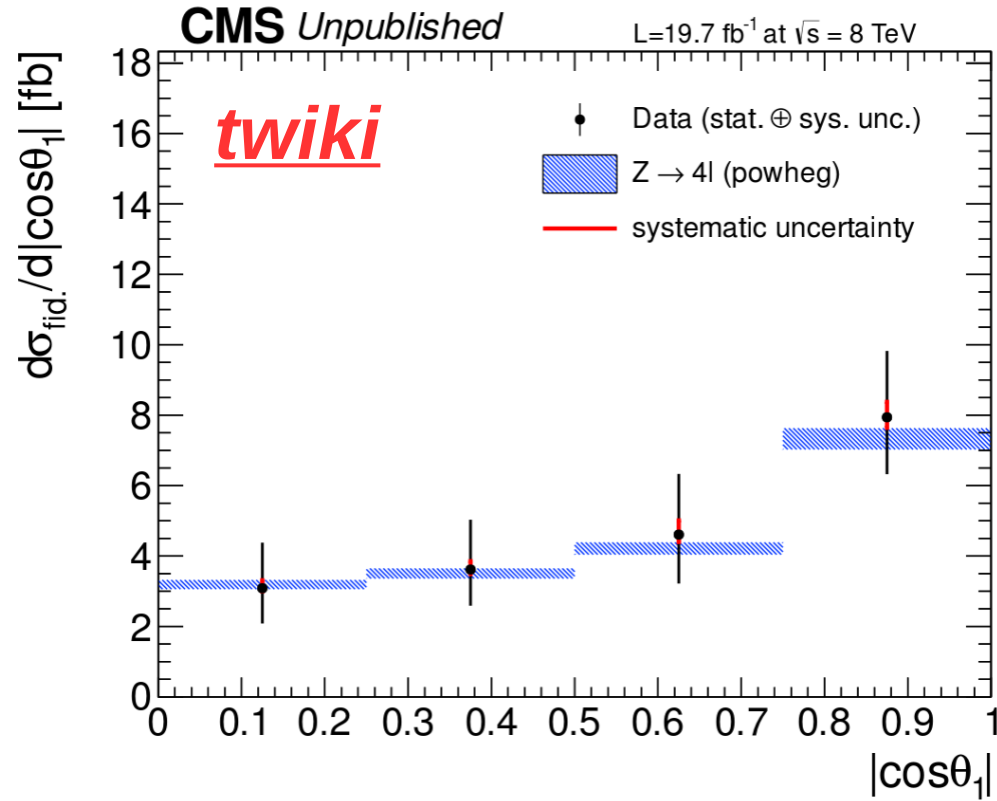
p-value = 0.55

$Z \rightarrow 4\ell$ Differential Results

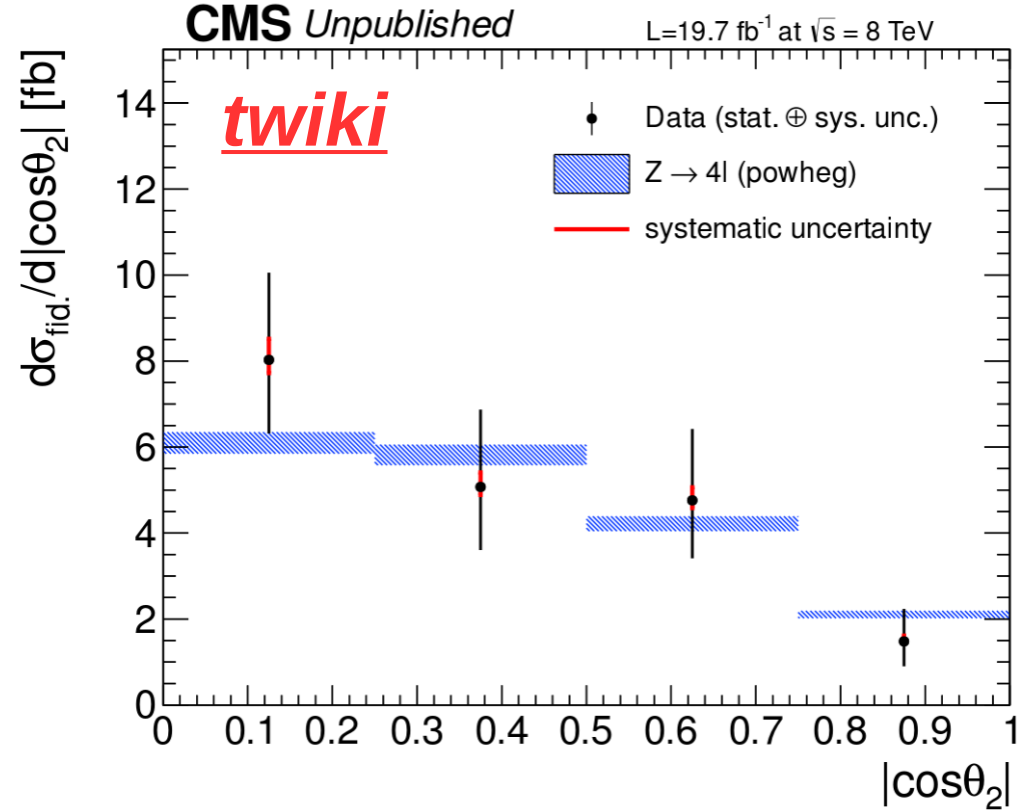


p-value = 0.84

$Z \rightarrow 4\ell$ Differential Results

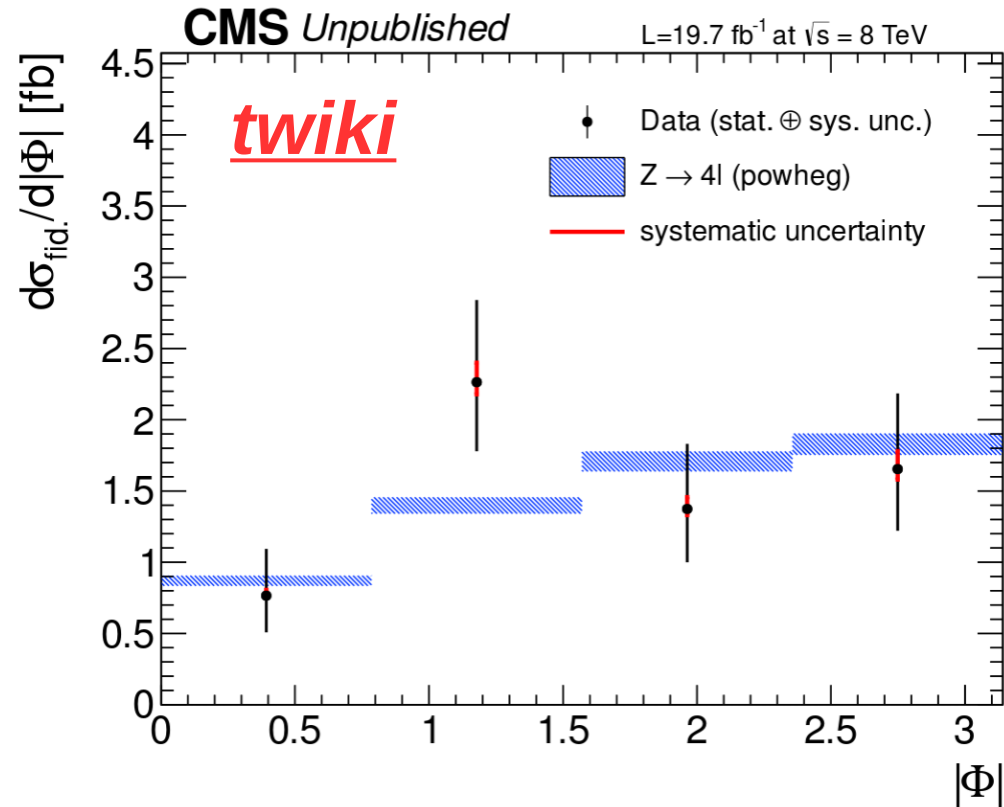


p-value = 0.99

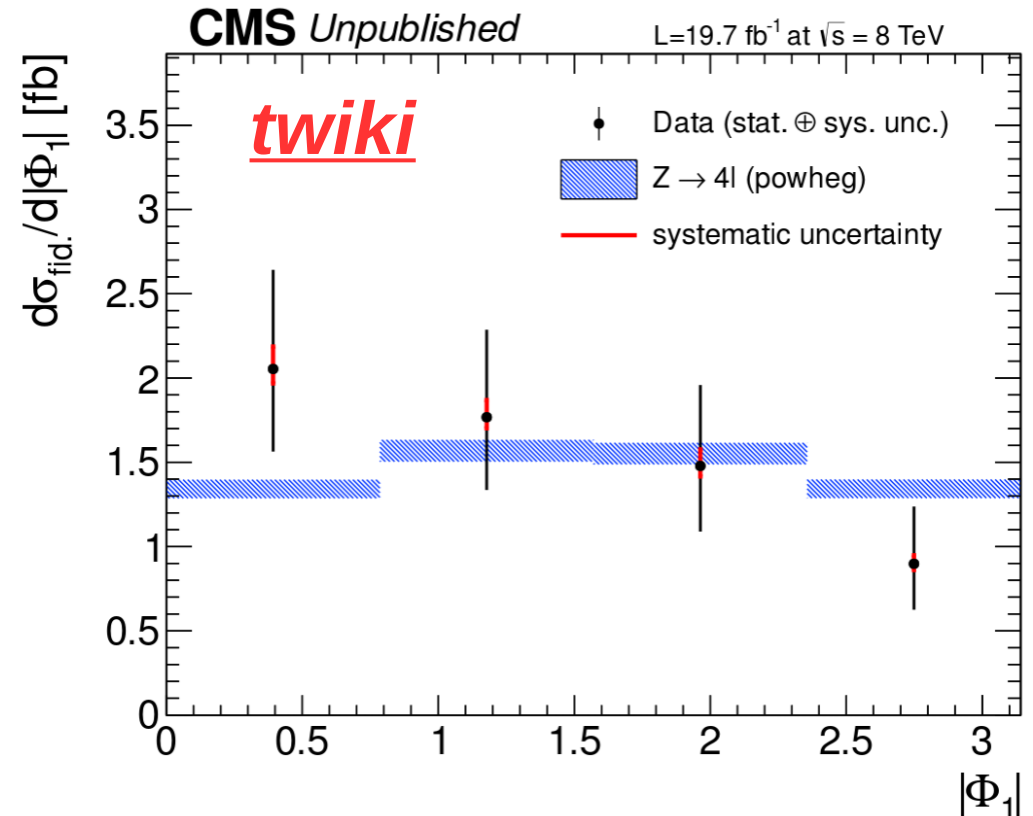


p-value = 0.69

$Z \rightarrow 4\ell$ Differential Results

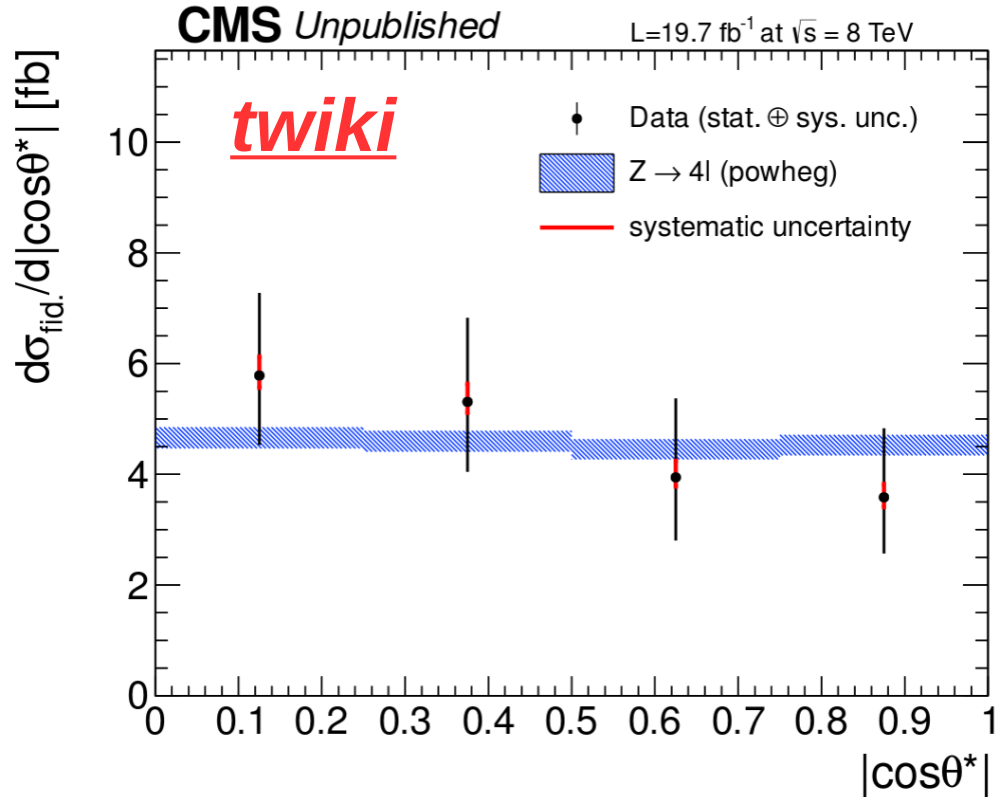


p-value = 0.34

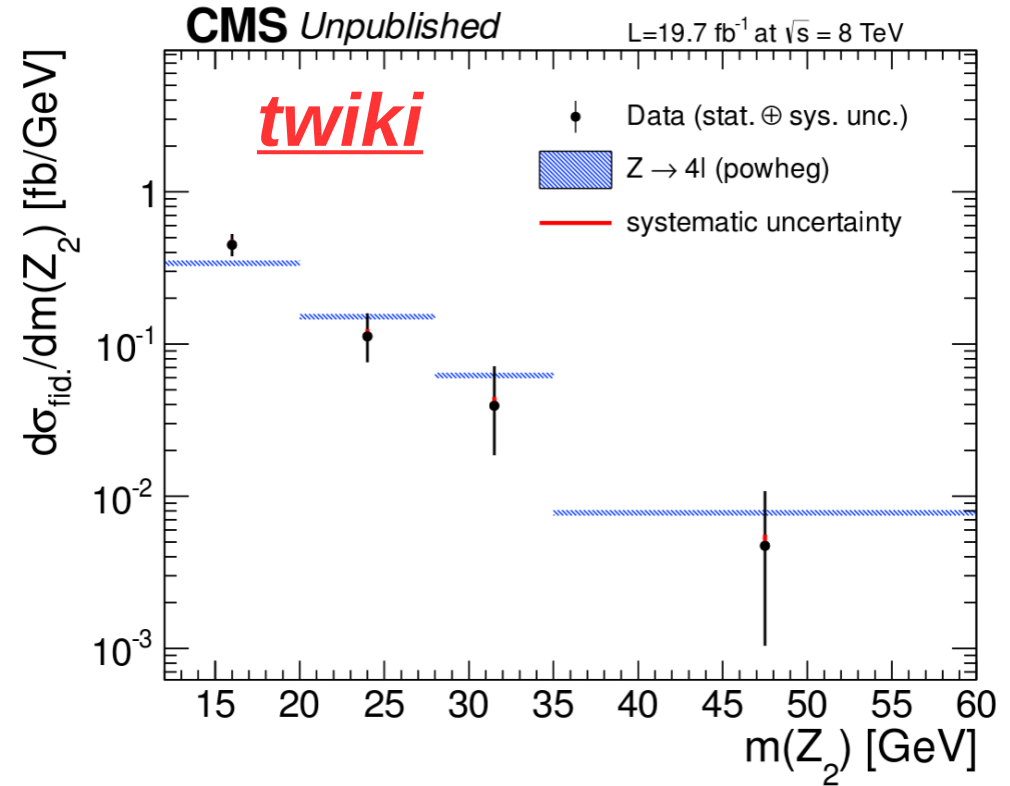


p-value = 0.37

$Z \rightarrow 4\ell$ Differential Results



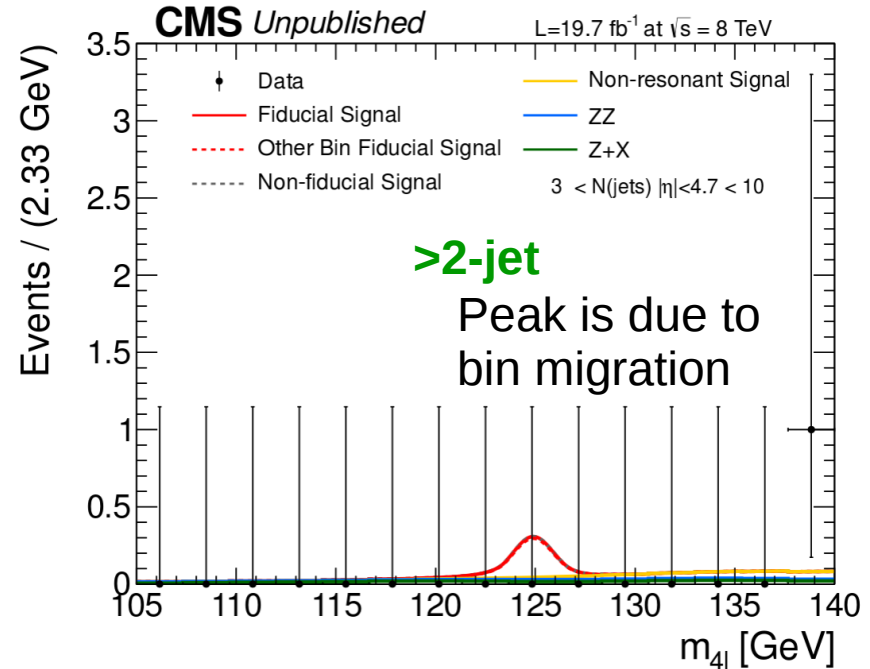
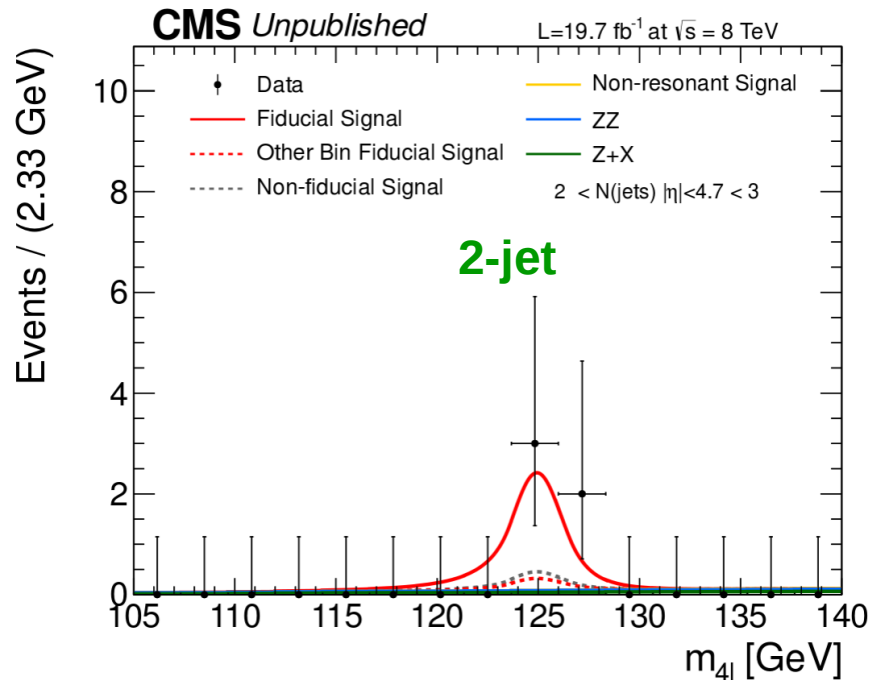
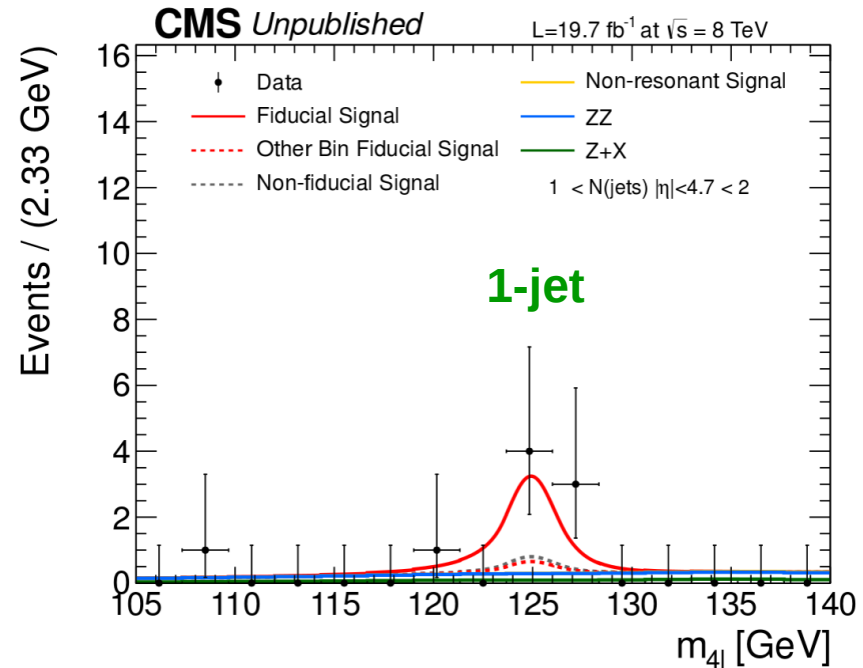
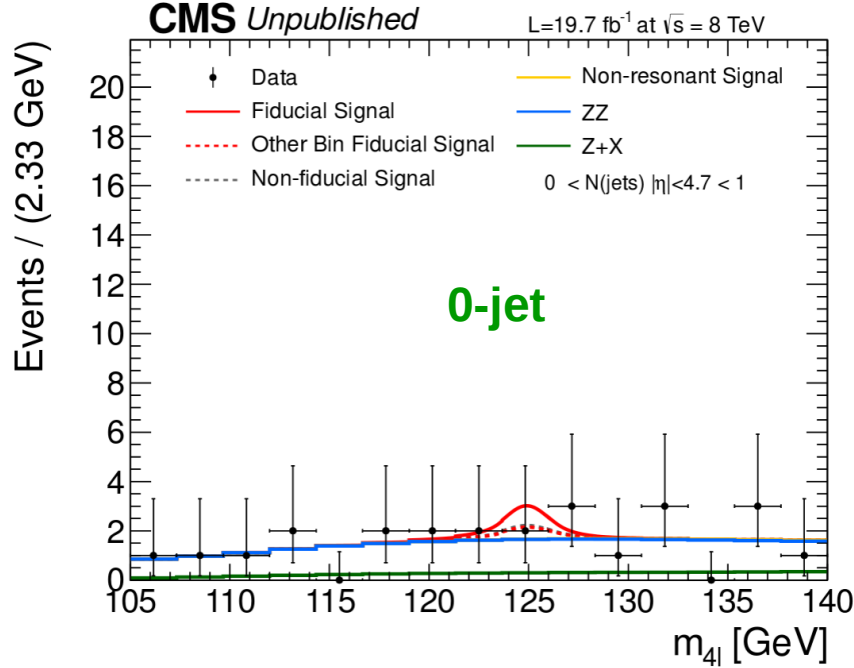
p-value = 0.76



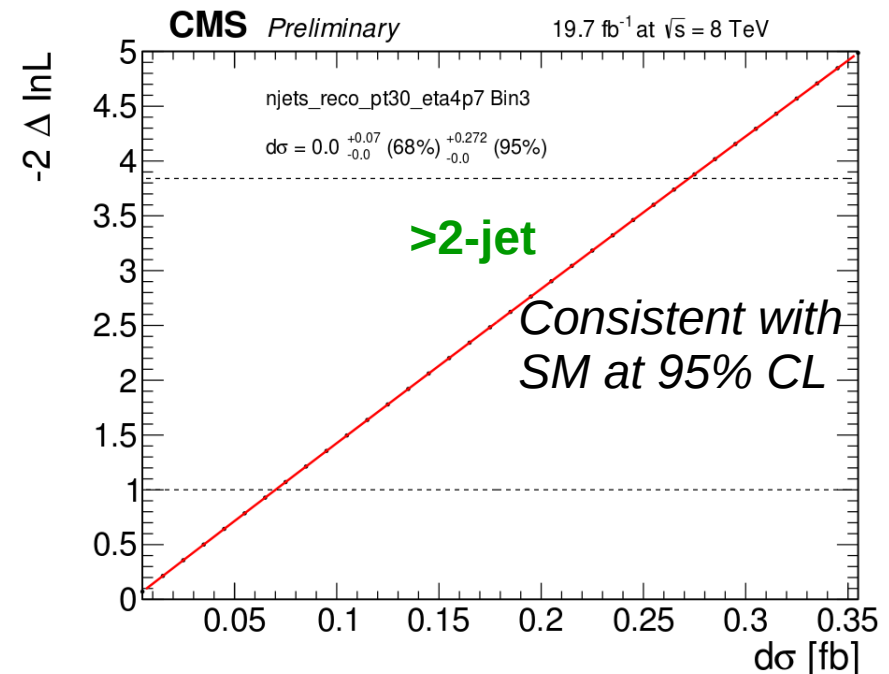
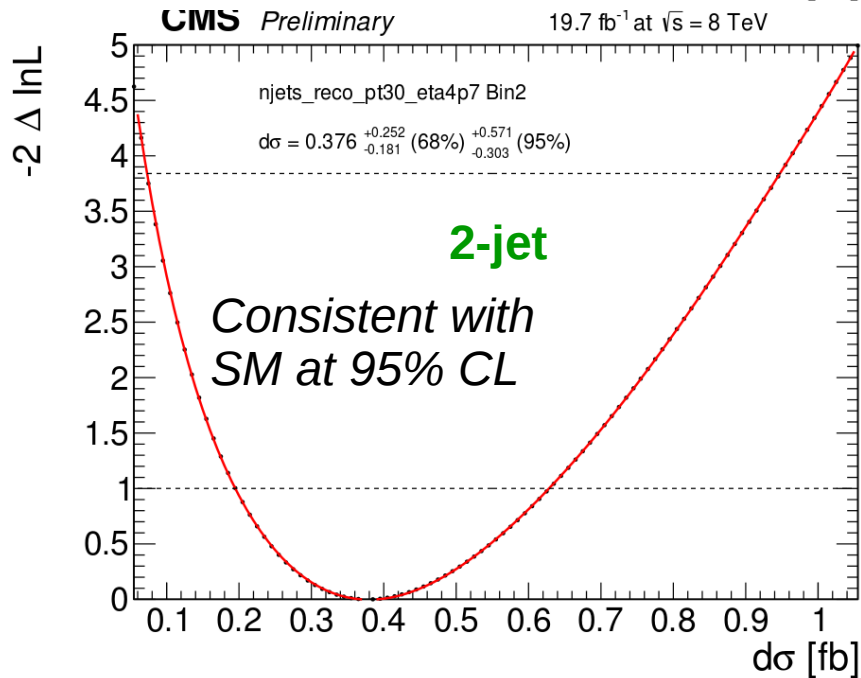
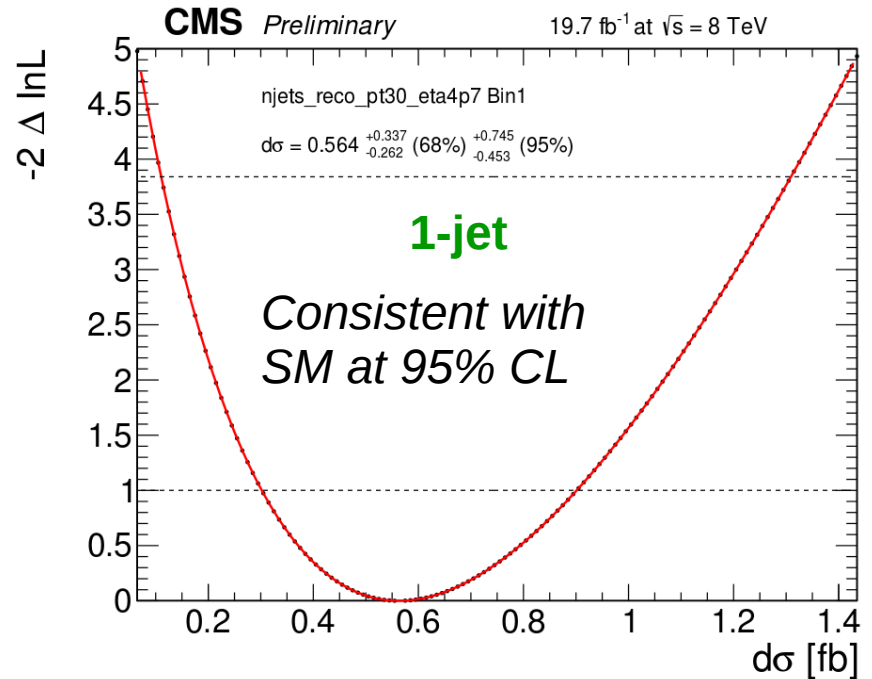
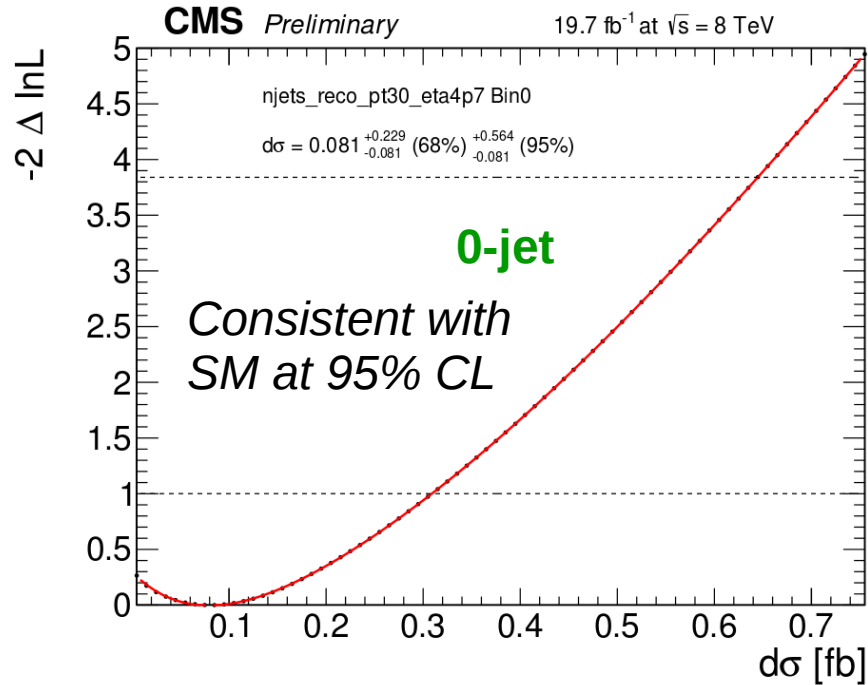
p-value = 0.38

Backup

Fit Results for N(jets)



Likelihood Scans for N(jets)

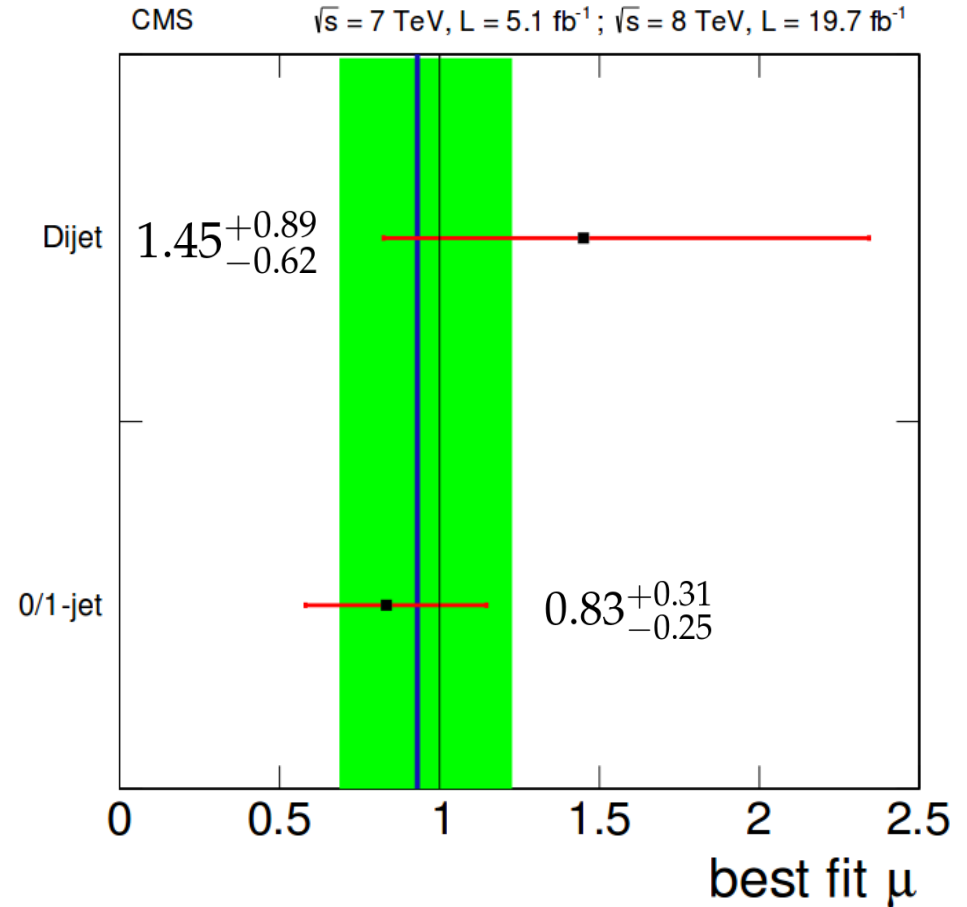


Comparison to Legacy N(jets) (3d)



- Direct Comparison:
 - Identical event counts / $m(4\ell)$ values in each jet category (0/1 or ≥ 2)
- Indirect Comparison:
 - Fiducial cross section is 8 TeV only, and only 1d, i.e. $m(4\ell)$
 - However...
 - Fiducial “ μ ” $\sigma_{\text{fid}}/\sigma_{\text{fid}}(\text{SM})$:
 - 0+1-jet: 0.652
 - 2+3-jet: 2.19
 - Within the uncertainty of the 7+8 TeV signal strength

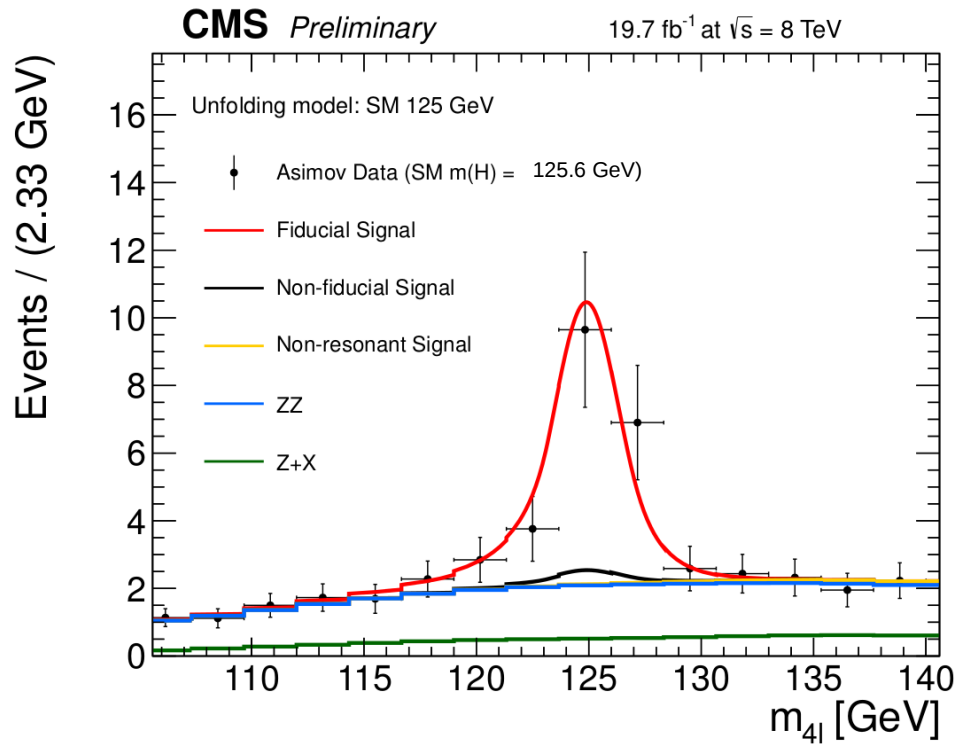
3d



Fixing vs. Floating the Mass

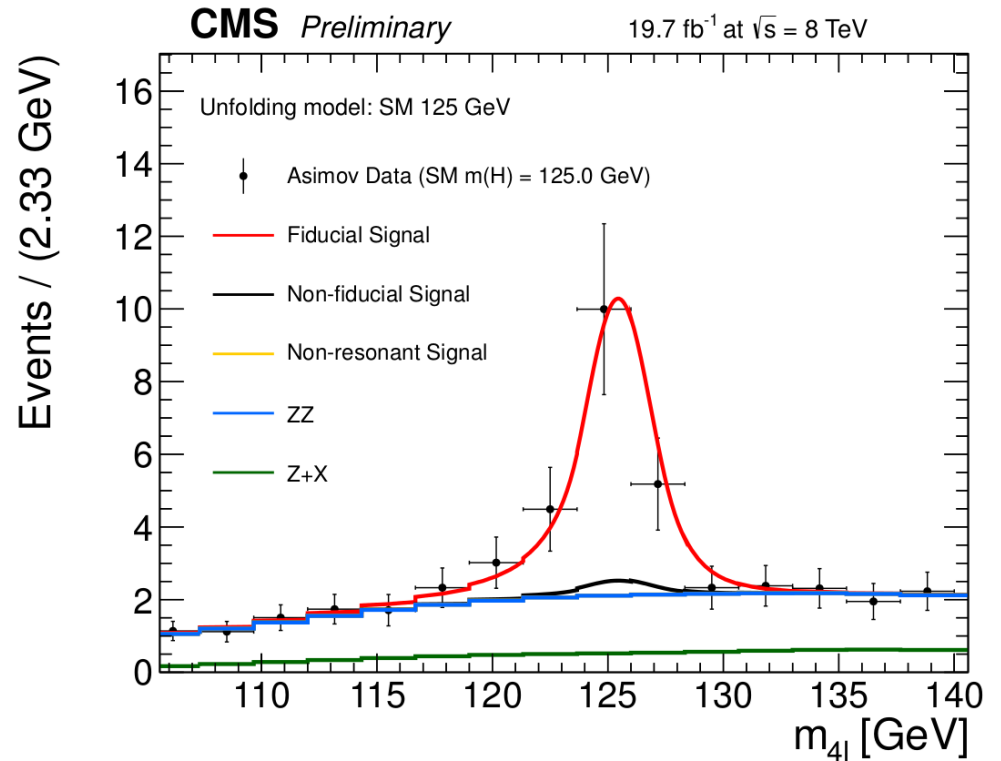
Asimov data 125.6, fix mass 125.0
(fractions floated)

$\sigma(4\ell) = 1.193 -0.383/+0.446 \text{ fb}$
Truth $m(H) = 125.6$: $\sigma(4\ell) = 1.215 \text{ fb}$



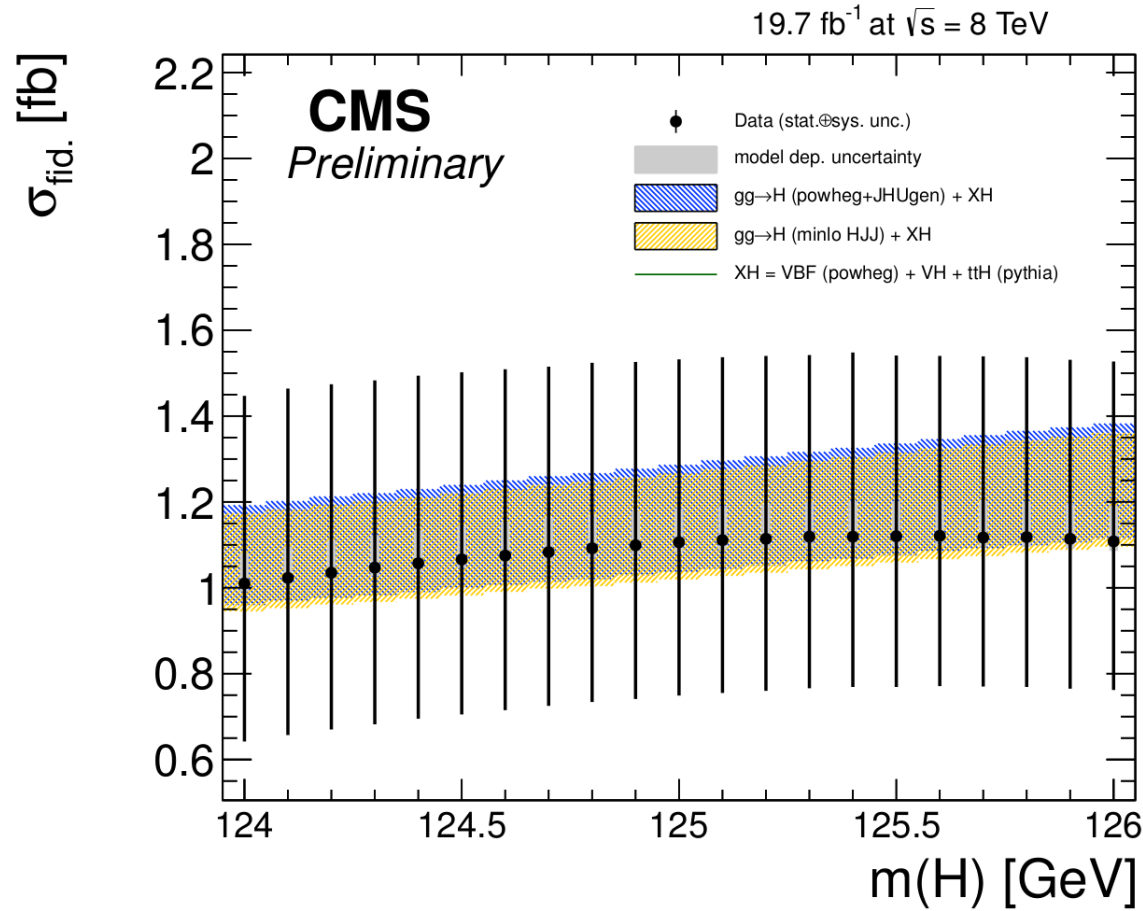
Asimov data 125.0, fix mass 125.6
(fractions floated)

$\sigma(4\ell) = 1.147 -0.381/+0.423 \text{ fb}$
Truth $m(H) = 125.0$: 1.162 fb



~2% bias for 0.6 GeV offset, $m(H)$ measured in ZZ and gamma gamma to ~0.3 GeV
→ Additional 1% uncertainty due to fixing the mass is negligible with respect to other systematic uncertainties

Mass Dependence



$$m(H) = 124.0 \text{ GeV: } \sigma_{\text{fid.}} = 1.010 +0.437/-0.368 \text{ fb } (\mu=0.938)$$

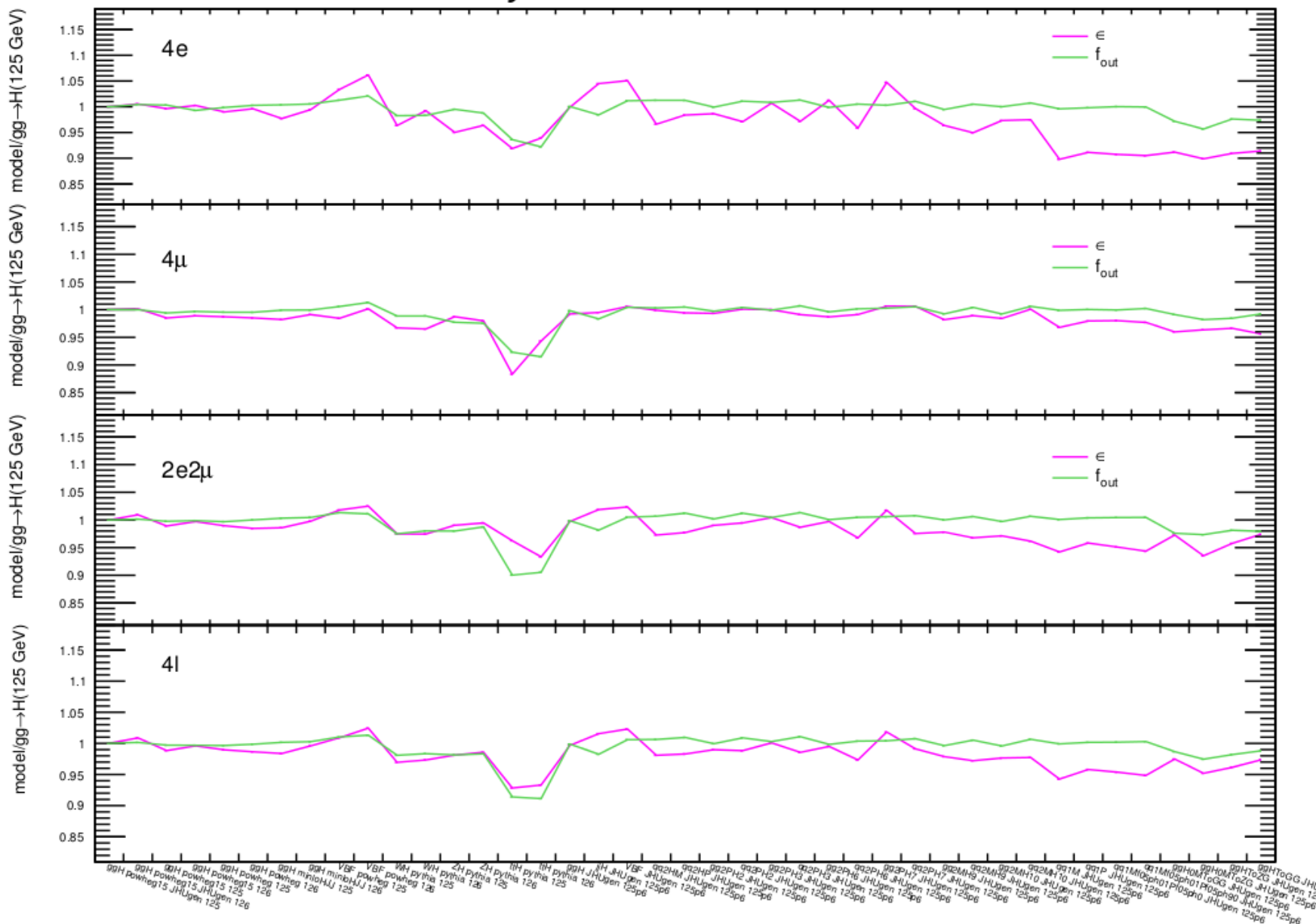
$$m(H) = 124.5 \text{ GeV: } \sigma_{\text{fid.}} = 1.066 +0.436/-0.361 \text{ fb } (\mu=0.953)$$

$$m(H) = 125.0 \text{ GeV: } \sigma_{\text{fid.}} = 1.106 +0.426/-0.357 \text{ fb } (\mu=0.952)$$

$$m(H) = 125.6 \text{ GeV: } \sigma_{\text{fid.}} = 1.121 +0.419/-0.350 \text{ fb } (\mu=0.922)$$

$$m(H) = 126.0 \text{ GeV: } \sigma_{\text{fid.}} = 1.108 +0.419/-0.346 \text{ fb } (\mu=0.888)$$

CMS *Preliminary*

 $\sqrt{s} = 8 \text{ TeV}$ 

Model Dependence

8 TeV:

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
$gg \rightarrow H$ (POWHEG+JHUGEN)	0.422 ± 0.001	0.647 ± 0.002	0.053 ± 0.001	0.681 ± 0.002
VBF (POWHEG)	0.476 ± 0.003	0.652 ± 0.005	0.040 ± 0.002	0.678 ± 0.005
WH (PYTHIA)	0.342 ± 0.002	0.627 ± 0.003	0.072 ± 0.002	0.672 ± 0.003
ZH (PYTHIA)	0.348 ± 0.003	0.634 ± 0.004	0.072 ± 0.003	0.679 ± 0.005
$t\bar{t}H$ (PYTHIA)	0.250 ± 0.003	0.601 ± 0.008	0.139 ± 0.008	0.685 ± 0.010
Some characteristic models of Higgs-like boson with exotic decays and properties				
$q\bar{q} \rightarrow H(J^{\text{CP}}=1^-)$ (JHUGEN)	0.238 ± 0.001	0.609 ± 0.002	0.054 ± 0.001	0.642 ± 0.002
$q\bar{q} \rightarrow H(J^{\text{CP}}=1^+)$ (JHUGEN)	0.283 ± 0.001	0.619 ± 0.002	0.051 ± 0.001	0.651 ± 0.002
$gg \rightarrow H \rightarrow Z\gamma^*$ (JHUGEN)	0.156 ± 0.001	0.622 ± 0.002	0.073 ± 0.001	0.667 ± 0.002
$gg \rightarrow H \rightarrow \gamma^*\gamma^*$ (JHUGEN)	0.188 ± 0.001	0.629 ± 0.002	0.066 ± 0.001	0.671 ± 0.002

7 TeV:

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs-boson production modes				
ggH_powheg15_JHUGen_125	0.438 ± 0.001	0.676 ± 0.002	0.049 ± 0.001	0.709 ± 0.002
ggH_minloHJJ_125	0.427 ± 0.001	0.675 ± 0.002	0.047 ± 0.001	0.707 ± 0.002
VBF_powheg_125	0.483 ± 0.003	0.674 ± 0.004	0.035 ± 0.002	0.697 ± 0.005
WH_pythia_125	0.357 ± 0.002	0.660 ± 0.004	0.061 ± 0.002	0.700 ± 0.004
ZH_pythia_125	0.372 ± 0.004	0.662 ± 0.006	0.059 ± 0.004	0.701 ± 0.007
$t\bar{t}H$ _pythia_125	0.248 ± 0.004	0.636 ± 0.010	0.158 ± 0.011	0.736 ± 0.013
Some characteristic models of Higgs-like boson with exotic decays and properties				
qq1M_JHUGen_126	0.254 ± 0.001	0.645 ± 0.002	0.049 ± 0.001	0.677 ± 0.002
qq1P_JHUGen_126	0.301 ± 0.001	0.657 ± 0.002	0.044 ± 0.001	0.686 ± 0.002
ggH0MToZG_JHUGen_125p6	0.115 ± 0.000	0.647 ± 0.002	0.086 ± 0.002	0.702 ± 0.002
ggH0MToGG_JHUGen_125p6	0.185 ± 0.001	0.659 ± 0.002	0.064 ± 0.001	0.701 ± 0.002

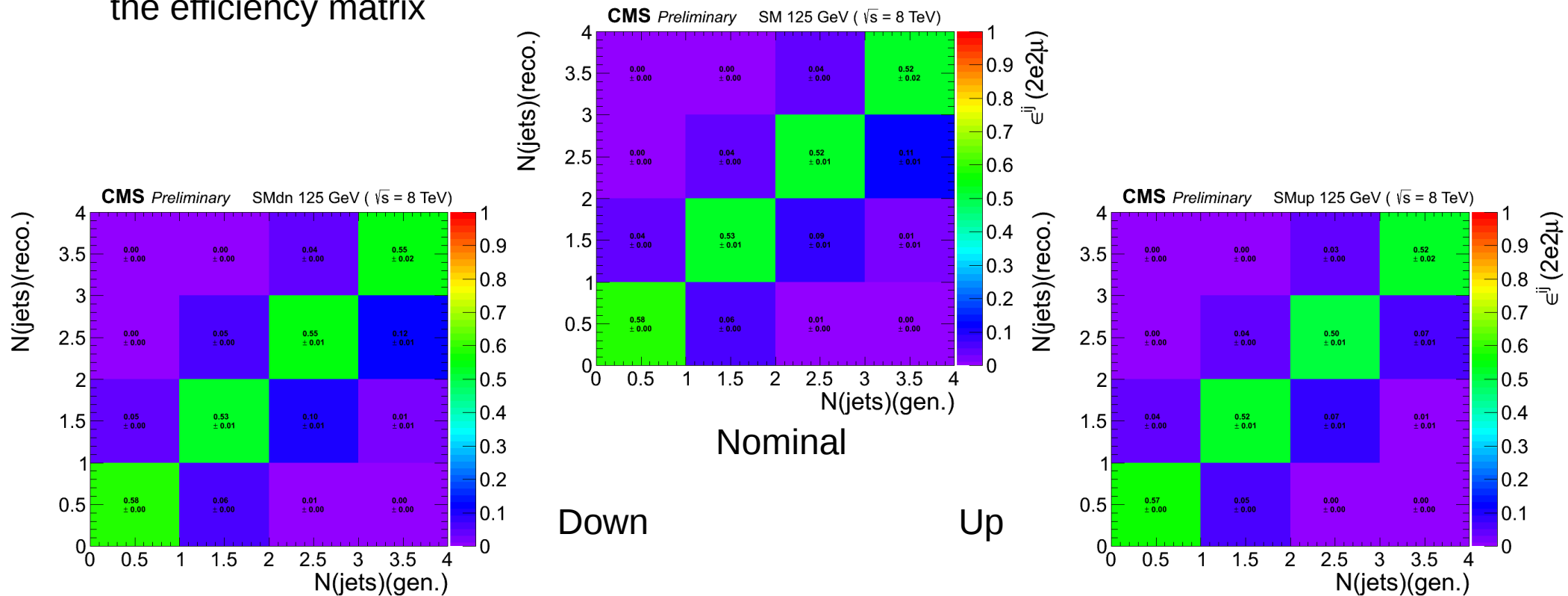
- weight efficiencies by their fiducial cross sections:

$$\text{eff}_{\text{SM}} = \sigma_{\text{ggH}}^* \text{eff}_{\text{ggH}} / \sigma_{\text{tot.}} + \sigma_{\text{VBF}}^* \text{eff}_{\text{VBF}} / \sigma_{\text{tot.}} + \sigma_{\text{WH}}^* \text{eff}_{\text{WH}} / \sigma_{\text{tot.}} + \sigma_{\text{ZH}}^* \text{eff}_{\text{ZH}} / \sigma_{\text{tot.}} + \sigma_{\text{ttH}}^* \text{eff}_{\text{ttH}} / \sigma_{\text{tot.}}$$

- Allow each prod. Mode to vary within some predefined range

$$\sigma_{ggH} \rightarrow \mu_{ggH} \sigma_{ggH}, \sigma_{VBF} \rightarrow \mu_{VBF} \sigma_{VBF} \dots$$

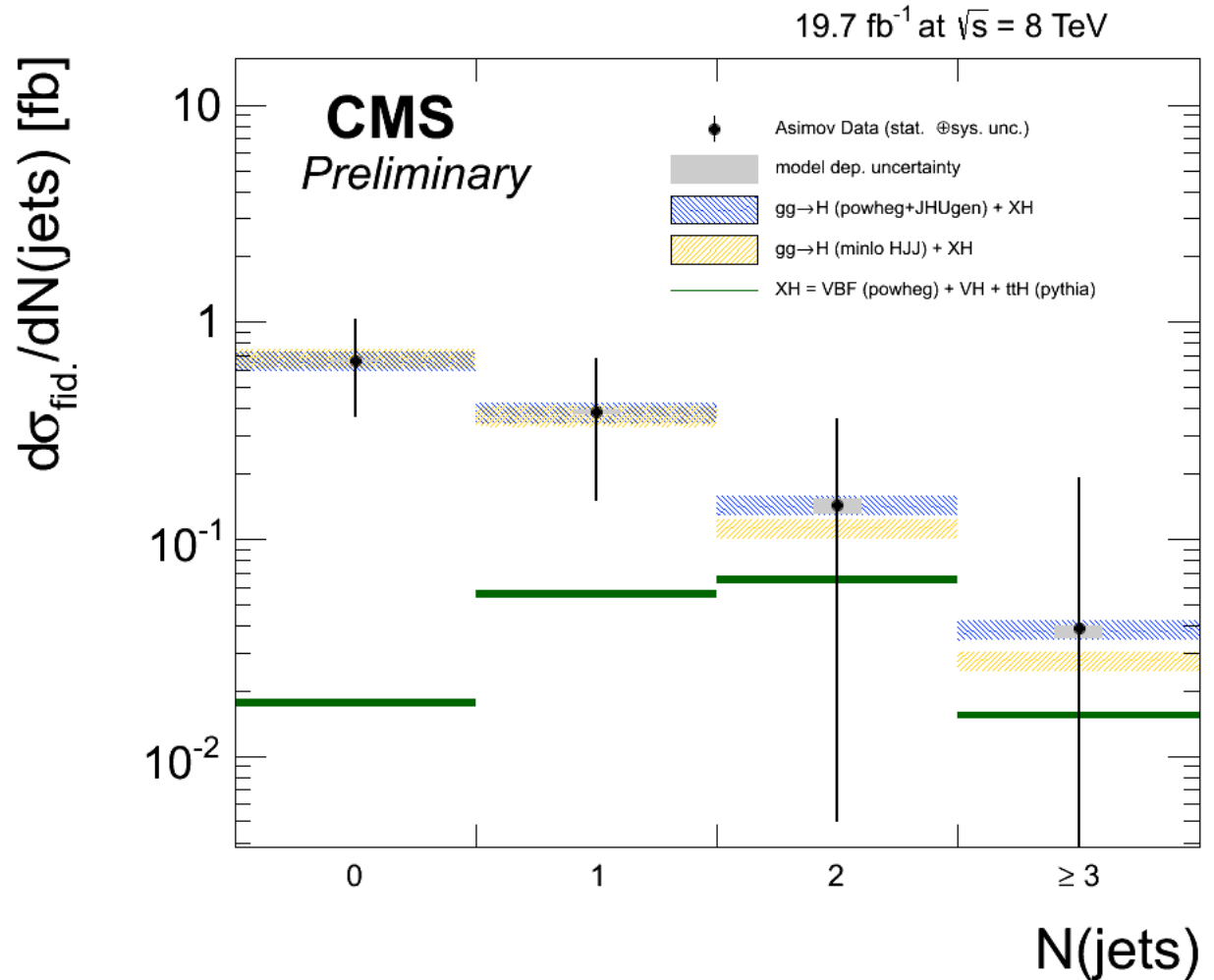
- Find combination of production modes that gives highest/lowest $(1-f_{\text{nonfid}})^* \text{eff}$ in each cell of the efficiency matrix
- CMS Preliminary SM 125 GeV ($\sqrt{s} = 8 \text{ TeV}$)



Constrained Model Dependence

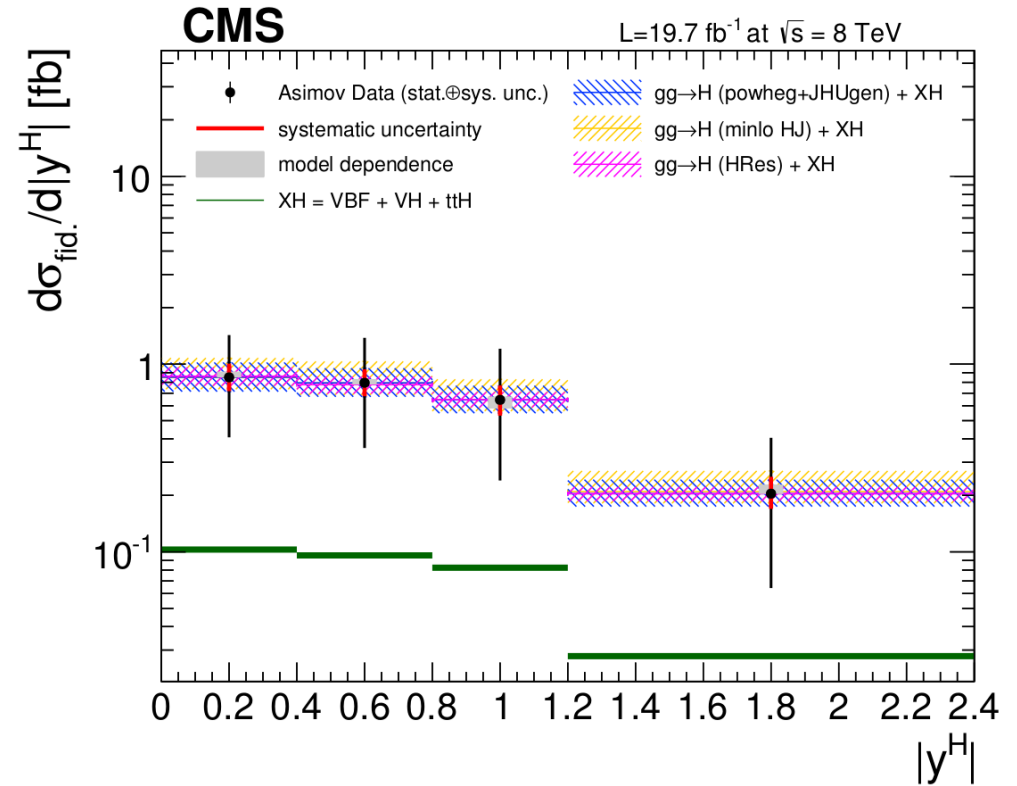
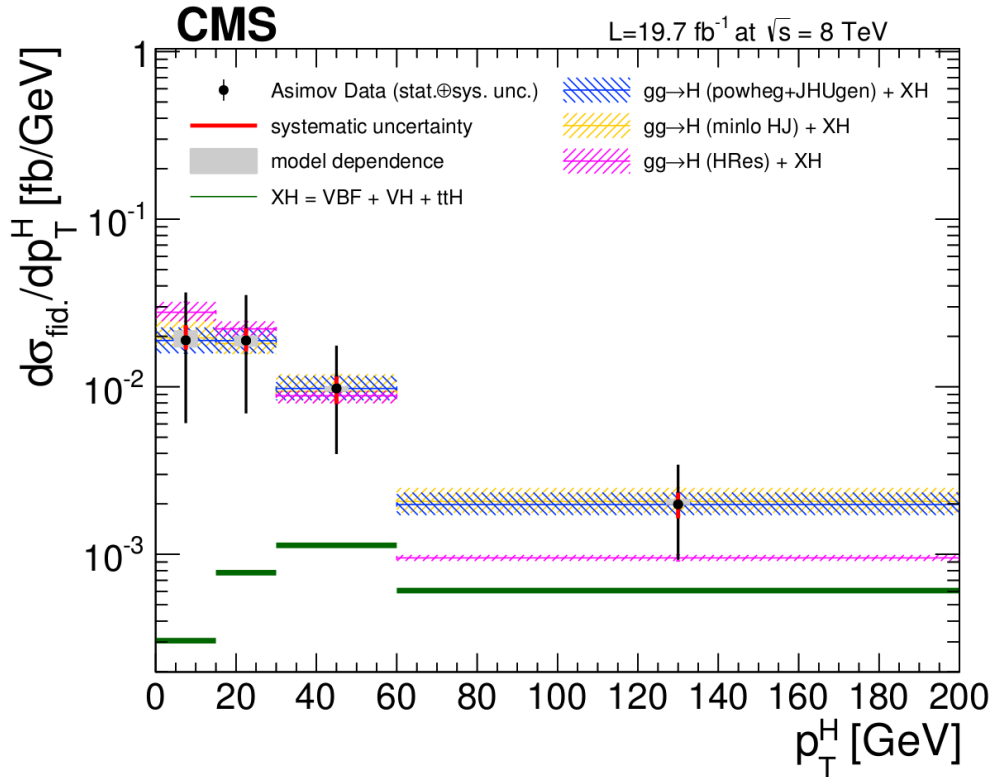


- Algorithm with constraints:
 - $0.5 < \mu(\text{ggH}) < 1.5$
 - $0.4 < \mu(\text{VBF}) < 2.0$
 - $0.2 < \mu(\text{VH}) < 1.8$
 - $0.5 < \mu(\text{ttH}) < 5.0$

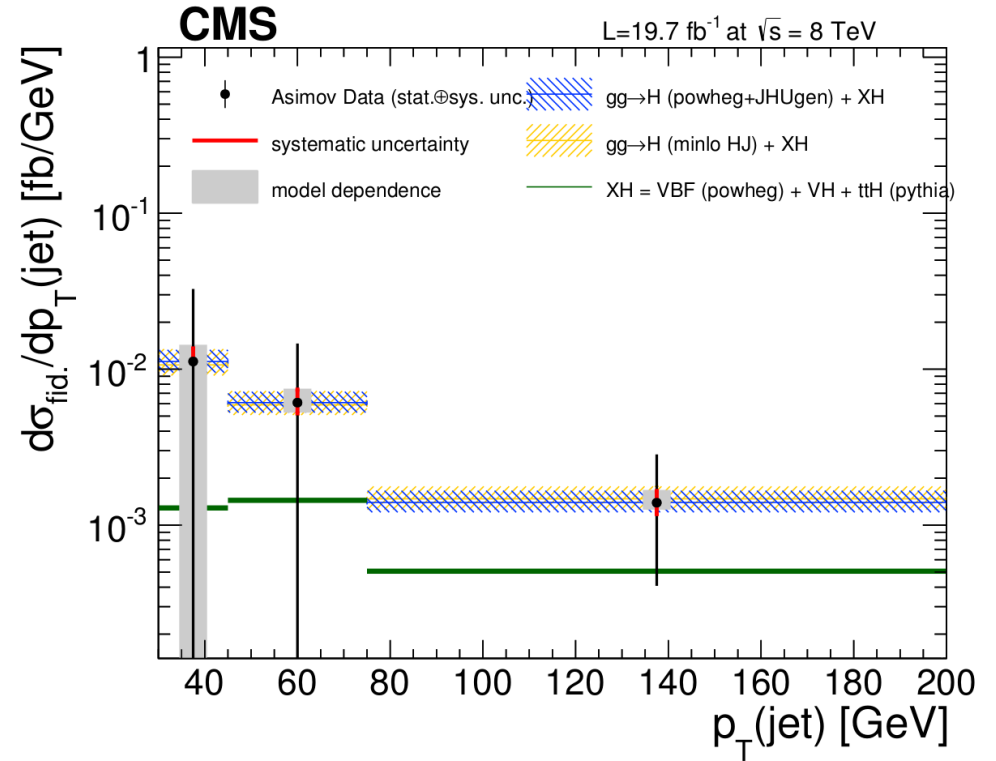
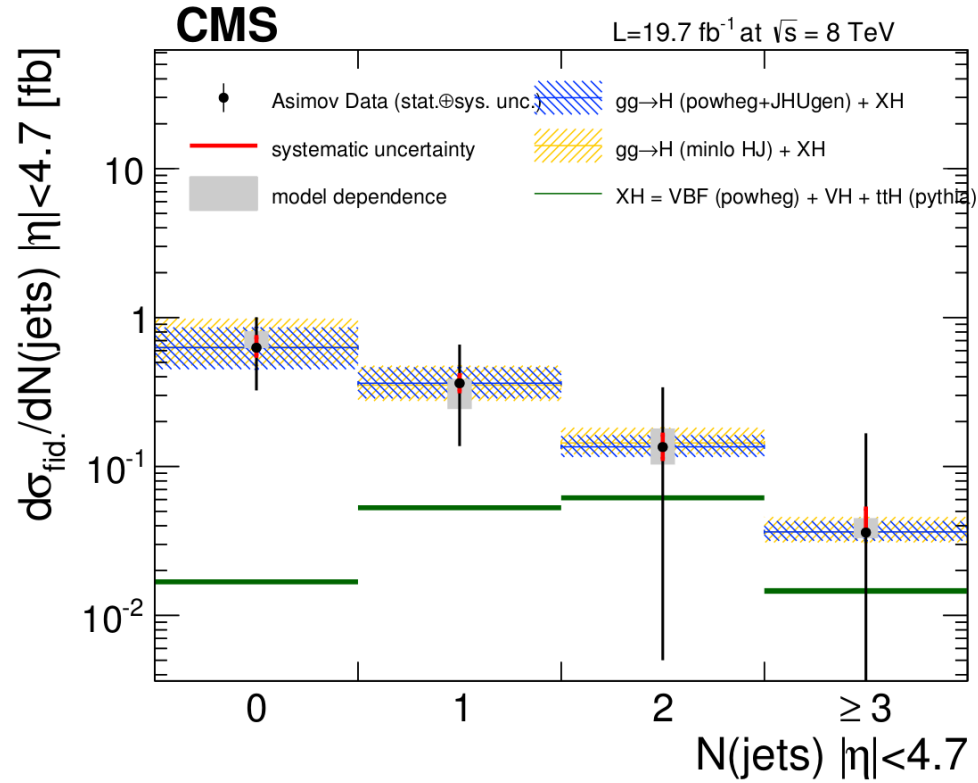


Asimov Data

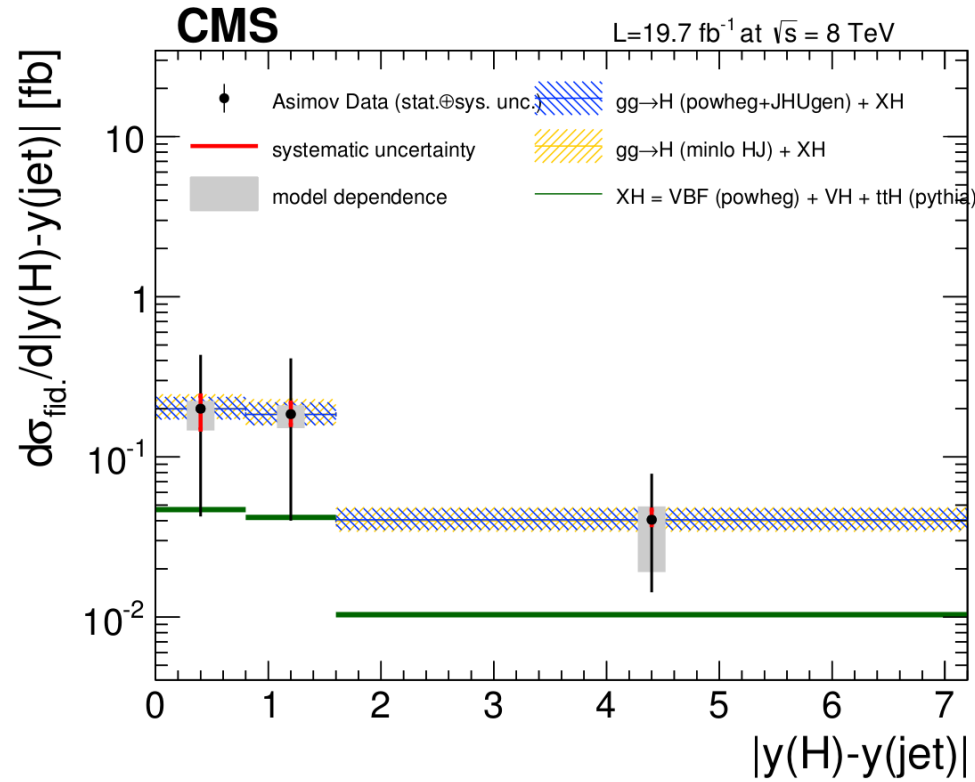
Results With Asimov Data



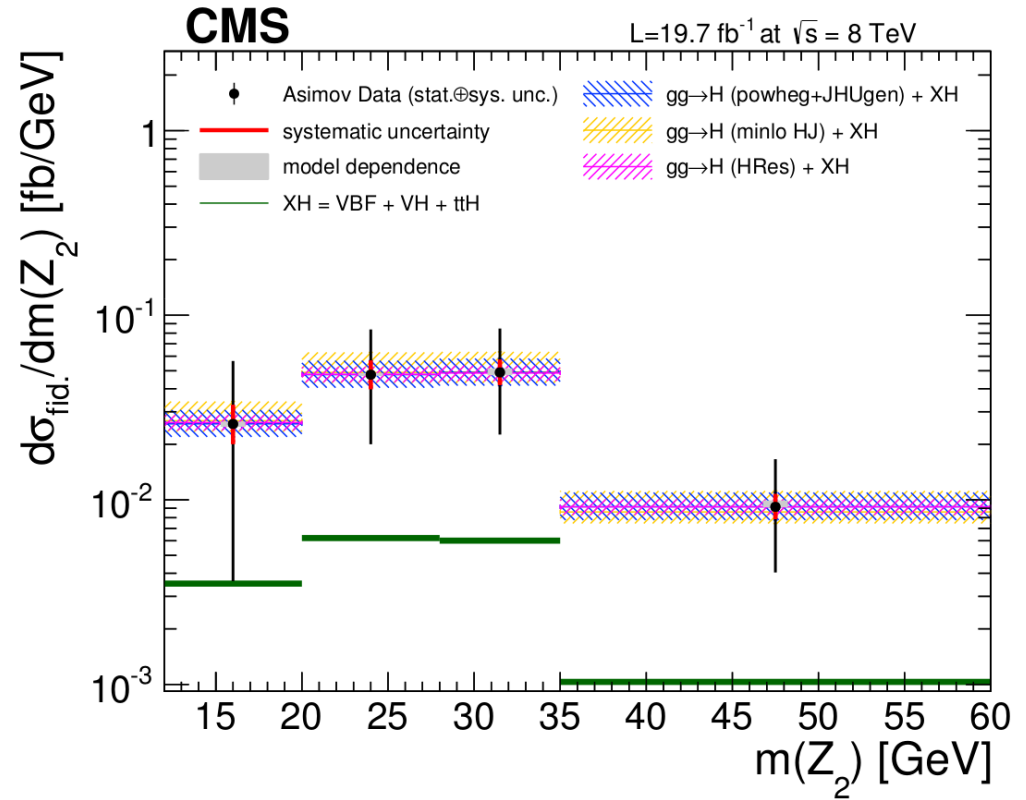
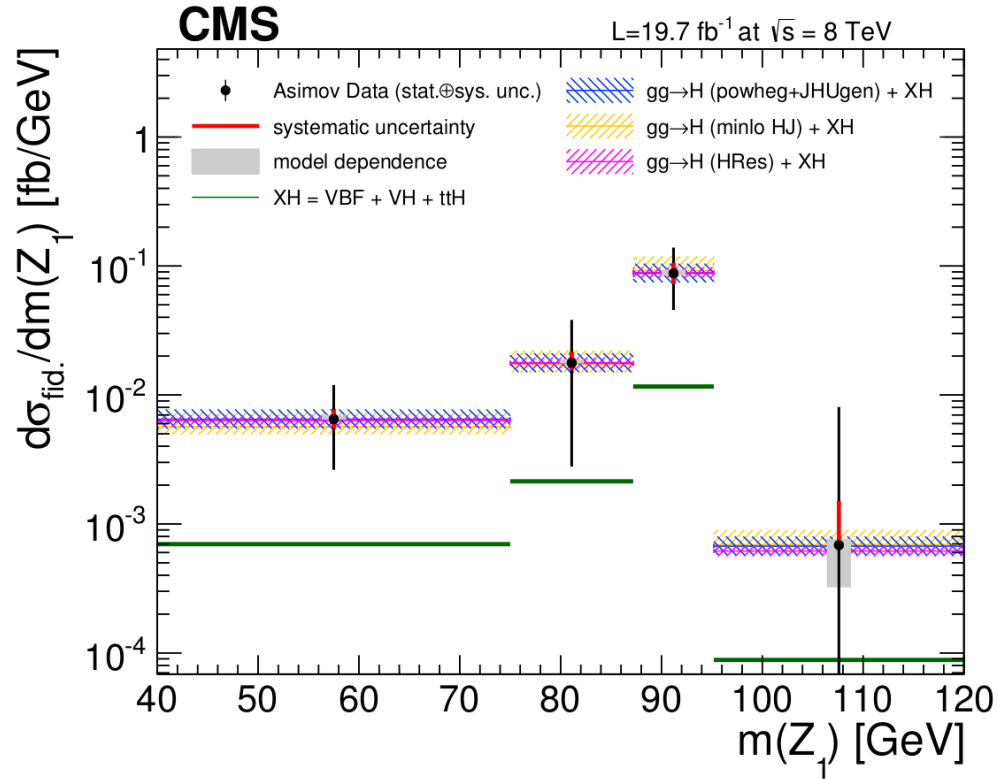
Results With Asimov Data



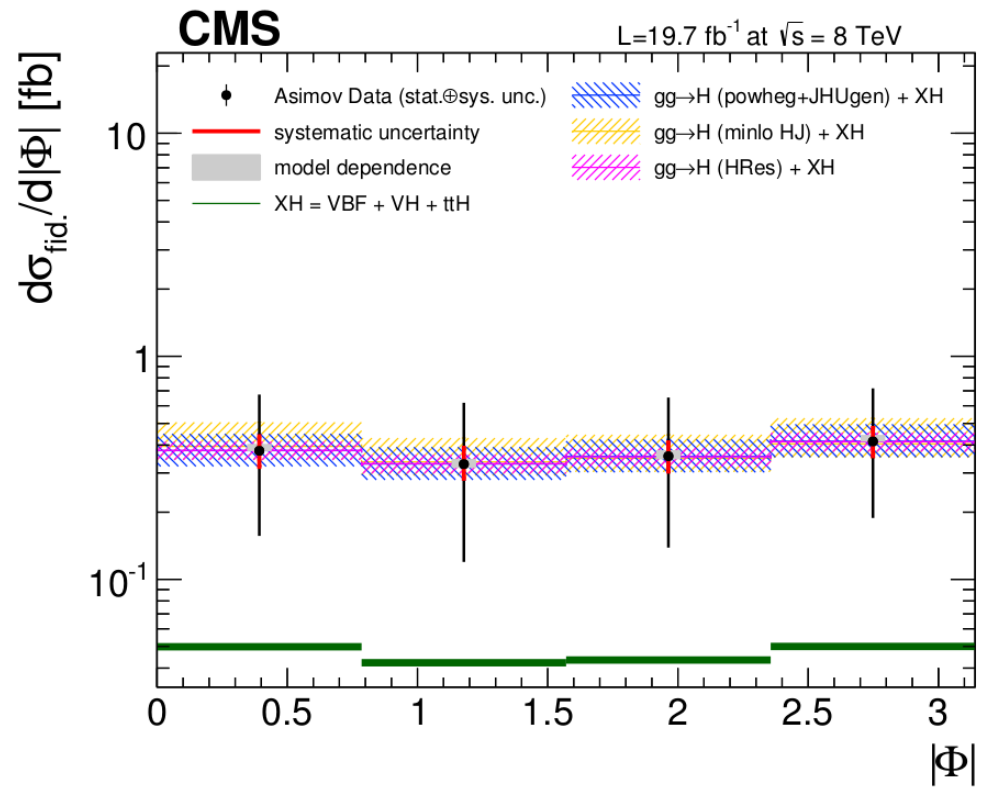
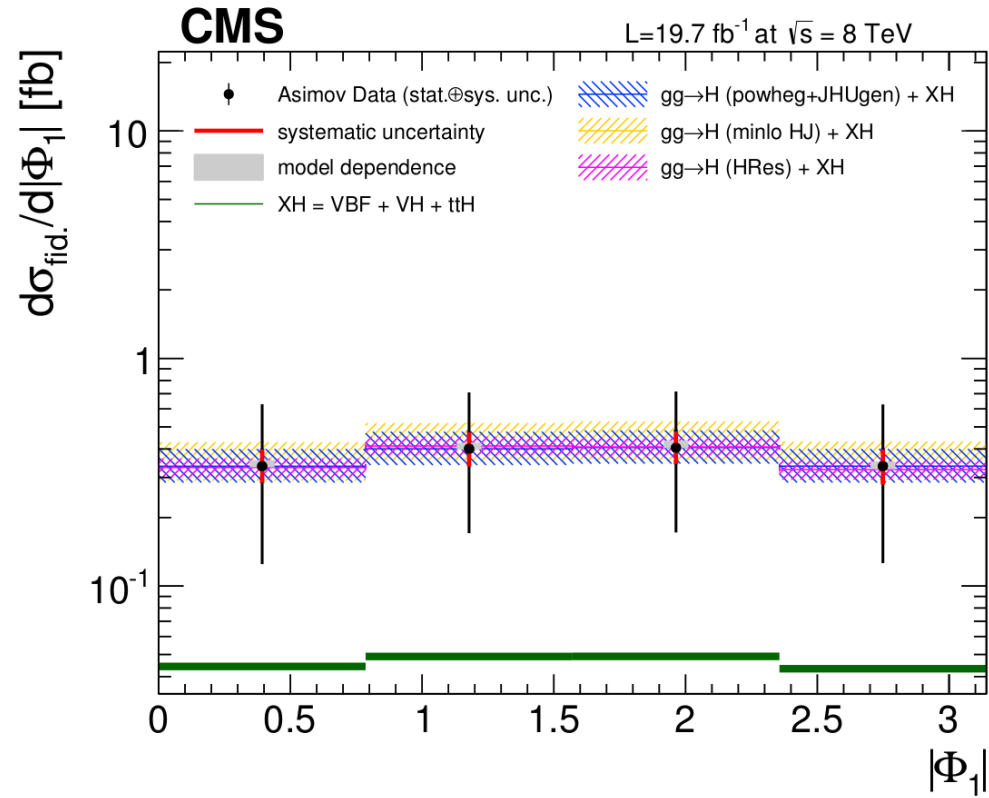
Results With Asimov Data



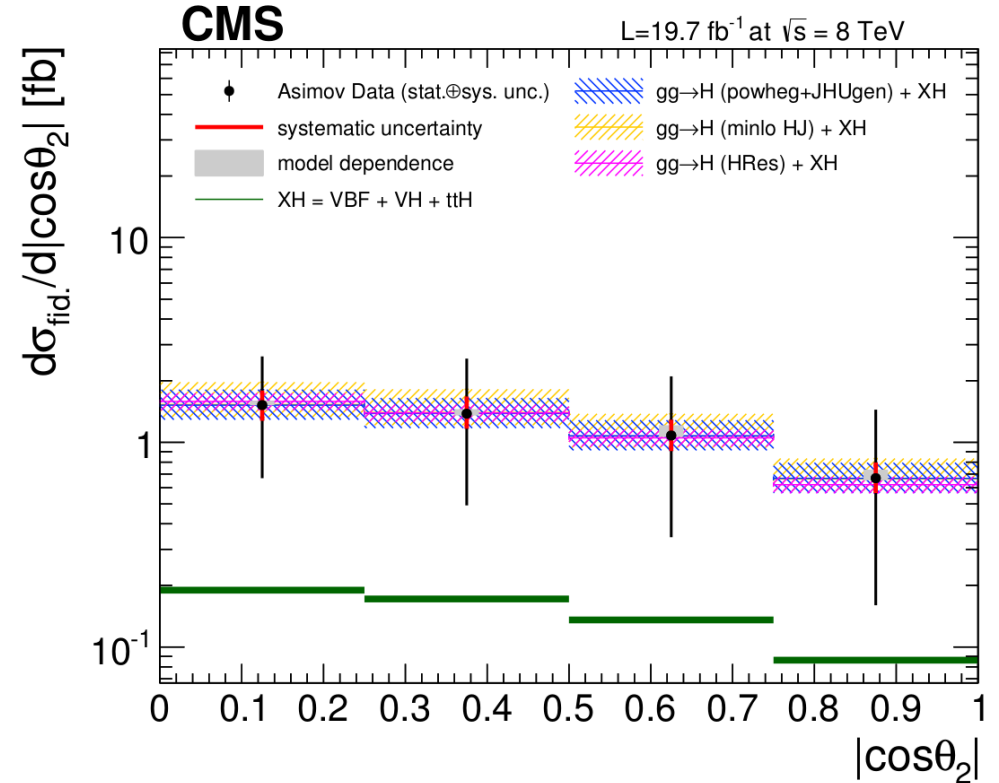
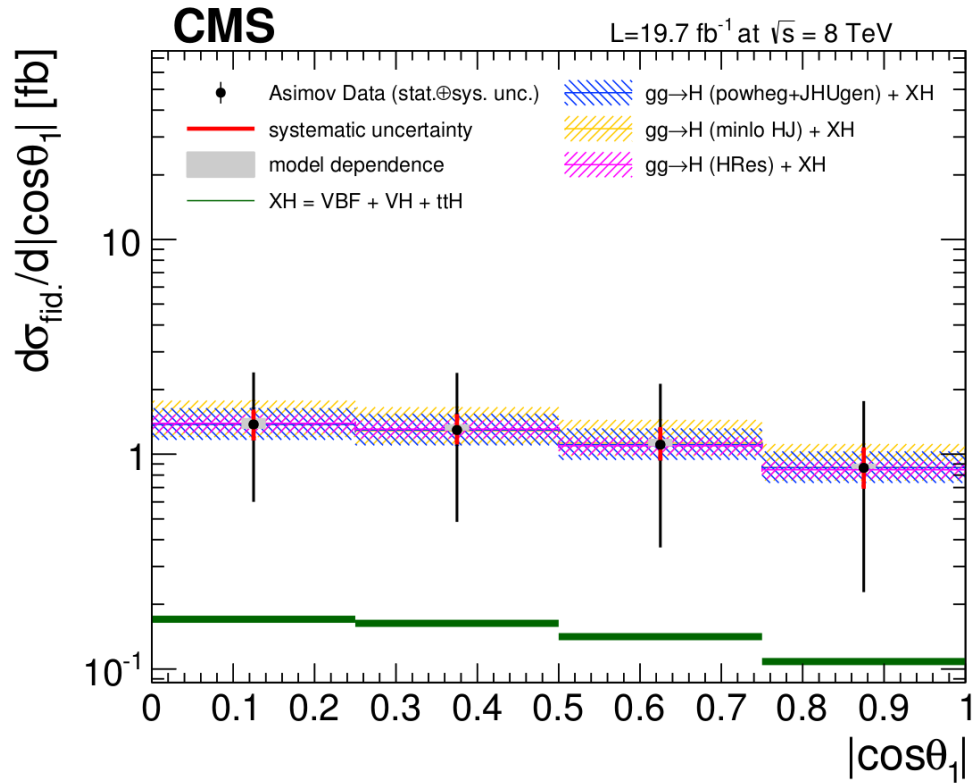
Results With Asimov Data



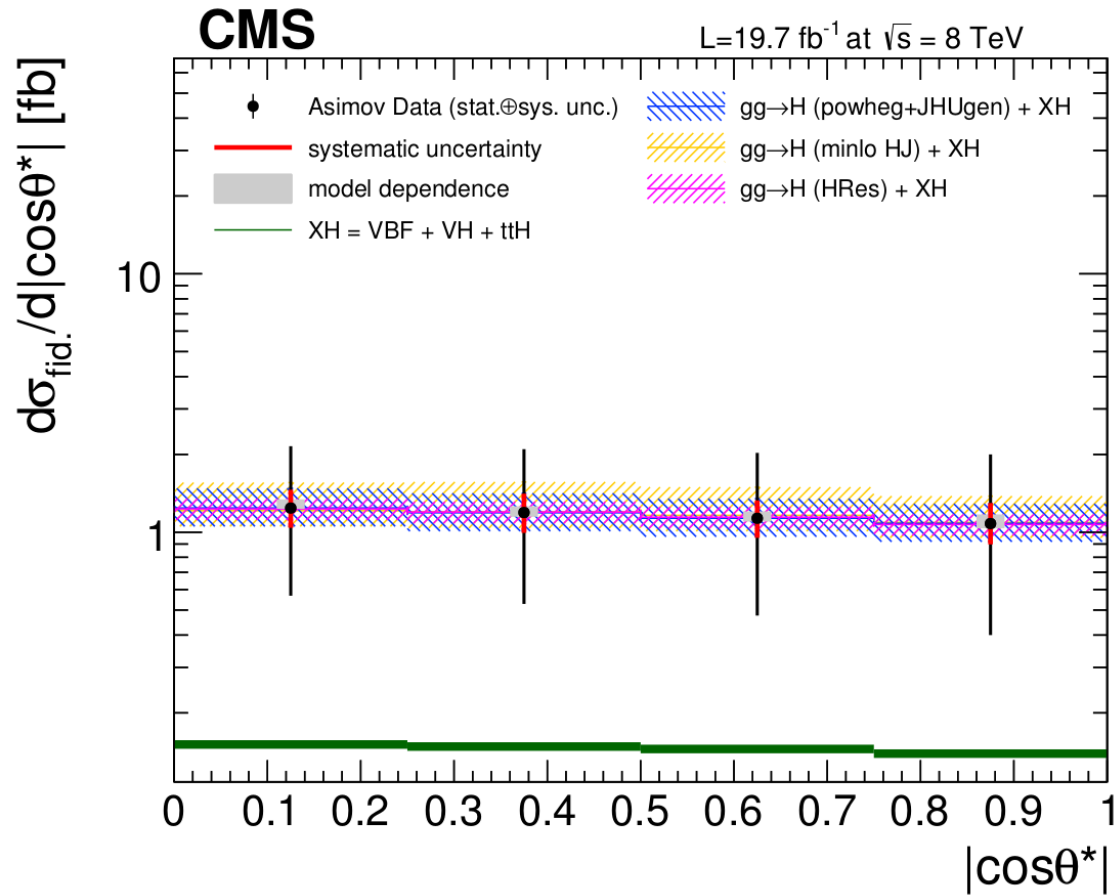
Results With Asimov Data



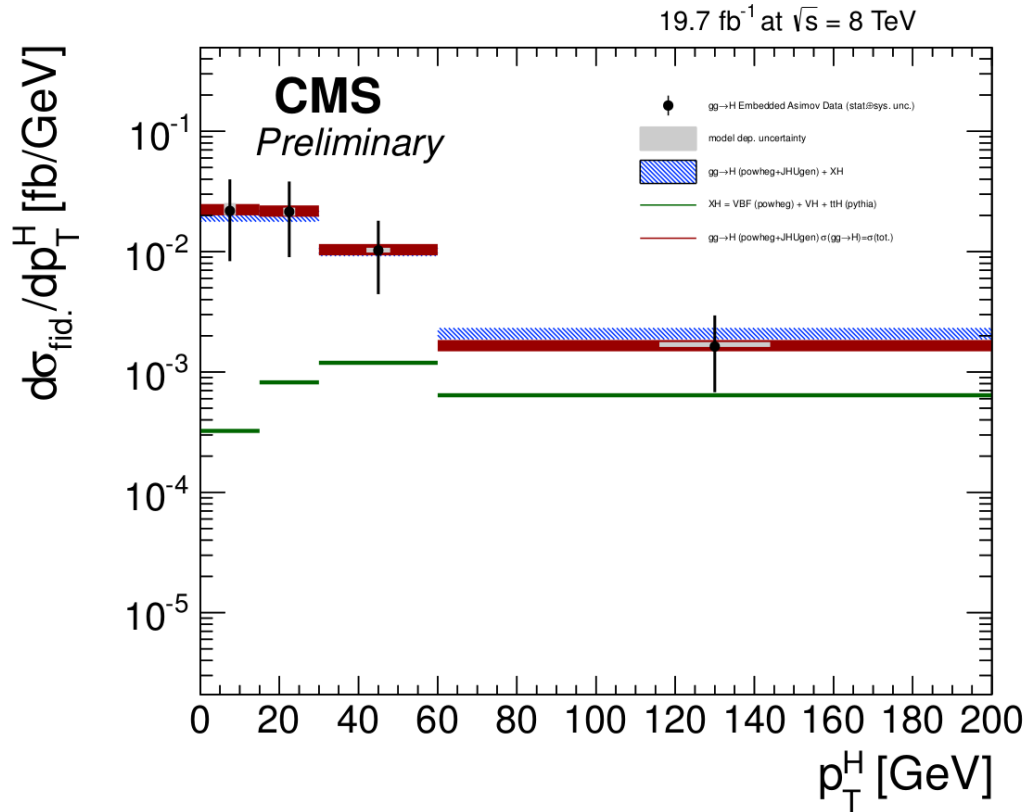
Results With Asimov Data



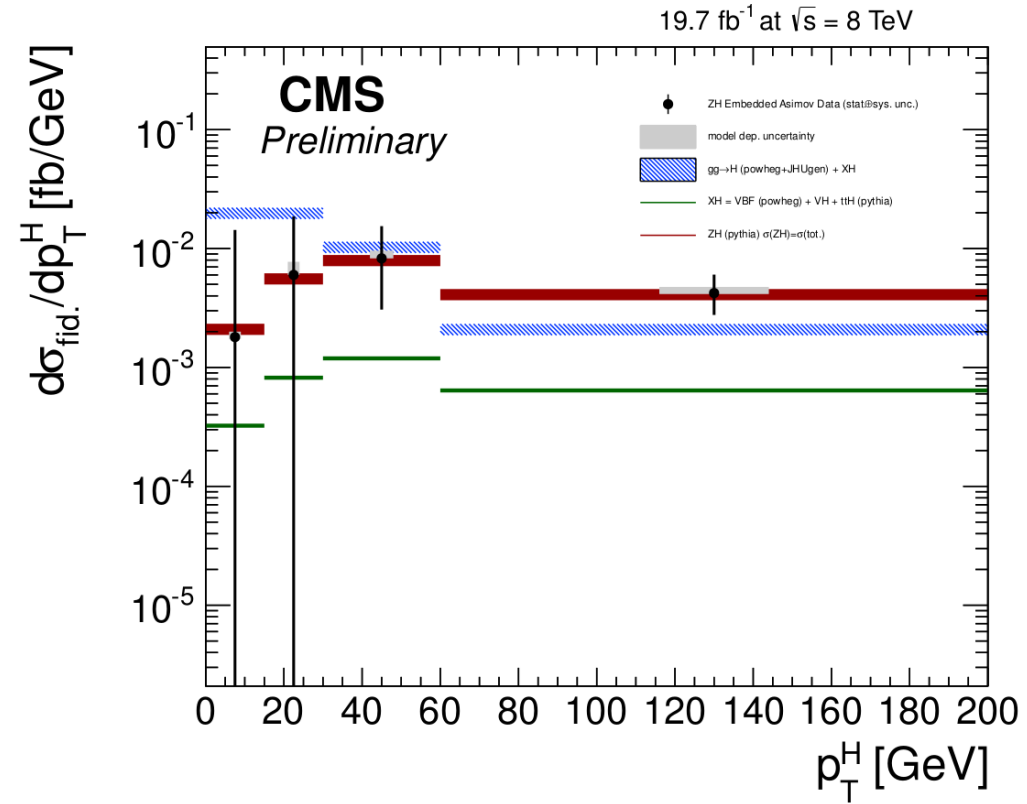
Results With Asimov Data



Closure Tests With Toys

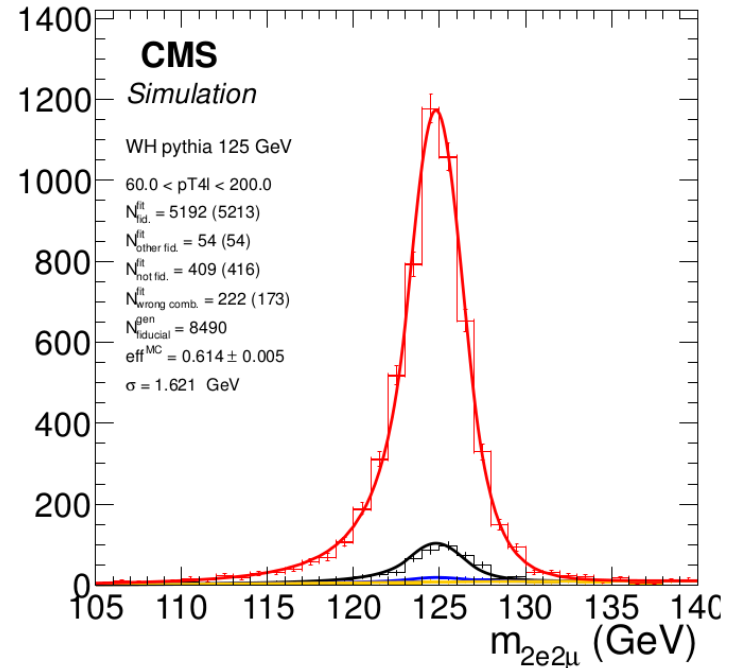
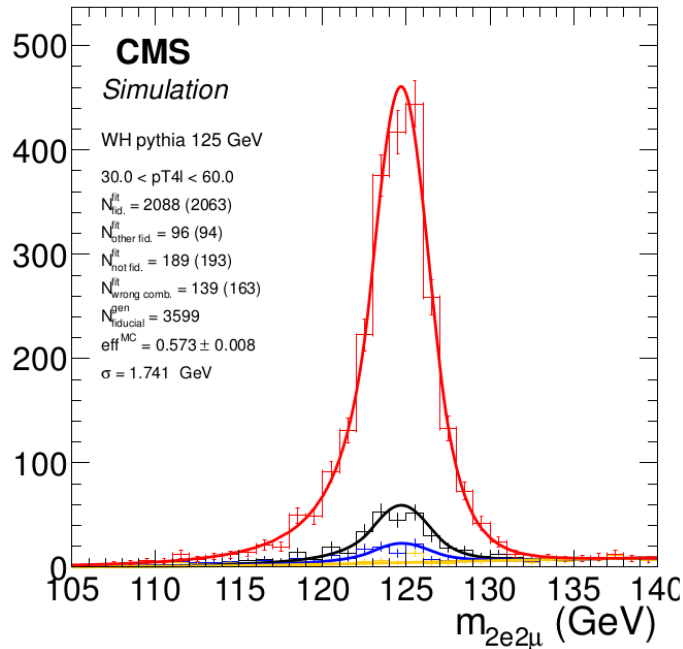
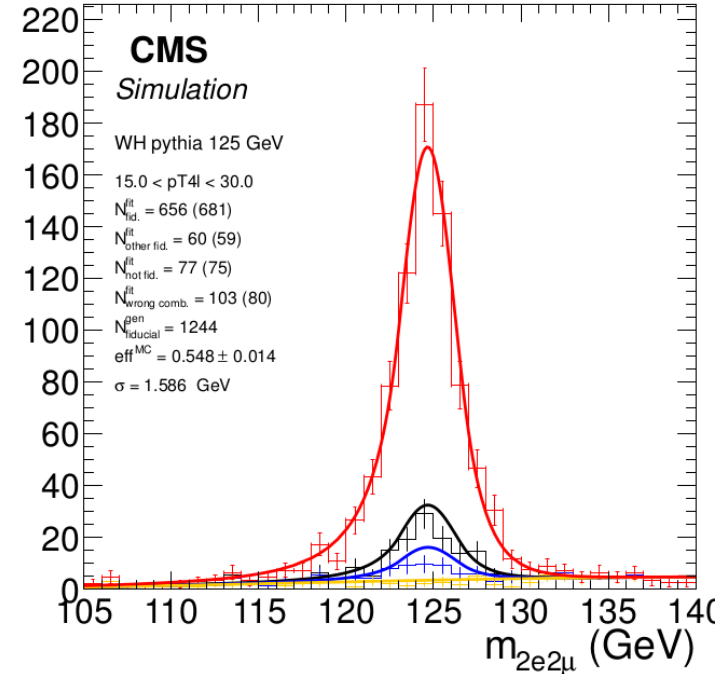
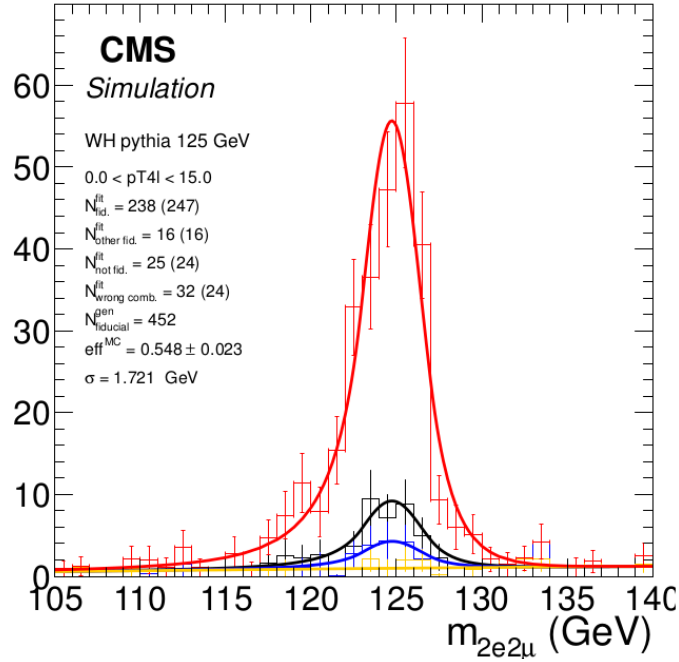


Closure test using embedded full-sim toy data (ggH production)



Closure test using embedded full-sim toy data (ZH production)

Full Sim $m(4\ell)$ shapes



No Isolation in Fiducial Volume

Table 11: Reconstruction efficiency (ϵ) for fiducial events per final state for different Standard Model signal models.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG+JHUGEN) 125 GeV	0.469 ± 0.004	0.834 ± 0.003	0.632 ± 0.003	0.647 ± 0.002
gg→H (POWHEG+JHUGEN)126 GeV	0.472 ± 0.004	0.834 ± 0.003	0.638 ± 0.003	0.652 ± 0.002
gg→H (POWHEG) 125 GeV	0.468 ± 0.004	0.821 ± 0.003	0.625 ± 0.003	0.639 ± 0.002
gg→H (POWHEG) 126 GeV	0.471 ± 0.004	0.825 ± 0.003	0.631 ± 0.003	0.644 ± 0.002
gg→H (POWHEG1.0) 125 GeV	0.465 ± 0.002	0.823 ± 0.002	0.626 ± 0.002	0.640 ± 0.001
gg→H (POWHEG1.0) 126 GeV	0.467 ± 0.004	0.821 ± 0.003	0.623 ± 0.003	0.638 ± 0.002
gg→H (MINLOHJJ) 125 GeV	0.459 ± 0.004	0.818 ± 0.003	0.623 ± 0.003	0.636 ± 0.002
gg→H (MINLOHJJ) 126 GeV	0.467 ± 0.004	0.827 ± 0.003	0.630 ± 0.002	0.644 ± 0.002
VBF (POWHEG) 125 GeV	0.485 ± 0.010	0.819 ± 0.007	0.644 ± 0.007	0.652 ± 0.005
VBF (POWHEG) 126 GeV	0.498 ± 0.010	0.836 ± 0.007	0.648 ± 0.007	0.663 ± 0.005
WH (PYTHIA) 125 GeV	0.452 ± 0.006	0.808 ± 0.004	0.615 ± 0.004	0.627 ± 0.003
WH (PYTHIA) 126 GeV	0.465 ± 0.006	0.807 ± 0.004	0.617 ± 0.004	0.630 ± 0.003
ZH (PYTHIA) 125 GeV	0.446 ± 0.009	0.822 ± 0.007	0.626 ± 0.006	0.634 ± 0.004
ZH (PYTHIA) 126 GeV	0.452 ± 0.009	0.817 ± 0.007	0.629 ± 0.006	0.638 ± 0.004
ttH (PYTHIA)125 GeV	0.431 ± 0.016	0.741 ± 0.013	0.609 ± 0.011	0.601 ± 0.008
ttH (PYTHIA)126 GeV	0.441 ± 0.016	0.783 ± 0.013	0.590 ± 0.011	0.602 ± 0.008

Efficiencies
with isolation:

Table 13: Reconstruction efficiency (ϵ) for fiducial events per final state for different Standard Model signal models without including isolation in the fiducial volume.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG+JHUGEN) 125 GeV	0.437 ± 0.004	0.764 ± 0.003	0.578 ± 0.003	0.594 ± 0.002
gg→H (POWHEG+JHUGEN)126 GeV	0.434 ± 0.004	0.768 ± 0.003	0.585 ± 0.003	0.599 ± 0.002
gg→H (POWHEG) 125 GeV	0.430 ± 0.004	0.745 ± 0.003	0.571 ± 0.003	0.584 ± 0.002
gg→H (POWHEG) 126 GeV	0.432 ± 0.004	0.752 ± 0.003	0.576 ± 0.003	0.589 ± 0.002
gg→H (MINLOHJJ) 125 GeV	0.430 ± 0.004	0.762 ± 0.003	0.581 ± 0.003	0.594 ± 0.002
gg→H (MINLOHJJ) 126 GeV	0.439 ± 0.004	0.771 ± 0.003	0.588 ± 0.002	0.602 ± 0.002
VBF (POWHEG) 125 GeV	0.451 ± 0.009	0.762 ± 0.008	0.600 ± 0.006	0.607 ± 0.004
VBF (POWHEG) 126 GeV	0.465 ± 0.009	0.774 ± 0.007	0.602 ± 0.006	0.616 ± 0.004
WH (PYTHIA) 125 GeV	0.387 ± 0.005	0.683 ± 0.005	0.519 ± 0.004	0.531 ± 0.003
WH (PYTHIA) 126 GeV	0.397 ± 0.005	0.683 ± 0.005	0.525 ± 0.004	0.536 ± 0.003
ZH (PYTHIA) 125 GeV	0.381 ± 0.008	0.697 ± 0.007	0.525 ± 0.006	0.536 ± 0.004
ZH (PYTHIA) 126 GeV	0.380 ± 0.008	0.695 ± 0.007	0.533 ± 0.006	0.540 ± 0.004
ttH (PYTHIA)125 GeV	0.240 ± 0.010	0.415 ± 0.011	0.338 ± 0.008	0.335 ± 0.005
ttH (PYTHIA)126 GeV	0.246 ± 0.010	0.434 ± 0.011	0.328 ± 0.007	0.335 ± 0.005

Efficiencies
without isolation:

No Isolation in Fiducial Volume

Table 12: Ratio of reconstructed events which are from outside the fiducial volume and reconstructed events which are from within the fiducial volume (f_{out}) per final state for different Standard Model signal models.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG+JHUGEN) 125 GeV	0.054 ± 0.003	0.049 ± 0.002	0.057 ± 0.002	0.053 ± 0.001
gg→H (POWHEG+JHUGEN)126 GeV	0.050 ± 0.003	0.051 ± 0.002	0.054 ± 0.002	0.052 ± 0.001
gg→H (POWHEG) 125 GeV	0.051 ± 0.003	0.058 ± 0.002	0.059 ± 0.002	0.058 ± 0.001
gg→H (POWHEG) 126 GeV	0.061 ± 0.003	0.053 ± 0.002	0.057 ± 0.002	0.057 ± 0.001
gg→H (POWHEG1.0) 125 GeV	0.056 ± 0.002	0.054 ± 0.001	0.060 ± 0.001	0.057 ± 0.001
gg→H (POWHEG1.0) 126 GeV	0.052 ± 0.003	0.054 ± 0.002	0.055 ± 0.002	0.054 ± 0.001
gg→H (MINLOHJJ) 125 GeV	0.051 ± 0.003	0.048 ± 0.002	0.052 ± 0.002	0.051 ± 0.001
gg→H (MINLOHJJ) 126 GeV	0.049 ± 0.002	0.049 ± 0.002	0.052 ± 0.002	0.051 ± 0.001
VBF (POWHEG) 125 GeV	0.042 ± 0.006	0.039 ± 0.004	0.039 ± 0.003	0.040 ± 0.002
VBF (POWHEG) 126 GeV	0.035 ± 0.005	0.031 ± 0.004	0.042 ± 0.004	0.037 ± 0.002
WH (PYTHIA) 125 GeV	0.071 ± 0.005	0.059 ± 0.003	0.081 ± 0.003	0.072 ± 0.002
WH (PYTHIA) 126 GeV	0.070 ± 0.005	0.058 ± 0.003	0.075 ± 0.003	0.068 ± 0.002
ZH (PYTHIA) 125 GeV	0.057 ± 0.007	0.075 ± 0.006	0.074 ± 0.005	0.072 ± 0.003
ZH (PYTHIA) 126 GeV	0.066 ± 0.007	0.072 ± 0.005	0.067 ± 0.004	0.068 ± 0.003
ttH (PYTHIA)125 GeV	0.115 ± 0.017	0.126 ± 0.013	0.157 ± 0.012	0.139 ± 0.008
ttH (PYTHIA)126 GeV	0.128 ± 0.018	0.143 ± 0.014	0.158 ± 0.012	0.148 ± 0.008

f_{out}
with isolation:

Table 14: Ratio of reconstructed events which are from outside the fiducial volume and reconstructed events which are from within the fiducial volume (f_{out}) per final state for different Standard Model signal models, without including isolation in the fiducial volume definition.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG+JHUGEN) 125 GeV	0.012 ± 0.001	0.004 ± 0.001	0.009 ± 0.001	0.008 ± 0.000
gg→H (POWHEG+JHUGEN)126 GeV	0.011 ± 0.001	0.004 ± 0.001	0.007 ± 0.001	0.007 ± 0.000
gg→H (POWHEG) 125 GeV	0.012 ± 0.001	0.004 ± 0.001	0.009 ± 0.001	0.008 ± 0.000
gg→H (POWHEG) 126 GeV	0.016 ± 0.002	0.004 ± 0.001	0.009 ± 0.001	0.008 ± 0.000
gg→H (MINLOHJJ) 125 GeV	0.013 ± 0.001	0.004 ± 0.001	0.008 ± 0.001	0.007 ± 0.000
gg→H (MINLOHJJ) 126 GeV	0.012 ± 0.001	0.004 ± 0.000	0.008 ± 0.001	0.008 ± 0.000
VBF (POWHEG) 125 GeV	0.008 ± 0.003	0.004 ± 0.001	0.006 ± 0.001	0.006 ± 0.001
VBF (POWHEG) 126 GeV	0.007 ± 0.002	0.003 ± 0.001	0.006 ± 0.001	0.005 ± 0.001
WH (PYTHIA) 125 GeV	0.023 ± 0.003	0.006 ± 0.001	0.016 ± 0.001	0.014 ± 0.001
WH (PYTHIA) 126 GeV	0.020 ± 0.002	0.006 ± 0.001	0.014 ± 0.001	0.012 ± 0.001
ZH (PYTHIA) 125 GeV	0.010 ± 0.003	0.010 ± 0.002	0.017 ± 0.002	0.013 ± 0.001
ZH (PYTHIA) 126 GeV	0.014 ± 0.003	0.009 ± 0.002	0.012 ± 0.002	0.011 ± 0.001
ttH (PYTHIA)125 GeV	0.039 ± 0.010	0.021 ± 0.005	0.040 ± 0.006	0.033 ± 0.004
ttH (PYTHIA)126 GeV	0.045 ± 0.010	0.021 ± 0.005	0.039 ± 0.006	0.034 ± 0.004

f_{out}
without isolation:

Fraction Parameterization

- Two parameters per bin, K1 and K2 with ranges:

K1 : range $[0, \text{maxK1}]$, where $\text{maxK1} = 1/\text{fracSM4e}$

K2 : range $[0, \text{maxK2}]$, where $\text{maxK2} = (1-\text{fracSM4e})/\text{fracSM4mu}$

- Physical cross sections expressed as:

$$\text{Sigma4e} = \text{Sigma} * \text{fracSM4e} * K1$$

$$\text{Sigma4mu} = \text{Sigma} * (1 - \text{fracSM4e} * K1) * K2 * \text{fracSM4mu} / (1 - \text{fracSM4e})$$

$$\text{Sigma2e2mu} = \text{Sigma} * (1 - \text{fracSM4e} * K1) * [1 - K2 * \text{fracSM4mu} / (1 - \text{fracSM4e})]$$

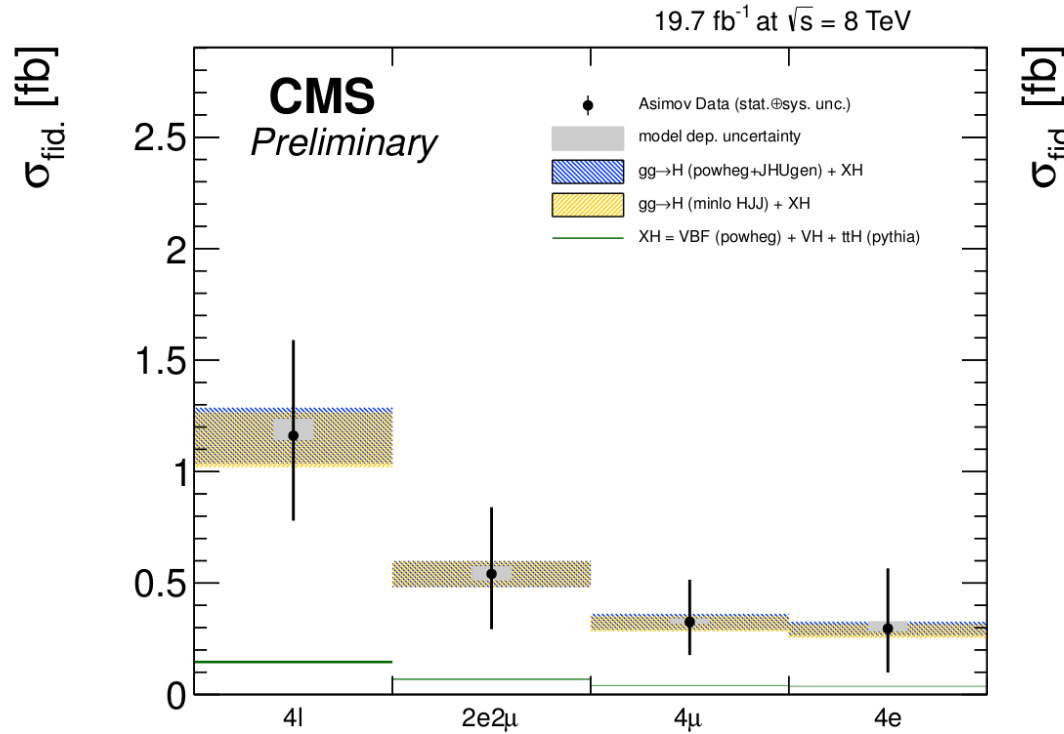
→ K1=K2=1.0 corresponds to SM values

→ Covers full parameter space

Expected precision: fixed/floated fractions

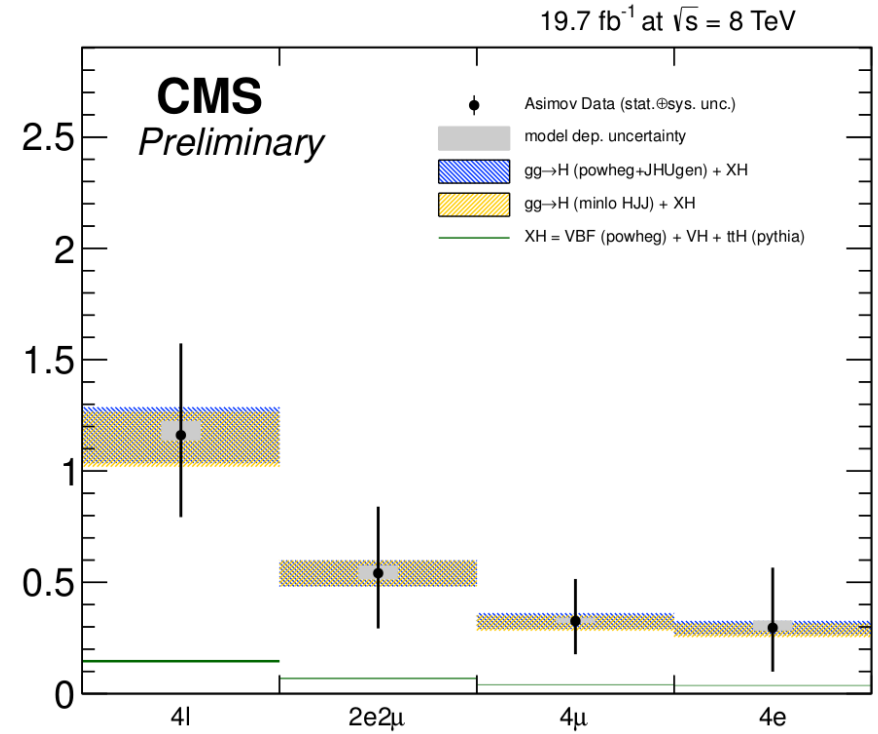
Inclusive
mass fixed, fractions floated:

$$\sigma(4\ell) = 1.161 -0.381/+0.429 \text{ fb}$$



Inclusive
mass fixed, fractions fixed:

$$\sigma(4\ell) = 1.161 -0.368/+0.412 \text{ fb}$$



Uncertainty on $\sigma(4\ell)$ very slightly reduced when fixing the fractions of 4e, 4μ, 2e2μ

Jet Energy Scale Uncertainty

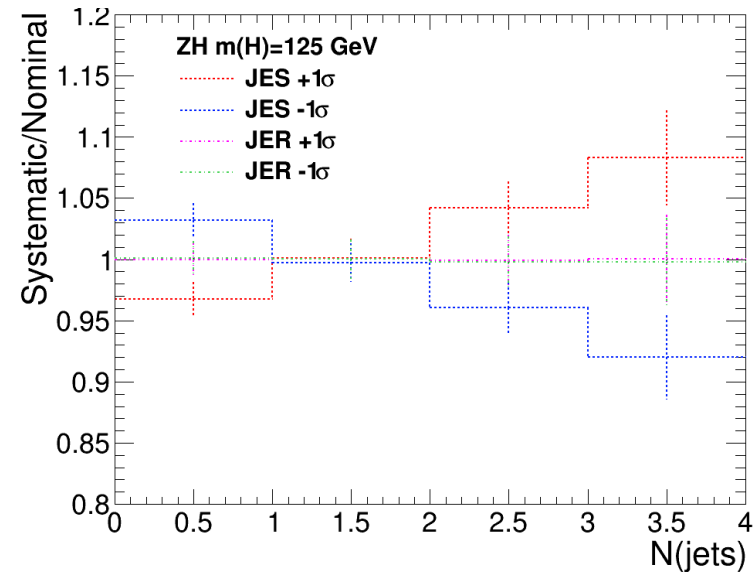
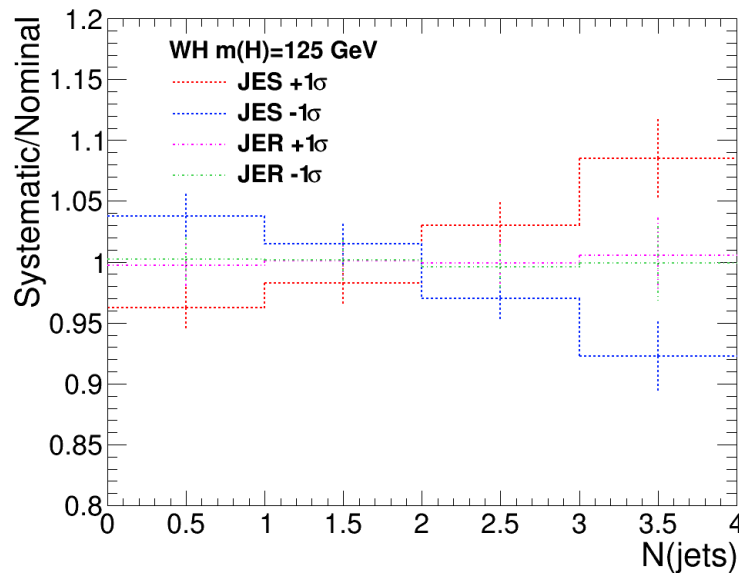
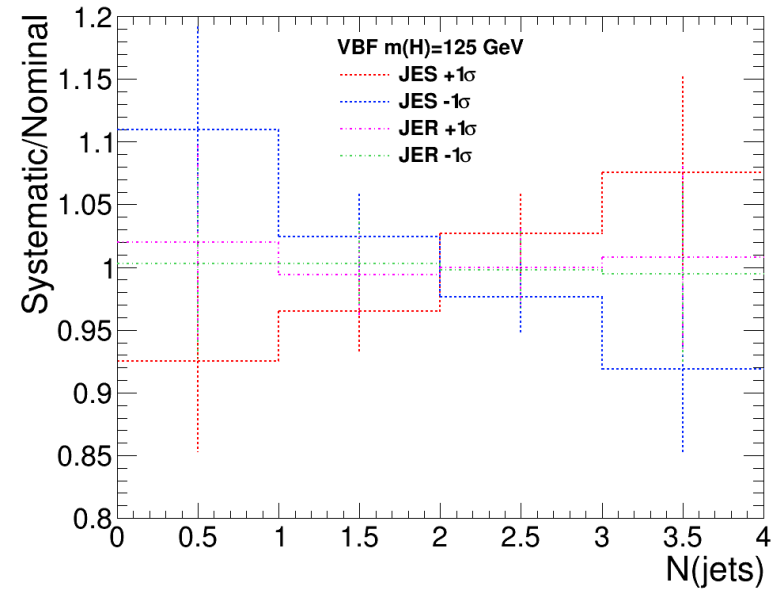
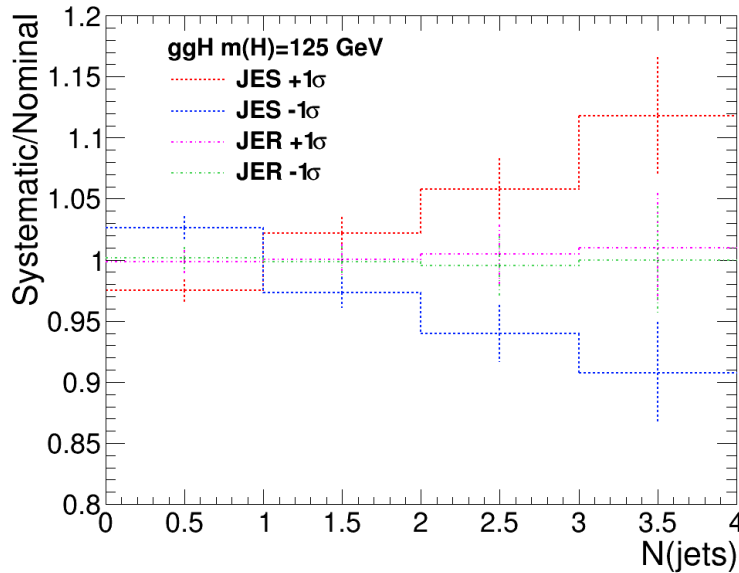
$$\text{eff}(N_{\text{reco-jet}}, N_{\text{gen-jet}} | \theta_{\text{JES}}) = (1 + \lambda(N_{\text{reco-jet}}, N_{\text{gen-jet}}) \theta_{\text{JES}}) \times \text{eff}(N_{\text{reco-jet}}, N_{\text{gen-jet}} | \theta_{\text{JES}} = 0)$$

$$\lambda(N_{\text{reco-jet}}, N_{\text{gen-jet}}) = \frac{\text{eff}(N_{\text{reco-jet}}, N_{\text{gen-jet}} | \theta_{\text{JES}} = \pm 1) - \text{eff}(N_{\text{reco-jet}}, N_{\text{gen-jet}} | \theta_{\text{JES}} = 0)}{\text{eff}(N_{\text{reco-jet}}, N_{\text{gen-jet}} | \theta_{\text{JES}} = 0)}$$

Sample	N(jets)=0	N(jets)=1	N(jets)=2	N(jets)≥3
gg→H (POWHEG+JHUGEN) 125 GeV	-0.027	0.024	0.075	0.104
gg→H (MINLOHJJ) 125 GeV	-0.023	0.028	0.073	0.120
VBF (POWHEG) 125 GeV	-0.088	-0.028	0.025	0.081
WH (PYTHIA) 125 GeV	-0.050	-0.028	0.029	0.097
ZH (PYTHIA) 125 GeV	-0.032	-0.034	0.032	0.071
ttH (PYTHIA) 125 GeV	0.000	-0.157	-0.053	0.010

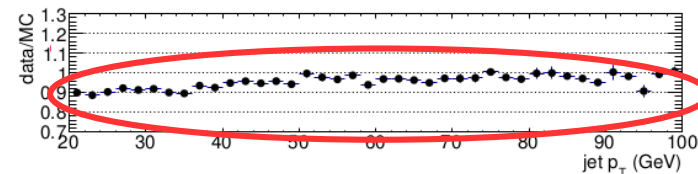
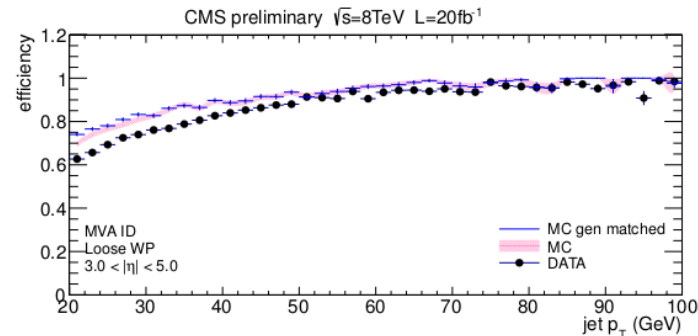
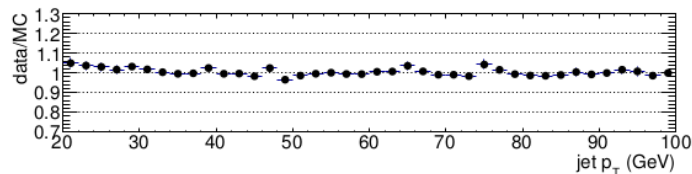
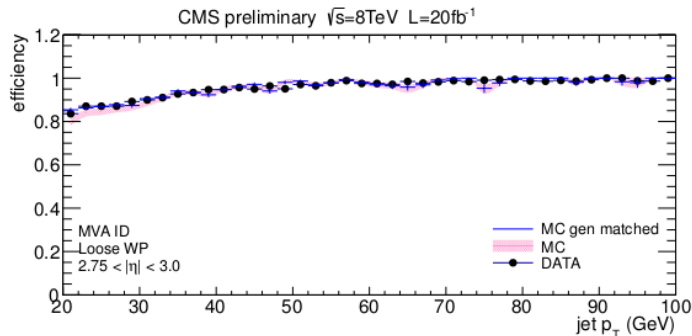
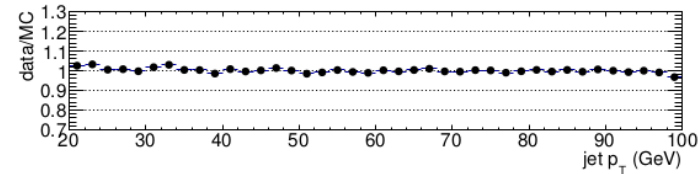
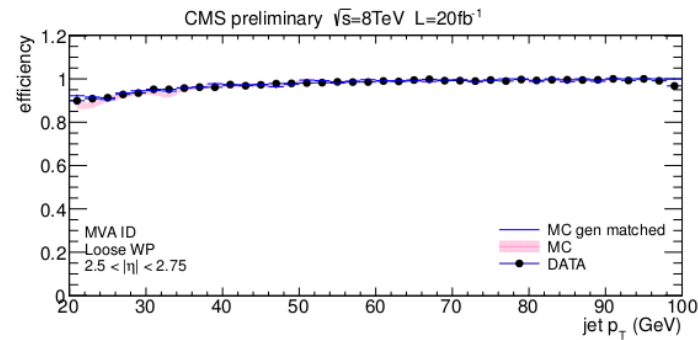
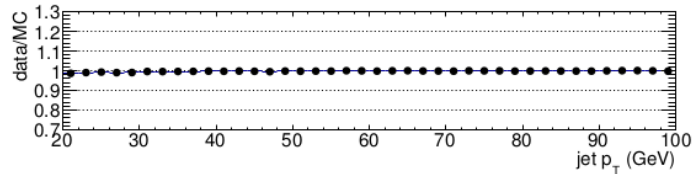
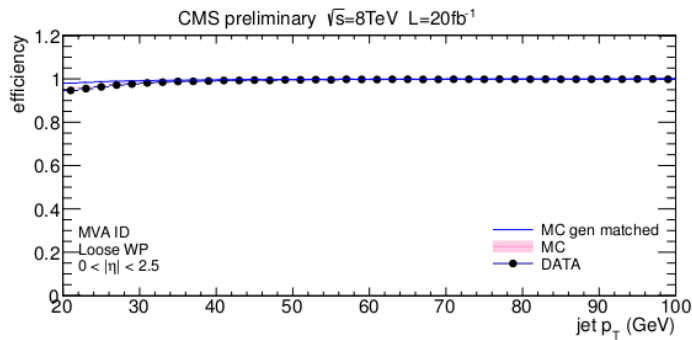
Jet Energy Resolution

- Implemented JER uncertainty according to Jet/MET recommendation
- Uncertainty due to JER is much smaller than (and covered by) JES $\rightarrow$ neglected



- Pileup jet ID efficiency measured in Data and agrees with simulation very well except for $|\eta| > 3.0$ (we use jets $p_T > 30$ GeV $|\eta| < 4.7$)
- Apply scale factor for $|\eta| > 3.0$ (0.90 for $30 < p_T < 36$, 0.94 for $36 < p_T < 50$, 0.97 for $p_T > 50$)
→ uncertainty is then small compared to JES

AN-2013-186 (JME-13-005)

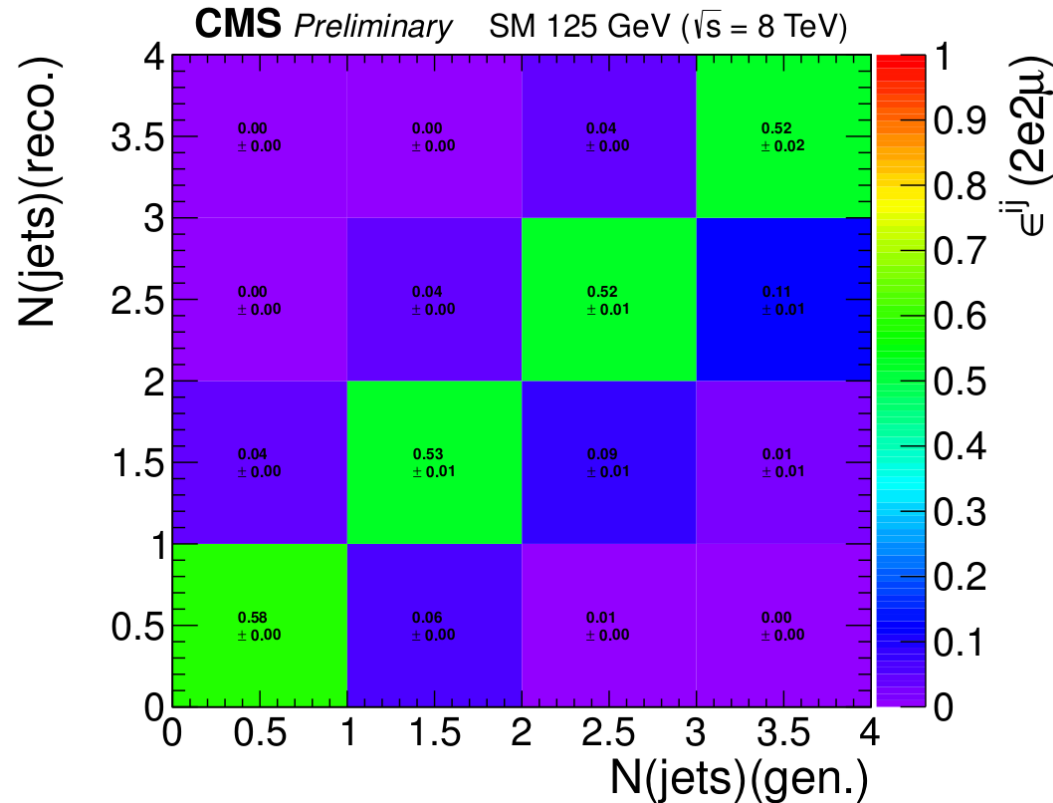


Effect JER / Pileup Jet ID Scale Factor

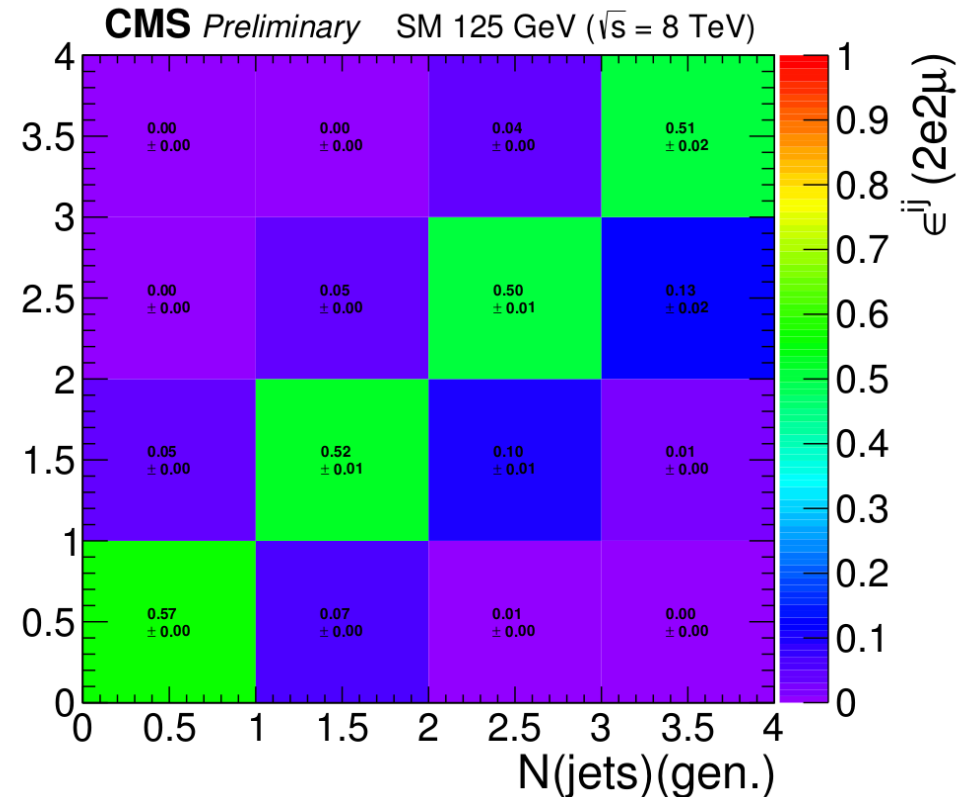


- MC smeared to match resolution in data
- Scale Factor for JetID applied for $|\eta| > 3.0$

OLD



NEW

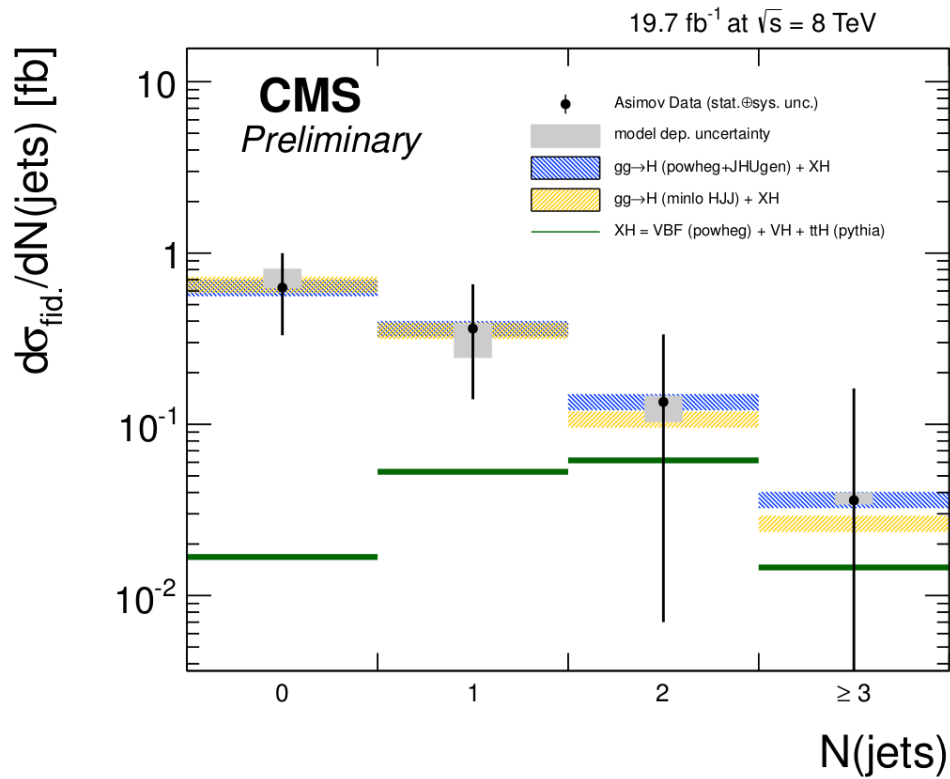


$pt > 30$ GeV , $|\eta| < 4.7$

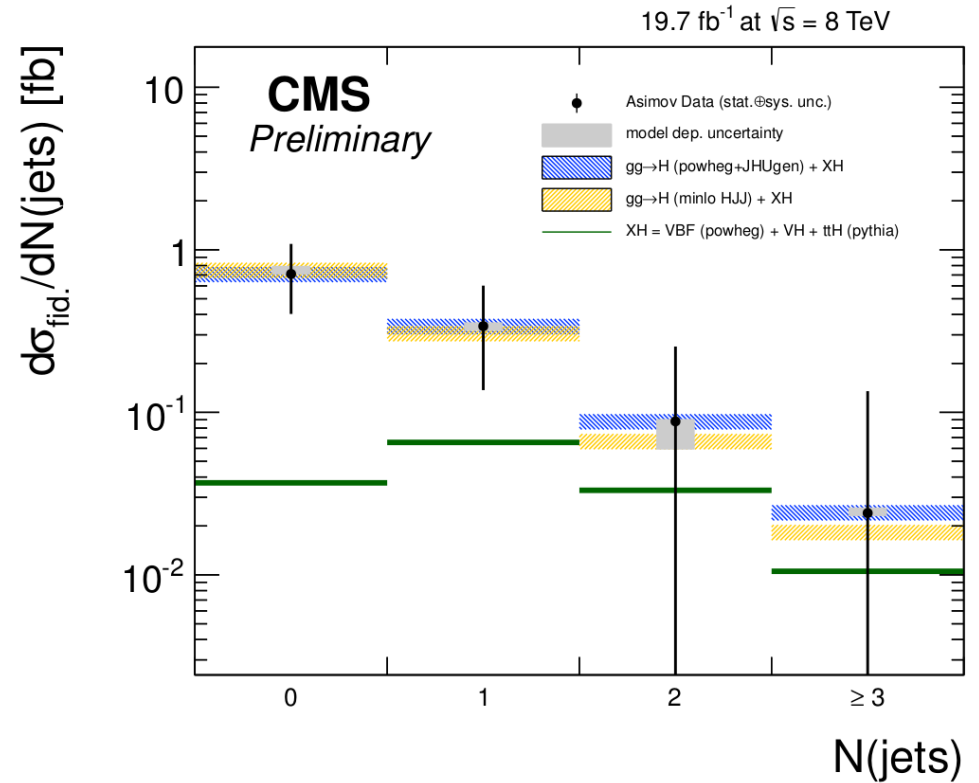
Effect of JER / Pileup Jet ID Scale Factor



- MC smeared to match resolution in data (no JER uncertainty)
- JetID scale factor for $|\eta| > 3.0$



pt > 30 GeV , $|\eta| < 4.7$



pt > 30 GeV , $|\eta| < 2.5$

- We have corrected a bug in the normalization of the qqZZ t-channel background prediction used in the $Z \rightarrow 4l$ and Ratio analysis
- Created asimov dataset from the H+Z model (105-140 GeV)
 - Fiducial cross section for H+Z mass window: $\sigma(4l)$: 1.162 fb
 - same as for smaller mass window because fiducial acceptance defined at generator level and Higgs has very small natural width
- Fit this asimov dataset with H model in the smaller mass window:
 - $\sigma(4l)$: 1.171 $-0.371/+0.446$ (68%) **Closure within 1%**
Well covered by uncertainties
- Potential reason for the difference:
 - Small differences in treatment of the background between the model
 - For $H \rightarrow 4l$, s+t channel qqZZ background normalization taken from integral of background pdf fitted to entire mass range (100-1000 GeV)
 - For $Z \rightarrow 4l$, for t-channel use predictions from MC (no fit)

Cross Section

Asimov $1.161 + 0.440 - 0.386$

All sys	= + 13.3 / - 13.5 %
Lepton eff	= + 9.3 / - 5.1 %
Bkg	= + 3.1 / - 1.8 %
Sig shape	= + 7.0 / - 5.5 %
Lumi	= + 4.4 / - 2.5 %
FakeH	= + 4.3 / - 10.6 %

Observed $1.106 + 0.433 - 0.360$

All sys:	= + 12.5 % / - 8.3 %
Lepton eff	= + 10.1 % / - 6.0 %
Bkg	= + 2.8 % / - 2.8 %
Sig shape	= + 5.6 % / - 3.9 %
Lumi	= + 4.4 % / - 2.8 %
FakeH	= + 0.4 % / - 0.6 %

Legacy

Asimov

All sys	= + 10.8 % / - 7.6 %
Lepton Eff	= + 7.6 % / - 5.0 %
Bkg	= + 2.6 % / - 2.4 %
Sig shape	= + 6.0 % / - 4.4 %
Lumi	= + 4.2 % / - 2.8 %

Observed

All sys	= + 10.3 % / - 7.3 %
Lepton eff	= + 7.6 % / - 5.0 %
Bkg	= + 2.6 % / - 2.4 %
Sig shape	= + 5.1 % / - 3.8 %
Lumi	= + 4.2 % / - 2.8 %

Systematics 8 TeV

Cross Section

Fixed Fractions

Asimov $1.161 + 0.420 - 0.3724$

All sys = + 12.1 % / - 12.8 %
 Lepton eff = + 7.9 % / - 4.5 %
 Bkg = + 2.9 % / - 1.6 %
 Sig shape = + 6.6 % / - 5.3 %
 Lumi = + 4.4 % / - 2.5 %
 FakeH = + 4.1 % / - 10.1 %

Observed $1.045 + 0.397 - 0.333$

All sys = + 10.8 % / - 7.5 %
 Lepton eff = + 8.0 % / - 5.1 %
 Bkg = + 2.9 % / - 2.8 %
 Sig shape = + 5.2 % / - 3.8 %
 Lumi = + 4.4 % / - 2.9 %
 FakeH = + 0.2 % / - 0.4 %

Legacy

Asimov

All sys = + 10.8 % / - 7.6 %
 Lepton Eff = + 7.6 % / - 5.0 %
 Bkg = + 2.6 % / - 2.4 %
 Sig shape = + 6.0 % / - 4.4 %
 Lumi = + 4.2 % / - 2.8 %

Observed

All sys = + 10.3 % / - 7.3 %
 Lepton eff = + 7.6 % / - 5.0 %
 Bkg = + 2.6 % / - 2.4 %
 Sig shape = + 5.1 % / - 3.8 %
 Lumi = + 4.2 % / - 2.8 %

Cross Section

Asimov

All sys	= + 22.3 % / - 18.2 %
Lepton eff	= + 18.5 % / - 6.4 %
Bkg	= + 3.6 % / - 2.7 %
Sig shape	= + 9.0 % / - 3.4 %
Lumi	= + 4.8 % / - 1.8 %
FakeH	= + 6.8 % / - 16.0 %

Observed

All sys	= + 38.2 % / - 11.1 %
Lepton eff	= + 21.5 % / - 6.6 %
Bkg	= + 7.8 % / - 6.2 %
Sig Shape	= + 31.0 % / - 8.1 %
Lumi	= + 6.7 % / - 2.1 %
FakeH	= + 3.6 % / -0.0%

Legacy

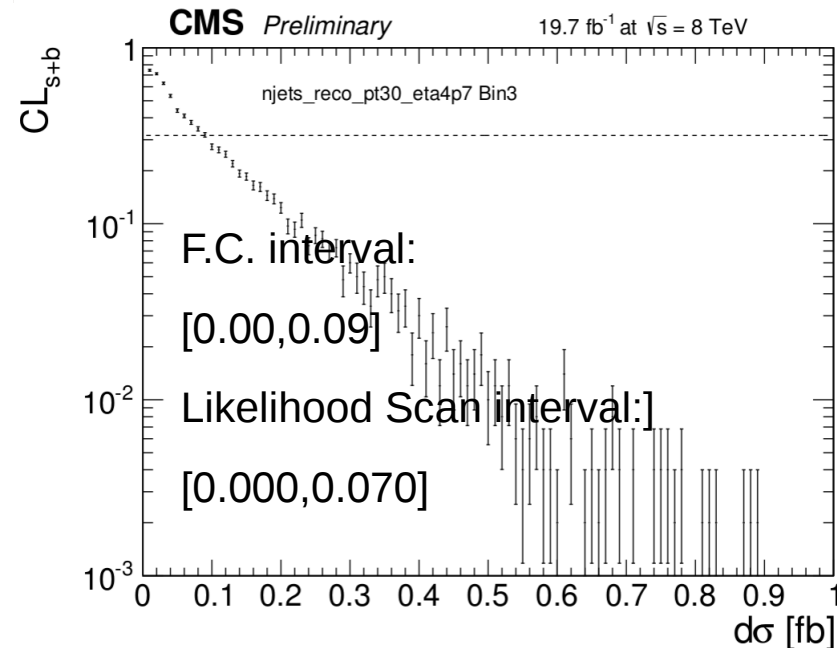
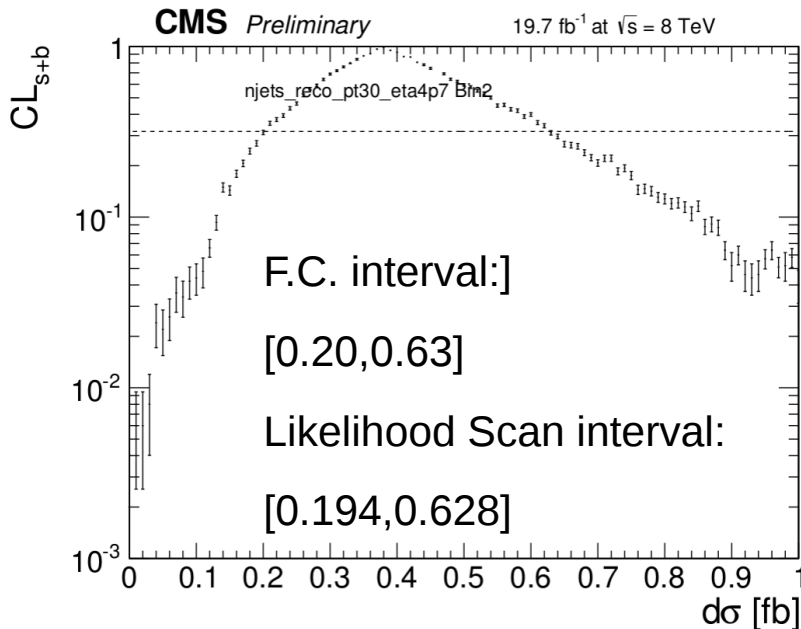
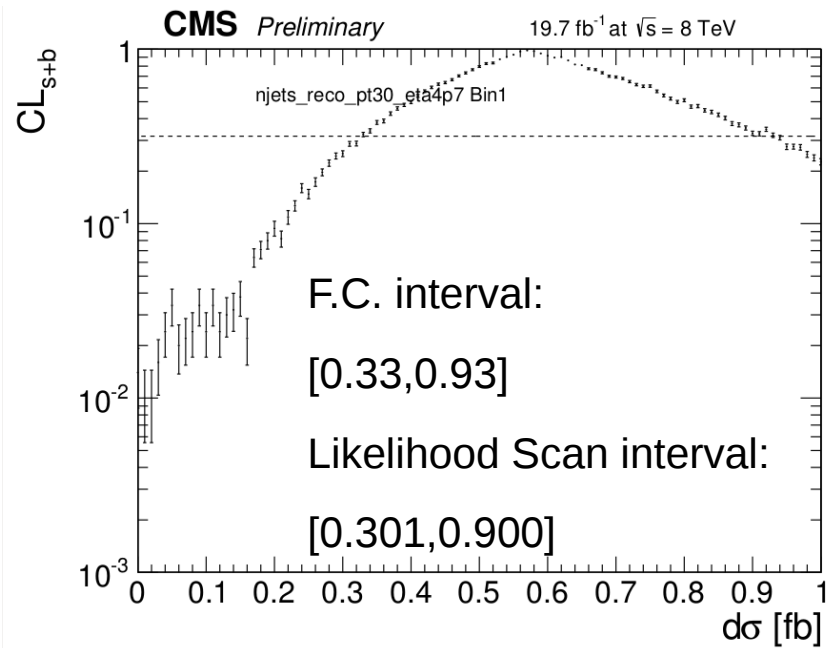
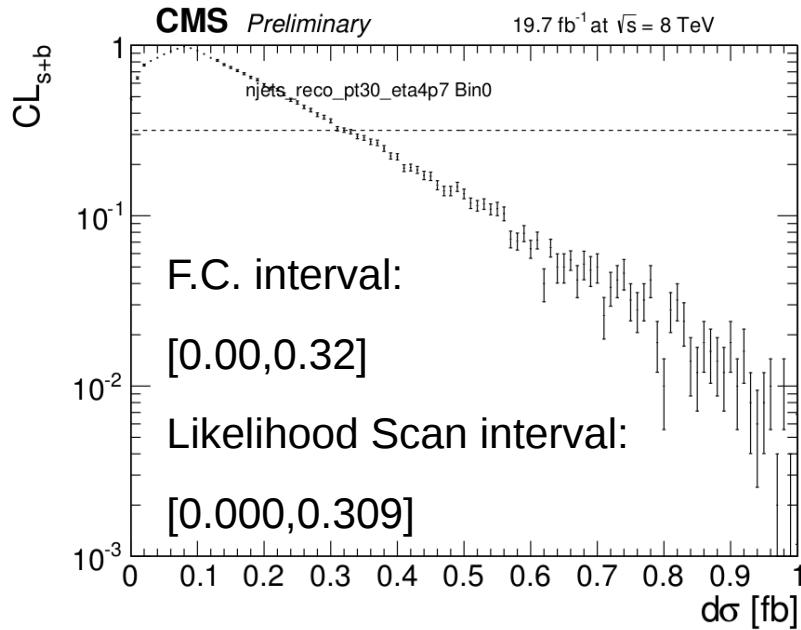
Asimov

All sys	= + 17.8 % / - 7.8 %
Lepton eff	= + 15.0 % / - 6.2 %
Bkg	= + 2.8 % / - 2.4 %
Sig shape	= + 7.9 % / - 3.4 %
Lumi	= + 4.6 % / - 1.0 %

Observed

All sys	= + 39.4 % / - 15.7 %
Lepton eff	= + 24.0 % / - 7.5 %
Bkg	= + 7.1 % / - 5.1 %
Sig Shape	= + 30.4 % / - 12.4 %
Lumi	= + 7.3 % / - 2.3 %

Feldman-Cousins Intervals



Step size= 0.01, CLs+b accuracy = 0.01

